Multi-pathway multi-echo (MPME) relaxometry: forward model, information limits, and efficient estimation

Technical note

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Abstract

This note gives a complete account of quantitative parameter mapping with the multi-pathway multi-echo (MPME) sequence of Cheng et al. [1]. We derive the steady-state forward model from the configuration-state description and prove that, for each scan, the signal depends on the flip angle and T_1 only through two invariants (Theorem 1); one scan therefore cannot separate M_0, B_1^+ and T_1, and for small flip angles two scans cannot either, to leading order. The published protocol escapes this through its near-360° second pulse. We implement the published analytic inverse and verify it, analyse the Fisher information and Cramér–Rao bounds (CRLB) of the model, identify its exact global aliases, and show that a joint maximum-likelihood (ML) fit attains the bound where the bound applies, while the analytic inverse is 3–7 times above it. For magnitude data we reduce the alias conditions to a square system and show that an exact alias exists inside the physiological B_1^+ range and explain why a magnitude fit needs the relative phase of the two FID images (one bit) while complex ML does not; with that bit, magnitude least squares performs like complex ML. Exact Jacobians obtained by implicit differentiation of the steady state make both fits 10–34 times faster, so that a 256×256 slice is fitted in seconds to minutes on a laptop CPU. A two-dimensional digital phantom with simulated acquisitions demonstrates all estimators, including the published low-resolution second scan, and validates per-voxel uncertainty maps; the BrainWeb phantom quantifies how partial volume with CSF biases every single-compartment estimator while leaving the residual and the uncertainty maps deceptively benign.

Summary of findings

1. **Two invariants per scan (exact).** Each pathway amplitude is $a_k = v\, G_k(u, E_2, \Delta\omega)$ with $u = (E_1 - \cos\alpha)/(1 - E_1\cos\alpha)$ and $v = \sin\alpha\,(1 - E_1)/(1 - E_1\cos\alpha)$ (Theorem 1). In terms of the Ernst angle $\alpha_E = \arccos E_1$, one scan measures only $\xi = \tan(\alpha/2)/\tan(\alpha_E/2)$, where the actual flip angle lies relative to the tissue's Ernst angle, and the amplitude $M_0 \tan(\alpha_E/2)$ (Proposition 2). A larger flip angle in a tissue with a shorter T_1 looks exactly like a smaller one in a tissue with a longer T_1, and the Fisher information about (M_0, B_1^+, T_1) is singular at every flip angle (Corollary 1).

2. **Small flip angles are degenerate to leading order.** In the joint limit of small α and T_R/T_1 the signal is invariant under $(M_0, B_1^+, T_1) \mapsto (M_0/\kappa, \kappa B_1^+, T_1/\kappa^2)$ (Proposition 4). With 15°/30° the information matrix has condition number 5×10^5; the parameters remain identifiable in principle but not usefully at realistic SNR.

3. **A single scan cannot be rescued by a smooth B_1^+ model.** For one scan, every physically plausible B_1^+ field—polynomial or otherwise—fits the data exactly, with compensating per-voxel T_1 and M_0 (Proposition 7); the profiled information of any parametric field is zero (Corollary 3). Voxels that share one B_1^+ slide together: their M_0 ratios are preserved exactly and their T_1 ratios to about 10^{-4}, but the common scale is undetermined (Proposition 8). A second acquisition with a different flip angle is required, but it can be short and coarse (Section 5).

4. **The near-360° pulse breaks the degeneracy.** At $B_1^+ = 1$, 330° and −30° carry the same information (Proposition 5); the benefit of 330° is the faster response of its angle to B_1^+. Two scans determine B_1^+ through $\tan(B_1^+\alpha_2/2)/\tan(B_1^+\alpha_1/2)$, in which T_1 and M_0 cancel, with the lever $\alpha_2/\sin\alpha_2 - \alpha_1/\sin\alpha_1 = -12.5$ (0.036 for 15°/30°). The condition number drops to 173 and the bound on $\log T_1$ is $40.5\,\sigma/M_0$.

5. **Exact global aliases.** Magnitudes admit an exact alias with $\alpha_2 = 360° + x$ inside the physiological range; complex data with a shared phase exclude it but admit one with $\alpha_2 > 540°$, removed by the range assumption $B_1^+ < 1.64$ (Proposition 6).

6. **The published inverse is inefficient; ML is efficient.** The analytic inverse is exact without noise but its T_1 standard deviation is 3.2–7.1 times the bound, with large bias and undefined estimates at moderate SNR. Joint ML attains the bound (ratio 0.99–1.03) where its assumptions hold.

7. **Magnitudes need one bit of phase.** Magnitudes carry essentially all the local information, but their likelihood has two equal maxima for every noise realisation; the FID-sign test resolves it, after which magnitude least squares matches complex ML.

8. **Exact Jacobians make it fast.** Implicit differentiation of the steady state gives exact Jacobians at the cost of about one model evaluation; fits are 10–12 times faster in double precision and up to 34 times in single precision.

9. **Phantom.** On a 256×256 digital phantom (37876 brain voxels) the estimators behave as the voxel-wise analysis predicts, the published low-resolution design costs precision in T_1 that a two-stage ML pipeline recovers in part, and the per-voxel uncertainty maps are calibrated (Section 9).

10. **Partial volume.** On BrainWeb, WM/GM mixtures are harmless (second-order model error), but voxels containing CSF are biased by far more than the noise, with uncertainty intervals that cover the truth in as few as 0.5% of voxels and residuals that a χ^2 test flags in only 3–15% at SNR 1000; the analytic inverse additionally jumps to a wrong flip-angle root in up to 38% of them (Section 10).

How to read this note

Section 1 fixes notation. Section 2 derives the forward model and proves its structural properties. Section 3 presents and verifies the published inverse. Section 4 analyses what the data can determine, locally (CRLB, conditioning) and globally (aliases); Section 5 shows why a second scan is indispensable and how small it can be. Sections 6 and 7 develop and evaluate maximum-likelihood estimation for complex data and least squares for magnitude data. Section 8 derives exact Jacobians and their consequences. Section 9 demonstrates everything on a digital phantom, and Section 10

measures the effect of partial volume on the BrainWeb phantom. Section 11 summarises which limitations belong to the forward model and which to an inverse, and lists future work (model mismatch, Rician ML, hierarchical estimation, learned estimators). Longer proofs are in Appendix A, algorithms in Appendix B, and reproduction instructions in Appendix C.

Contents

1 Notation and conventions **5**

2 The MPME sequence and its forward model **6**
2.1 Sequence . . . 6
2.2 Configuration states . . . 7
2.3 Exact steady state . . . 7
2.4 Two invariants per scan . . . 8
2.5 What the invariants mean, and what one scan measures . . . 9
2.6 Echo evolution and the voxel signal . . . 12
2.7 Closed forms for $k = 0, -1$ . . . 13
2.8 Imaging and noise . . . 13
2.9 Validation . . . 13

3 The published inverse **14**
3.1 Steps . . . 14
3.2 Consistency with the forward model . . . 15
3.3 Implementation and behaviour . . . 15

4 What the data determine **16**
4.1 Fisher information and the Cramér–Rao bound . . . 16
4.2 Logarithmic parameters . . . 16
4.3 One scan, and small flip angles . . . 16
4.4 How the published protocol escapes . . . 17
4.5 Bounds and conditioning . . . 18
4.6 Global aliases . . . 21
4.7 Information in magnitude data . . . 22

5 Why two scans are required **22**
5.1 Any B_1^+ field fits a single scan exactly . . . 23
5.2 Voxels that share B_1^+ . . . 24
5.3 What a prior would do instead . . . 26
5.4 Physics outside the ideal model . . . 27
5.5 What a second acquisition does . . . 28
5.6 How much second scan is needed . . . 28

6 Maximum-likelihood estimation **29**
6.1 Estimator . . . 29
6.2 Efficiency . . . 30

7 Least squares on magnitude images **34**
7.1 Why a magnitude fit needs the branch test and complex ML does not 34
7.2 Comparison on identical data 35

8 Exact Jacobians by implicit differentiation **36**
8.1 Differentiating the steady state 36
8.2 Chain rule 37
8.3 Levenberg–Marquardt with exact Jacobians 37
8.4 Verification and benchmark 37
8.5 Implications 37

9 Digital phantom **38**
9.1 Phantom and simulated acquisition 38
9.2 Reconstructions 40
9.3 Results 41
9.4 Uncertainty maps 46
9.5 Run times 46

10 Partial volume: the BrainWeb phantom **46**
10.1 The mixed-voxel signal 46
10.2 Phantom and protocol 47
10.3 Results 48
10.4 What partial volume does to the model and the estimators 50

11 Discussion **51**
11.1 Forward model or inverse? 51
11.2 Limitations 53
11.3 Future work 53

A Proofs and derivations **54**
A.1 Direct small-angle expansion of the isochromat solution 54
A.2 Fisher information of a Rician amplitude 54
A.3 Implicit differentiation 55
A.4 The pathway amplitudes are imaginary; the FID-sign identity 55

B Algorithms **55**

C Reproducibility **57**

1 Notation and conventions

Table 1: Principal symbols.

Symbol	Meaning
$i \in \{1,2\}$, j, k	scan, echo, pathway (configuration order)
T_R, t_{ijk}	repetition time; echo time after the RF pulse (ms)
α_i^{nom}, $\alpha_i = B_1^+ \alpha_i^{\mathrm{nom}}$	nominal and actual flip angle of scan i
B_1^+	transmit-field scale (dimensionless)
M_0	equilibrium magnetisation times receive sensitivity ($C \times M_0$)
T_1, T_2, T_2^*	relaxation times (ms); $T_2^* = 1/(1/T_2 + R_2')$
R_2'	reversible (Lorentzian) decay rate (1/ms), $R_2' \geq 0$
$\Delta\omega$, ϕ_0	off-resonance (rad/ms), global receive phase (rad)
E_1, E_2, ρ	e^{-T_R/T_1}, e^{-T_R/T_2}, T_R/T_1
F_k, Z_k	transverse and longitudinal configuration states
a_k	steady-state amplitude of F_k just after the pulse, per unit M_0
u, v	the two invariants of Theorem 1
S_{ijk}	noise-free voxel signal
σ	noise standard deviation per real/imaginary component
θ	fitted parameters (log of positive ones, see below)
$\mathcal{I}$	Fisher information matrix

Transverse magnetisation and phases. $M^+ = M_x + iM_y$. An RF pulse of flip α and phase φ is the rotation $R_z(\varphi)R_x(\alpha)R_z(-\varphi)$; from equilibrium it gives $M^+ = -i\mathrm{e}^{i\varphi}\sin\alpha$. One repetition of the unbalanced gradient multiplies M^+ by $\mathrm{e}^{i\psi}$, where ψ is the intravoxel phase, and free precession by $\Delta\omega$ for a time t multiplies it by $\mathrm{e}^{i\Delta\omega t}$ (the same sense). The RF phase is constant ($\varphi = 0$ unless stated).

Units. Times in ms, rates in 1/ms, $\Delta\omega$ in rad/ms ($1\,\mathrm{Hz} = 2\pi \times 10^{-3}$ rad/ms), angles in radians in formulas and degrees in text.

Parameters. The fitted vector is

$$\theta = (\log M_0,\ \log B_1^+,\ \log T_1,\ \log T_2,\ \log T_2^*,\ \Delta\omega,\ \phi_0), \tag{1}$$

with the last two absent for magnitude data. Positive scale parameters are logged (Section 4.2); $\Delta\omega$ and ϕ_0 are not. Bounds and spreads of log-parameters are *log-parameter standard deviations*: for a small value s this is the relative standard deviation; in general a spread s corresponds to a one-sigma factor $\mathrm{e}^{\pm s}$.

Signal-to-noise ratio. We write $\mathrm{SNR}(M_0) = M_0/\sigma$ with $M_0 = 1$ for the reference tissue. The brightest MPME image of the reference tissue has $|S| \approx 0.088\,M_0$, so the image SNR is about $0.088\,\mathrm{SNR}(M_0)$.

Reference tissue and protocol. Unless stated otherwise: $T_1/T_2/T_2^* = 1500/70/60$ ms (the tissue used for protocol optimisation in [1]), $B_1^+ = 1$, and the published in-vivo protocol (Table 2).

2 The MPME sequence and its forward model

2.1 Sequence

The MPME sequence [1] is a 3D gradient-echo sequence without RF spoiling: the RF phase is constant and the repetition time T_R is fixed. Only the readout gradient G_x is unbalanced; its net area per repetition, $A = \int_0^{T_R} G_x\, dt$, dephases the magnetisation by many cycles across a voxel (the *fully dephased* regime). Configuration states of different order are therefore displaced in k_x by multiples of γA, and a state of order k created by an RF pulse is refocused when the moment accumulated since the pulse equals $-kA$.

In the published design each of three readout windows of alternating polarity sweeps the moment across three adjacent multiples of A (lobe areas $+1.5 : -3 : +3 : -3 : +0.5$ in units of $-A$), so that every window samples the pathways $k \in \{+1, 0, -1\}$, separated by the time needed to traverse one A, which is $1/\mathrm{BW} \approx 2\,\mathrm{ms}$ (Fig. 1). The visiting order reverses in alternate windows. Two acquisitions with the same gradients but very different flip angles are performed; the second covers only the central part of k-space (Table 2). The actual flip angles are

$$\alpha_i = B_1^+\, \alpha_i^{\mathrm{nom}}, \tag{2}$$

so B_1^+ scales all pulses by the same factor.

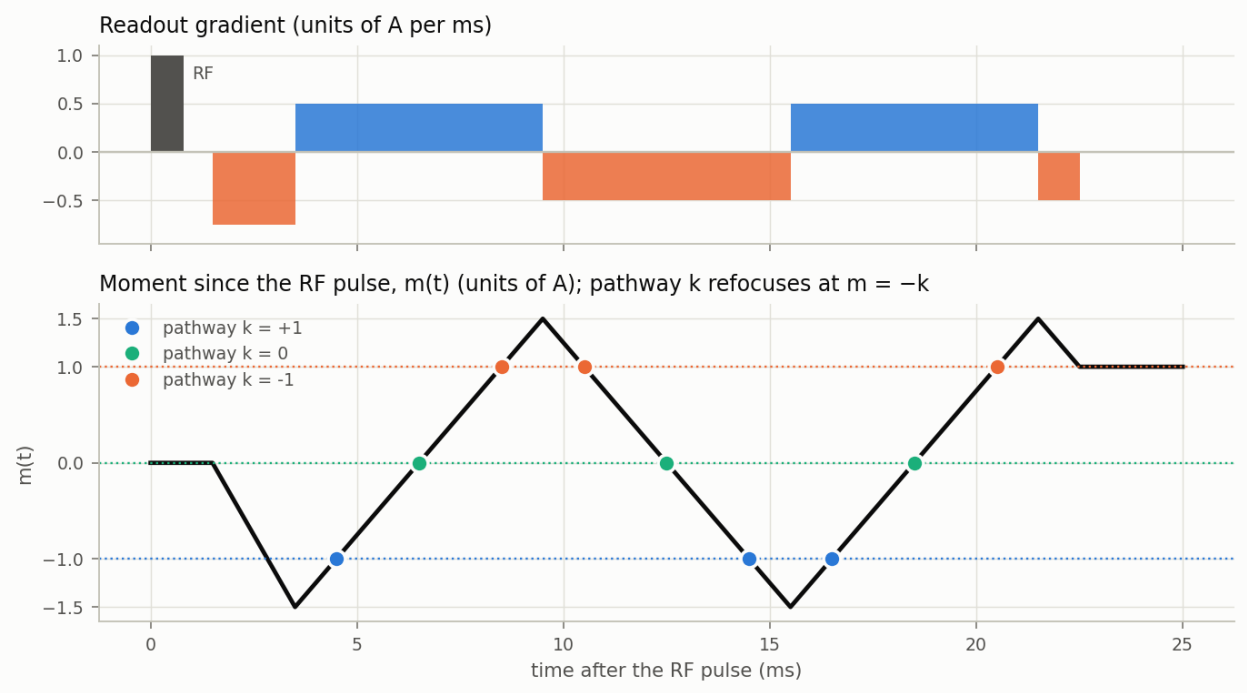


Figure 1: One repetition of the $[+1, 0, -1]$ scheme. Top: readout gradient (prephaser, three readout windows of alternating polarity, rewinder); net area A per repetition. Bottom: the moment accumulated since the RF pulse; pathway k is refocused (circles) whenever $m(t) = -k$. The time from the pulse to the first echo (4.5 ms) is an assumption; the paper does not state it.

Table 2: Published in-vivo protocol [1] and the values used in this note.

Repetition time	$T_R = 25\,\text{ms}$ (both scans)
Nominal flip angles	$\alpha_1^{\text{nom}} = 15°$, $\alpha_2^{\text{nom}} = 330°$ (hard pulses)
Pathways	$k \in \{+1, 0, -1\}$; three readout windows
Pathway spacing	2.0 ms in vivo (1.86 ms phantom); 501 Hz/pixel
Echo times (ms)	$k = +1$: 4.5, 14.5, 16.5; $k = 0$: 6.5, 12.5, 18.5; $k = -1$: 8.5, 10.5, 20.5
Scan 2 coverage	central 25% × 25% of k_y–k_z
Resolution, matrix	$1.2 \times 1.0 \times 1.2\,\text{mm}^3$, $160 \times 192 \times 160$, 11.5 min

2.2 Configuration states

Let $\psi \in [0, 2\pi)$ be the intravoxel phase imparted by one repetition of the unbalanced gradient and expand

$$M^+(\psi) = \sum_k F_k\, \mathrm{e}^{ik\psi}, \qquad M_z(\psi) = \sum_k Z_k\, \mathrm{e}^{ik\psi}, \qquad Z_{-k} = \overline{Z_k}. \tag{3}$$

One repetition consists of [4, 5]:

$$\begin{aligned}
\text{RF:}\quad & F_k' = \cos^2\tfrac{\alpha}{2} F_k + \mathrm{e}^{2i\varphi}\sin^2\tfrac{\alpha}{2}\overline{F_{-k}} - i\mathrm{e}^{i\varphi}\sin\alpha\, Z_k, \\
& Z_k' = -\tfrac{i}{2}\mathrm{e}^{-i\varphi}\sin\alpha\, F_k + \tfrac{i}{2}\mathrm{e}^{i\varphi}\sin\alpha\,\overline{F_{-k}} + \cos\alpha\, Z_k, \\
\text{relaxation:}\quad & F_k \to E_2 F_k, \quad Z_k \to E_1 Z_k\ (k \neq 0), \quad Z_0 \to E_1 Z_0 + M_0(1 - E_1), \\
\text{gradient:}\quad & F_k \to F_{k+1}.
\end{aligned} \tag{4}$$

We write $a_k(\alpha, T_1, T_2, \Delta\omega)$ for the steady-state amplitude of F_k immediately after the pulse, per unit M_0.

2.3 Exact steady state

The steady state can be computed without truncating the configuration space. An isochromat whose total precession per repetition is $\beta = \psi + \Delta\omega T_R$ has the post-pulse steady state

$$\big(I - R\,R_z(\beta)\,\mathrm{diag}(E_2, E_2, E_1)\big)\,\mathbf{M}^+ = (1 - E_1)\,R\,\hat{\mathbf{z}}, \qquad R = R_z(\varphi)R_x(\alpha)R_z(-\varphi), \tag{5}$$

and a_k is the k-th Fourier coefficient of $M^+(\psi) = M_x^+ + iM_y^+$ over ψ. On a uniform grid of n isochromats this is a discrete Fourier transform; orders with $|k| < n/2$ are exact up to aliasing of orders differing by n, which decay as E_2^n-like factors. For every tissue in this note, including CSF ($T_2 = 1.5\,\text{s}$) at 330°, $n = 128$ reproduces $n = 16\,384$ to 10^{-15}.

Proposition 1 (Off-resonance only adds phase)**.** *In the fully dephased regime*

$$a_k(\Delta\omega) = a_k(0)\,\mathrm{e}^{ik\Delta\omega T_R};$$

in particular $|a_k|$ does not depend on $\Delta\omega$.

Proof. By (5) the steady state depends on ψ and $\Delta\omega$ only through $\beta = \psi + \Delta\omega T_R$, so $M^+_{\Delta\omega}(\psi) = M^+_0(\psi + \Delta\omega T_R)$. Shifting a 2π-periodic function by θ multiplies its k-th Fourier coefficient by $\mathrm{e}^{ik\theta}$. □

2.4 Two invariants per scan

Theorem 1 (Exact two-invariant reduction)**.** *Let $\varphi = 0$ and, for one scan with flip angle α, define*

$$u = \frac{E_1 - \cos\alpha}{1 - E_1\cos\alpha}, \qquad v = \frac{\sin\alpha\,(1 - E_1)}{1 - E_1\cos\alpha}. \tag{6}$$

Then for every pathway k and every $\Delta\omega$

$$a_k(\alpha, T_1, T_2, \Delta\omega) = v\; G_k(u, E_2, \Delta\omega), \tag{7}$$

where G_k does not depend on α or T_1 otherwise.

Proof. Let (x, y, z) be the pre-pulse state of one isochromat and $w = x + iy$. The pulse $R_x(\alpha)$ maps it to $(x,\ cy - sz,\ sy + cz)$ with $c = \cos\alpha$, $s = \sin\alpha$; relaxation and precession by β then give the next pre-pulse state. The steady state therefore satisfies

$$w = E_2 e^{i\beta}\big(x + i(cy - sz)\big), \qquad z = E_1(sy + cz) + 1 - E_1. \tag{8}$$

The second equation gives $z = (E_1 sy + 1 - E_1)/(1 - E_1 c)$; note $1 - E_1 c > 0$. Substituting,

$$cy - sz = y\,\frac{c - E_1 c^2 - E_1 s^2}{1 - E_1 c} - \frac{s(1 - E_1)}{1 - E_1 c} = -u\,y - v.$$

The post-pulse transverse magnetisation is $m_\perp = x + i(cy - sz) = x - iuy - iv$, and $w = E_2 e^{i\beta} m_\perp$. Eliminating $x = \operatorname{Re} w$ and $y = \operatorname{Im} w$,

$$m_\perp = \operatorname{Re}\big(E_2 e^{i\beta} m_\perp\big) - iu\,\operatorname{Im}\big(E_2 e^{i\beta} m_\perp\big) - iv. \tag{9}$$

This is a real-linear equation for $m_\perp$ (two real unknowns) whose coefficients involve only u, E_2 and β, and whose inhomogeneous term is $-iv$. Its 2×2 real matrix is $I - E_2\, D_u\, R(\beta)$ with $D_u = \operatorname{diag}(1, u)$ and $R(\beta)$ a planar rotation; since $|u| \le 1$ and $E_2 < 1$, $\|E_2 D_u R(\beta)\| \le E_2 < 1$, so it is invertible. Hence $m_\perp = v\, g(u, E_2, \beta)$ with g independent of α and T_1, and taking Fourier coefficients over ψ (with $\beta = \psi + \Delta\omega T_R$) gives (7). □

Lemma 1. *$|u| \le 1$ and $|v| = \sqrt{1 - u^2}\,\sqrt{\tanh\big(T_R/(2T_1)\big)}$.*

Proof. From (6), $1 - u = (1 - E_1)(1 + c)/(1 - E_1 c)$ and $1 + u = (1 + E_1)(1 - c)/(1 - E_1 c)$, both non-negative, so $|u| \le 1$ and $1 - u^2 = (1 - E_1^2)\, s^2/(1 - E_1 c)^2$. Hence $v^2 = s^2(1 - E_1)^2/(1 - E_1 c)^2 = (1 - u^2)(1 - E_1)/(1 + E_1)$, and $(1 - E_1)/(1 + E_1) = \tanh(T_R/2T_1)$ for $E_1 = e^{-T_R/T_1}$. □

The invariants take their simplest form in terms of the Ernst angle $\alpha_E = \arccos E_1$, the flip angle that maximises the spoiled-gradient-echo signal of the tissue.

Proposition 2 (Ernst-angle form of the invariants)**.** *Let $\zeta = \tan(\alpha_E/2)$ and, for a flip angle α that is not an odd multiple of π,*

$$\xi = \frac{\tan(\alpha/2)}{\tan(\alpha_E/2)}, \qquad \vartheta = \arctan\xi \in (-\tfrac{\pi}{2}, \tfrac{\pi}{2}). \tag{10}$$

Then $\zeta^2 = (1 - E_1)/(1 + E_1) = \tanh(T_R/2T_1)$ and

$$u = \frac{\xi^2 - 1}{\xi^2 + 1} = -\cos 2\vartheta, \qquad v = \zeta\,\frac{2\xi}{1 + \xi^2} = \zeta \sin 2\vartheta. \tag{11}$$

Proof. Write $q = \tan(\alpha/2)$, so $\cos\alpha = (1-q^2)/(1+q^2)$ and $\sin\alpha = 2q/(1+q^2)$, and likewise $E_1 = \cos\alpha_E = (1-\zeta^2)/(1+\zeta^2)$, which gives $\zeta^2 = (1-E_1)/(1+E_1) = \tanh(T_R/2T_1)$. Then $E_1 - \cos\alpha = 2(q^2-\zeta^2)/D$, $1 - E_1\cos\alpha = 2(q^2+\zeta^2)/D$ and $\sin\alpha\,(1-E_1) = 4q\zeta^2/D$ with $D = (1+q^2)(1+\zeta^2)$. Substituting in (6), $u = (q^2-\zeta^2)/(q^2+\zeta^2)$ and $v = 2q\zeta^2/(q^2+\zeta^2)$; dividing numerator and denominator by ζ^2 gives (11), and $\xi = \tan\vartheta$ gives the trigonometric forms. □

Lemma 1 is the identity $u^2 + (v/\zeta)^2 = 1$: the pair $(u, v/\zeta)$ lies on the unit circle at angle 2ϑ. ξ measures where the actual flip angle lies relative to the Ernst angle: $\xi < 1$ ($u < 0$) below it, $\xi = 1$ ($u = 0$, $v = \zeta$, the largest value of v) at it, and $\xi \to \infty$ ($u \to 1$, $v \to 0$) as $\alpha \to \pi$. For $\alpha \ll 1$ and $T_R \ll T_1$,

$$\xi \approx \frac{\alpha}{\alpha_E} \approx \alpha\sqrt{\frac{T_1}{2T_R}}, \qquad \zeta \approx \sqrt{\frac{T_R}{2T_1}}, \tag{12}$$

so ξ^2 is the exact counterpart of the similarity variable $c = \alpha^2 T_1/2T_R$ of Proposition 4. At 15° and $T_R = 25$ ms the Ernst angles of WM, GM and CSF are 13.8°, 11.0° and 6.4°, so $\xi = 1.09$, 1.37 and 2.36 ($u = 0.08$, 0.30 and 0.69). The small-angle forms are close but not exact: for WM, $\xi^2 = 1.1787$, $(\alpha/\alpha_E)^2 = 1.1767$ and $\alpha^2 T_1/2T_R = 1.1652$.

On a logarithmic half-angle axis the invariants take an even simpler form. With $s = \ln|\tan(\alpha/2)|$ and $s_E = \ln\zeta = \ln\tan(\alpha_E/2)$, so that $|\xi| = \mathrm{e}^{s-s_E}$, (11) reads

$$u = \tanh(s - s_E), \qquad |v| = \zeta\ \mathrm{sech}(s - s_E), \qquad \operatorname{sign} v = \operatorname{sign}\tan(\alpha/2). \tag{13}$$

As a function of s, the spoiled-gradient-echo amplitude $|v|$ is a bump of fixed shape (sech), centred at the tissue's Ernst angle s_E and of height ζ. T_1 sets the position and height of the bump, and the actual flip angle sets the point s at which a scan samples it. Every log-derivative with respect to B_1^+ follows from $\mathrm{d}\ln|\tan(x/2)|/\mathrm{d}\ln x = x/\sin x$.

Corollary 1 (One scan cannot separate M_0, B_1^+, T_1)**.** *For a single scan, with T_2, T_2^*, $\Delta\omega$, ϕ_0 fixed, the signal depends on (M_0, B_1^+, T_1) only through the two functions $(u,\ M_0 v)$. The Jacobian of the signal with respect to $(\log M_0, \log B_1^+, \log T_1)$ has rank at most two, and the Fisher information about these three parameters is singular at every flip angle.*

Proof. By Theorem 1 and the echo factors below (which do not involve M_0, α or T_1 except through the prefactor M_0), $S = M_0 v\, h(u)$ for a function h independent of (M_0, B_1^+, T_1). By the chain rule, the 3 columns of $\partial S/\partial(\log M_0, \log B_1^+, \log T_1)$ are linear combinations of the 2 vectors $\partial S/\partial(u, M_0 v)$, so they span at most a two-dimensional space. The Fisher information restricted to these parameters is $J^{\mathsf{H}} J/\sigma^2$ with J of rank ≤ 2, hence singular. □

2.5 What the invariants mean, and what one scan measures

Intuition. Figure 2 draws (13). Plot the amplitude $M_0|v|$ against $s = \ln|\tan(\alpha/2)|$: it is a sech bump of height $M_0\zeta$ centred at the Ernst angle, and a scan samples it at $s = \ln|\tan(B_1^+\alpha^{\mathrm{nom}}/2)|$. The MPME sequence is not spoiled, but by Theorem 1 each of its nine signals is v times a function of u alone, so all of them report the same sample. The ratios between pathways give $u = \tanh(s - s_E)$, the offset of the sample from the centre of the bump, and the overall scale then gives the height $M_0\zeta$. Neither tells where the sample and the bump lie in absolute terms. Raising B_1^+ moves the sample to the right by $\ln[\tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)/\tan(B_1^+\alpha^{\mathrm{nom}}/2)] \approx \ln(B_1^{+\prime}/B_1^+)$; shortening T_1 by the right amount (about the square of the same factor, since $\zeta \approx \sqrt{T_R/2T_1}$) moves the bump by the same distance; and lowering M_0 by the factor restores the height. The whole picture translates rigidly, and the scan cannot tell. In words: a larger flip angle in a tissue with a larger Ernst angle (shorter T_1) is indistinguishable from a smaller flip angle in a tissue with a smaller Ernst angle.

A second scan samples the same bump at a different s. A 330° pulse is −30° modulo 360°, but its sample moves with B_1^+ at the rate $\mathrm{d}\ln|\tan(\alpha_2/2)|/\mathrm{d}\ln B_1^+ = \alpha_2/\sin\alpha_2 = -11.5$, against +1.01 for the 15° pulse. When the family translates the picture to keep scan 1, the scan-2 sample therefore lands elsewhere on the bump (Fig. 2a). A second small angle such as 30° moves at almost the same rate as the first (1.05), so its sample translates with the bump and nearly preserves the data (Fig. 2b). This is the leading-order degeneracy of Proposition 4 for small-angle protocols.

For small angles, $\xi^2 \approx \alpha^2 T_1/2T_R$ is the ratio of the saturation per pulse ($\propto \alpha^2$) to the recovery per repetition ($\propto T_R/T_1$), a useful mnemonic for the spoiled part of the signal. The exact invariant uses half-angle tangents: along the family of Table 4, the plain ratio of the flip angle to the Ernst angle drifts from 1.0851 to 1.0836, while $\xi = 1.0857$ is constant.

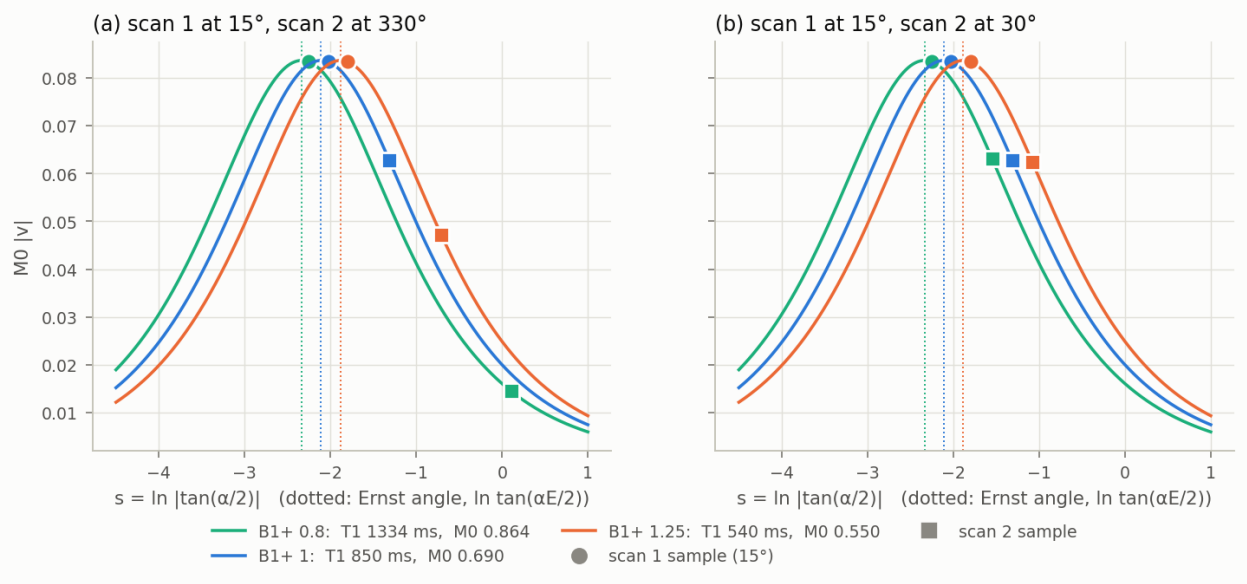


Figure 2: The single-scan degeneracy on the log half-angle axis (WM, $T_R = 25\,\mathrm{ms}$). Curves: the amplitude $M_0|v| = M_0\zeta\,\mathrm{sech}(s - s_E)$ for three members of the family (16) ($B_1^{+\prime} = 0.8$, 1, 1.25); dotted lines: their Ernst angles s_E. Circles: the 15° scan-1 sample of each member, at the same offset from its bump and the same height, hence identical scan-1 data. Squares: the scan-2 sample. (a) For 330° the scan-2 samples land at different heights (and, for $B_1^{+\prime} = 1.25$, with the opposite sign of v). (b) For 30° they almost coincide: a second small flip angle hardly separates the members.

Relation to the spoiled gradient echo. Let $m_z = (1 - E_1)/(1 - E_1\cos\alpha)$ be the longitudinal magnetisation (per unit M_0) just before each pulse in the steady state of a spoiled gradient echo with the same α, T_R and T_1. Then

$$v = \sin\alpha\, m_z, \qquad u = 1 - (1 + \cos\alpha)\, m_z, \tag{14}$$

since $1 - u = (1 - E_1)(1 + \cos\alpha)/(1 - E_1\cos\alpha)$. So v is the spoiled (Ernst) signal: the transverse magnetisation that each pulse creates from recovered longitudinal magnetisation (the *source*). $u \in [-1, 1]$ is the factor by which one pulse-and-relaxation cycle carries existing transverse magnetisation into the next repetition (the *retention*, an effective cosine). In the proof of Theorem 1 they appear as the inhomogeneous term and the coefficient of the transverse steady-state equation (9). Unbalanced SSFP differs from the spoiled sequence only in keeping the transverse magnetisation, and it sees α and T_1 only through this source and retention.

What one scan measures. A scan yields 3 pathways × 3 echoes = 9 complex numbers (18 real values). By Theorem 1 and Proposition 2 and (21), they depend on the parameters as listed in Table 3.

Table 3: Information content of one scan.

Feature of the data	Determined by
amplitude ratios between pathways, $a_k/a_0 = G_k(u)/G_0(u)$	u, i.e. ξ (and E_2)
overall scale, $M_0 v = M_0\zeta \sin 2\vartheta$	$M_0\zeta$, given ξ
decay across the echoes of each pathway	T_2 and R_2' (hence T_2^*)
phase change across echoes	$\Delta\omega$
common phase	ϕ_0

The 18 values therefore determine six quantities, (ξ, $M_0\zeta$, T_2, R_2', $\Delta\omega$, ϕ_0), and are redundant with respect to them (which averages noise and allows model checking). The deficiency is specific: the three unknowns (M_0, B_1^+, T_1) enter only through the two combinations ξ and $M_0\zeta$ (equivalently u and $M_0 v$). More echoes or more pathways from the same scan do not help, because every one of them depends on (α, T_1) through the same pair.

The data model for (M_0, B_1^+, T_1). Write ξ_i and A for the values of ξ and $M_0\zeta$ that the data of scan i determine. With $\alpha_i = B_1^+ \alpha_i^{\mathrm{nom}}$, the equations that (M_0, B_1^+, T_1) must satisfy are

$$\frac{\tan(B_1^+\alpha_i^{\mathrm{nom}}/2)}{\sqrt{\tanh(T_R/2T_1)}} = \xi_i, \qquad M_0\sqrt{\tanh(T_R/2T_1)} = A, \tag{15}$$

equivalent to $u(\alpha_i, E_1) = U_i$ and $M_0 v(\alpha_i, E_1) = V_i$ with $U_i = (\xi_i^2-1)/(\xi_i^2+1)$ and $V_i = 2A\xi_i/(1+\xi_i^2)$. Because all scans share T_R, the amplitude equation is the same for every scan; only ξ_i is new in each. For *one* scan, (15) is two equations in three unknowns, and its solutions form a one-parameter family. With the trial transmit scale $B_1^{+\prime}$ as the parameter, $\tan(\alpha_E'/2) = \tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)/\xi$, that is,

$$T_1'(B_1^{+\prime}) = \frac{T_R}{2\,\operatorname{artanh}\!\big(\tan^2(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)/\xi^2\big)}, \qquad M_0'(B_1^{+\prime}) = M_0\,\frac{\xi\,\zeta}{\tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)}. \tag{16}$$

In terms of E_1 this is $E_1' = (U+c')/(1+Uc')$ with $c' = \cos(B_1^{+\prime}\alpha^{\mathrm{nom}})$, the form used in Proposition 7. A member exists whenever $|\tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)| < |\xi|$, equivalently $c' > -U$ (the Ernst angle must stay below 90°). Every member reproduces the scan exactly (Proposition 7). With $x = B_1^{+\prime}\alpha^{\mathrm{nom}}$ and $\tau' = T_R/T_1'$, differentiating (16) gives

$$\frac{\mathrm{d}\ln M_0'}{\mathrm{d}\ln B_1^{+\prime}} = -\frac{x}{\sin x}, \qquad \frac{\mathrm{d}\ln T_1'}{\mathrm{d}\ln B_1^{+\prime}} = -2\,\frac{x}{\sin x}\,\frac{\sinh\tau'}{\tau'}, \tag{17}$$

the exact forms of the factors -1 and -2 of Proposition 4. For WM at 15° they are -1.0115 and -2.023, the slope measured in Table 9.

What a second flip angle adds. Dividing the first equation of (15) for $i = 2$ by that for $i = 1$ eliminates T_1 and M_0:

$$f(B_1^+) \equiv \frac{\tan(B_1^+\alpha_2^{\mathrm{nom}}/2)}{\tan(B_1^+\alpha_1^{\mathrm{nom}}/2)} = \frac{\xi_2}{\xi_1}. \tag{18}$$

This is one equation in B_1^+ alone, the half-angle analogue of the double-angle method. Its roots are isolated, and T_1 and M_0 then follow from (16). Its sensitivity to B_1^+ is the exact lever

$$L = \frac{\mathrm{d}\ln|f|}{\mathrm{d}\ln B_1^+} = \frac{\alpha_2}{\sin\alpha_2} - \frac{\alpha_1}{\sin\alpha_1} \qquad \text{(actual angles)}, \tag{19}$$

-12.53 for $15°/330°$ and $+0.036$ for $15°/30°$ at $B_1^+ = 1$ (Section 4).

The data fix ξ_i^2 through u_i. Complex data with a phase ϕ_0 shared by the two scans also fix the relative sign of v_1 and v_2, hence the sign of ξ_2/ξ_1; magnitude data fix only $|\xi_2/\xi_1|$. The aliases of Proposition 6 are the other roots of $f(B_1^{+\prime}) = f(B_1^+)$ (complex data) or $|f(B_1^{+\prime})| = |f(B_1^+)|$ (magnitude data). Their B_1^+ values depend on the true B_1^+ alone, not on the tissue; the tissue enters only through the alias T_1', from (16), and through admissibility ($\zeta' < 1$). For the published protocol and true $B_1^+ = 1$ ($f = -2.04$), the magnitude alias is $B_1^{+\prime} = 1.20$ ($f = +2.04$, scan 2 negated); it lies inside the physiological range and is removed only by the shared phase (the FID-sign test of Section 7). The complex alias is $B_1^{+\prime} = 2.01$, which the bound $B_1^+ \le 540/330$ excludes. Within the fitting range $B_1^+ \in [\mathrm{e}^{-1}, 540/330]$, the complex equation has a single root for every true B_1^+ between 6/11 and 1.592, i.e. true α_2 between 180° and 525°. Because f is the same function in every voxel, a B_1^+ shared by many voxels has the same aliases (Section 5.2).

A concrete example. Table 4 evaluates (16) for one white-matter voxel ($T_1 = 850\,\mathrm{ms}$, $M_0 = 1$, $B_1^+ = 1$, scan 1 at 15°, $T_R = 25\,\mathrm{ms}$; $\xi = 1.086$). As $B_1^{+\prime}$ runs from 0.8 to 1.5, the flip angle and the Ernst angle move together (their half-angle tangents keep the ratio ξ). T_1' falls from 1334 to 372 ms and M_0' from 1.25 to 0.66. Every member reproduces all nine complex scan-1 signals to machine precision, while the scan-2 (330°) signals differ by 41% to 200% (of the largest scan-2 signal) for every member except the truth. The predicted ratio ξ_2/ξ_1 for a 330° scan changes from -10.6 to $+12.1$ across the family, whereas for a 30° scan it changes only from 2.022 to 2.082. The second scan fixes B_1^+ when its angle moves differently with B_1^+; more scan-1 data cannot.

Table 4: The single-scan solution family (16) for one white-matter voxel (truth: $B_1^+ = 1$, $T_1 = 850\,\mathrm{ms}$, $M_0 = 1$). $B_1^{+\prime}\alpha$: actual scan-1 flip angle; α_E': Ernst angle of T_1'. Misfit: largest difference of the nine complex signals from the true ones, relative to the largest true signal of that scan. Predicted ξ_2/ξ_1: the ratio $f(B_1^{+\prime})$ of (18) that the member predicts for a second scan at 330° or 30°; the data have $f(1)$. Generated by `scripts/single_scan_family.py`.

					misfit		predicted ξ_2/ξ_1	
$B_1^{+\prime}$	$B_1^{+\prime}\alpha$ (°)	α_E' (°)	T_1' (ms)	M_0'	scan 1 (15°)	scan 2 (330°)	330°	30°
0.80	12.0	11.06	1334	1.253	1×10^{-15}	0.69	−10.57	2.022
0.90	13.5	12.44	1052	1.112	2×10^{-15}	0.41	−5.18	2.028
1.00	15.0	13.83	850	1.000	2×10^{-16}	0.00	−2.04	2.035
1.10	16.5	15.21	701	0.908	3×10^{-16}	1.36	+0.18	2.043
1.20	18.0	16.60	587	0.831	5×10^{-16}	2.00	+2.05	2.051
1.50	22.5	20.76	372	0.662	1×10^{-15}	1.27	+12.14	2.082

2.6 Echo evolution and the voxel signal

Pathway k read at time t after the pulse carries, for one isochromat, the off-resonance phase $\Delta\omega(t + kT_R)$ (Proposition 1 plus precession since the pulse). An intravoxel frequency distribution

with Lorentzian density $\ell(\delta) = (R_2'/\pi)/(R_2'^2 + \delta^2)$ averages this to

$$\int_{-\infty}^{\infty} \ell(\delta)\, \mathrm{e}^{i\delta\tau}\, d\delta = \mathrm{e}^{-R_2'|\tau|}, \tag{20}$$

the characteristic function of the Cauchy distribution. Together with transverse relaxation since the pulse, the noise-free signal of scan i, echo j, pathway k is

$$\boxed{S_{ijk} = M_0\, a_k\big(B_1^+\alpha_i^{\mathrm{nom}}, T_1, T_2, 0\big)\, \mathrm{e}^{-t_{ijk}/T_2}\, \mathrm{e}^{-R_2'|t_{ijk}+kT_R|}\, \mathrm{e}^{i(\phi_0+\Delta\omega(t_{ijk}+kT_R))}} \tag{21}$$

For $k \geq 0$ the magnitude decays at $1/T_2 + R_2' = 1/T_2^*$; for $k < 0$ and $t < |k|T_R$ at $1/T_2 - R_2'$, a spin echo forming at $t = |k|T_R$. Echo times differ between pathways, and $t + kT_R$ differs in sign between them, which is what separates T_2 from R_2'.

2.7 Closed forms for $k = 0, -1$

For the two lowest pathways the steady-state magnitudes have the closed forms of unbalanced FISP and PSIF,

$$|a_0| = \Big|\tan\tfrac{\alpha}{2}\Big[1 - \frac{(E_1 - \cos\alpha)(1 - E_2^2)}{r}\Big]\Big|, \qquad |a_{-1}| = \Big|\frac{\tan\frac{\alpha}{2}}{E_2}\Big[1 - \frac{(1 - E_1\cos\alpha)(1 - E_2^2)}{r}\Big]\Big|, \tag{22}$$
$$r^2 = p^2 - q^2, \quad p = 1 - E_1\cos\alpha - E_2^2(E_1 - \cos\alpha), \quad q = E_2(1 - E_1)(1 + \cos\alpha).$$

The PSIF formula in the literature is the echo amplitude at $t = T_R$, hence the factor $1/E_2$. Dividing p and q by $1 - E_1\cos\alpha$ shows that both brackets depend on (α, E_1) only through u, and $\tan(\alpha/2) = v/(1 - u)$, in agreement with Theorem 1.

2.8 Imaging and noise

Image formation $y_{ijkc} = \mathcal{M}_i\mathcal{F}(C_c S_{ijk}) + n_{ijkc}$ (coil maps C_c, Fourier transform $\mathcal{F}$, sampling $\mathcal{M}_i$) is linear. Voxel-wise, the coil-combined complex images are modelled as $S_{ijk} + n_{ijk}$ with independent circular Gaussian noise of standard deviation σ per real and imaginary part; magnitudes are then Rician. The published low-resolution scan 2 is modelled in Section 9 by truncation of k-space.

2.9 Validation

Table 5: Numerical checks of the forward model (all are unit tests).

Check	Agreement
Isochromat solver vs truncated EPG ($K = 400$), pathways $-3 \ldots 2$, four tissues incl. CSF	$< 10^{-5}$ absolute
$\lvert a_0\rvert$, $\lvert a_{-1}\rvert$ vs closed forms (22)	10^{-6} relative
Theorem 1: a_k at (α', E_1') with equal u equals $(v'/v)\, a_k$, six pathways, incl. 330°, $\Delta\omega \neq 0$	10^{-9} relative
Proposition 1: $a_k(\Delta\omega) = a_k(0)\mathrm{e}^{ik\Delta\omega T_R}$	10^{-10} absolute
Echo log-slopes $-(1/T_2 + R_2')$ ($k \geq 0$), $-(1/T_2 - R_2')$ ($k = -1$)	exact
128 vs 16 384 isochromats, CSF at 15° and 330°	10^{-15} relative

3 The published inverse

Cheng et al. [1] map the parameters by a sequence of closed-form steps applied to magnitude images. We present their sequential version (scan 2 acquired after scan 1), using their equation numbers, and verify it against the model of Section 2.

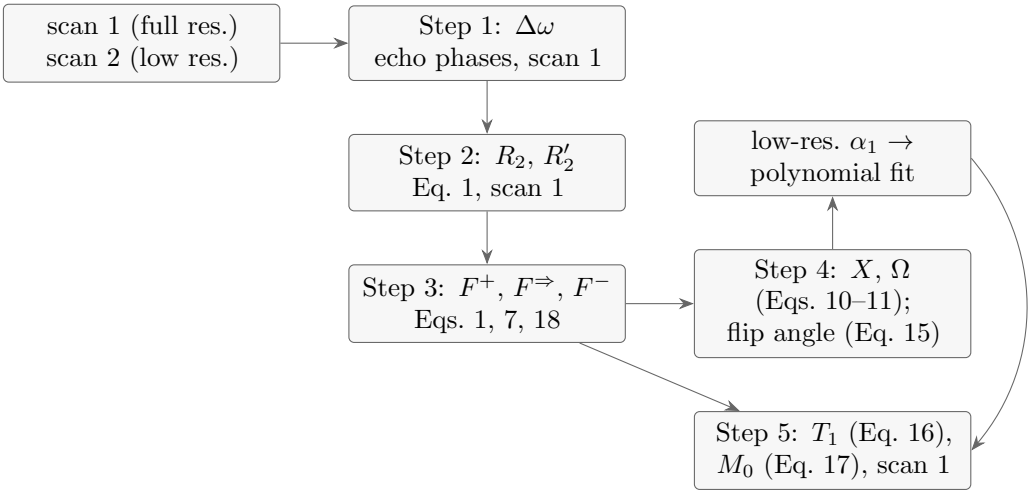


Figure 3: The published reconstruction [1]. The flip angle is computed at the resolution of scan 2 and smoothed by a polynomial; T_1 and M_0 are then computed voxel-wise from scan 1.

3.1 Steps

1. **Off-resonance.** $\Delta\omega$ from the phase evolution between echoes, all pathways jointly, scan 1.

2. **Transverse decay.** Their Eq. 1, $S_{i,j,k} = F^+_{i,k} \mathrm{e}^{-(R_2 \pm R_2')\mathrm{TE}_{i,j,k}}$ with + for $k \geq 0$ and − for $k < 0$, fitted log-linearly to scan 1 with weights $|S|$.

3. **Real-valued states.** $F^+_{i,k}$ (Eq. 1 at $\mathrm{TE} = 0$) and $F^{\Rightarrow}_{i,k}$ (at $\mathrm{TE} = T_R$), signed by their Eq. 18 ($+|F|$ for $k \geq 0$, $-|F|$ for $k < 0$). States before the next pulse follow from the gradient shift, $F^-_k = F^{\Rightarrow}_{k-1}$, and the missing one from their Eq. 7, $F^+_k - F^+_{-k} - F^-_k + F^-_{-k} = 0$.

4. **Flip angle.** The scaling factor $\Omega_{i,k} = F^-_{i,k} + F^-_{i,-k}$ (Eq. 10) and the *mixing factor* $X_{i,k} = 1 - 2(F^-_{i,k} - F^+_{i,k})/\Omega_{i,k}$ (Eq. 11), with $k = +1$ for the $[+1, 0, -1]$ scheme or $k = -1$ for $[0, -1, -2]$, then Eq. 15,

$$\left(\frac{X_1 c_1 - 1}{X_1 - c_1}\right)^{T_{R,2}/T_{R,1}} = \frac{X_2 c_2 - 1}{X_2 - c_2}, \qquad c_i = \cos\alpha_i, \quad \alpha_2 = \beta\,\frac{\alpha_2^{\mathrm{nom}}}{\alpha_1^{\mathrm{nom}}}\,\alpha_1, \tag{23}$$

   solved for α_1, with $\beta = 1$ in phantoms and $\beta = 1.24$ in vivo (their Eq. 19). The branch $\alpha_2 < 360°$ or $> 360°$ is chosen from the relative phase of F^+_0 in the two scans; $\alpha_2^{\mathrm{nom}} B_1^+ \in [180°, 540°]$ is assumed.

5. T_1 **and** M_0**.** Their Eq. 16, read as

$$T_1 = T_R \Big/ \ln\frac{X_1 c_1 - 1}{X_1 - c_1}, \tag{24}$$

   and Eq. 17 for M_0, both from scan 1.

Equation (24) follows from their own derivation ($Z^{+}/Z^{\Rightarrow} = \mathrm{e}^{T_R/T_1}$); the extracted text of the author manuscript prints $T_R \ln(\cdot)$, which does not reproduce T_1. Their Eq. 8, a nutation function $\nu(\alpha, \Delta f)$ that corrects $\cos\alpha$ for off-resonance during a finite hard pulse, is not used here because our pulses are instantaneous.

3.2 Consistency with the forward model

The intercepts of Eq. 1 are not the configuration amplitudes a_k of Section 2: by (21) the magnitude at TE = 0 is $M_0|a_k|\mathrm{e}^{-R_2'|k|T_R}$. The next proposition shows that this does not matter.

Proposition 3 (The published relations hold for the forward model)**.** *Define $\tilde{F}_k = F_k\,\mathrm{e}^{-R_2'|k|T_R}$ for transverse and $\tilde{Z}_k = Z_k\,\mathrm{e}^{-R_2'|k|T_R}$ for longitudinal states. (i) The quantities that Eq. 1 provides are $\tilde{F}_k^{+}$ and $\tilde{F}_k^{\Rightarrow}$. (ii) Every relation used in Steps 3–5 (Eqs. 2–7 and 10–17) holds for the tilde states whenever it holds for the configuration states. Consequently the published inverse recovers the exact parameters from noise-free data of* (21)*.*

Proof. (i) At TE = 0, (21) gives magnitude $M_0|a_k|\mathrm{e}^{-R_2'|k|T_R}$, i.e. $|\tilde{F}_k^{+}|$; extrapolating the same fit to TE = T_R gives $M_0|a_k|\mathrm{e}^{-T_R/T_2}\mathrm{e}^{-R_2'(|k|T_R \pm T_R)}$ with + for $k \geq 0$ and − for $k < 0$, which is $|\tilde{F}_{k+1}^{-}|$ because the state of order k after a repetition becomes order $k+1$ and $|k| \pm 1 = |k+1|$ for those signs. (ii) Each relation in Eqs. 2–7 and 10–17 is linear in states of a single order $|k|$ (Eq. 2 and the steady-state conditions relate Z_k before and after a repetition at the same order; Eqs. 4–7 relate $F_{\pm k}$ and $Z_{\pm k}$ across one pulse; Eqs. 10–11 combine $F_{\pm k}$), or is a ratio of such quantities. Multiplying all states of order $\pm k$ by the common factor $\mathrm{e}^{-R_2'|k|T_R}$ preserves linear relations among them and leaves ratios unchanged; in particular $X_{i,k}$ is unchanged. Eq. 17 uses $k = 0$, where the factor is 1. The equations are exact identities of the noise-free steady state, so the inverse returns the true parameters. □

Numerically, Eqs. 7, 16 and 17 hold to 10^{-12} on the steady states of (5) for $\alpha = 15°$ and 330° and three tissues including CSF, for both $k = \pm 1$. That real-valued states suffice is a consequence of Lemma 2 (Appendix A): all configuration amplitudes are purely imaginary, i.e. share the phase $\pm\pi/2$. The specific sign pattern of Eq. 18 (all $k \geq 0$ states with one sign, all $k < 0$ with the other) is observed for every parameter set we examined.

3.3 Implementation and behaviour

We implemented Steps 1–5 equation by equation (`paper.py`). The flip angle is found by a dense grid over the admissible branch interval followed by bisection on the sign change of (23). On noise-free data the implementation recovers all parameters to relative error 10^{-11}, for both pathway schemes, for $B_1^{+} \in [0.7, 1.4]$ (so that α_2 crosses 360°) and $|\Delta f| \leq 100\,\mathrm{Hz}$.

Two properties matter under noise. First, (24) requires $(X_1 c_1 - 1)/(X_1 - c_1) > 1$; noise can violate it, and T_1 is then undefined. Second, the inverse is sequential: R_2 and R_2' are fixed from scan 1 before the flip angle is estimated, and every later step combines extrapolated, differenced and ratioed estimates whose noise is not controlled. Both effects are quantified in Sections 6 and 9.

4 What the data determine

4.1 Fisher information and the Cramér–Rao bound

For complex data with independent Gaussian noise of variance σ^2 per component,

$$\mathcal{I}(\theta) = \frac{1}{\sigma^2} \sum_{i,j,k} \mathrm{Re}\Big[\overline{\partial_\theta S_{ijk}}\; \partial_\theta S_{ijk}^{\mathsf{T}}\Big], \qquad \mathrm{Cov}(\hat{\theta}) \succeq \mathcal{I}^{-1} \tag{25}$$

for every estimator that is (locally) unbiased at θ. Biased estimators—constrained, regularised or learned ones—are not covered; they can have smaller variance and mean squared error, at the price of a bias that the CRLB does not quantify (Bayesian bounds such as the van Trees inequality address that case). With σ in units of M_0 every bound scales as σ/M_0; we report bounds per unit σ/M_0. Jacobians are computed exactly (Section 8) and agree with automatic differentiation and finite differences.

Three diagnostics describe conditioning: the condition number and smallest eigenpair (λ, u) of the normalised information $\tilde{\mathcal{I}} = D^{-1/2}\mathcal{I}D^{-1/2}$, $D = \mathrm{diag}\,\mathcal{I}$, where the direction in original coordinates is $w \propto D^{-1/2}u$ (the generalised eigenvector of $\mathcal{I}w = \lambda D w$); the variance inflation factors $(\mathcal{I}^{-1})_{ll}\mathcal{I}_{ll}$; and the fixed-parameter bias $\delta\theta_r = -\mathcal{I}_{rr}^{-1}\mathcal{I}_{rm}\delta$ that results from holding θ_m at a wrong value.

4.2 Logarithmic parameters

$M_0, B_1^+, T_1, T_2, T_2^*$ are positive scale parameters, and the model depends on them through dimensionless combinations: by Theorem 1 and Lemma 1 and (21), S/M_0 is a function of $B_1^+\alpha^{\mathrm{nom}}$, T_R/T_1, T_R/T_2, t/T_2, $R_2' t$ and $\Delta\omega\, t$. In log coordinates a change of units is a translation, derivatives are dimensionless elasticities, and the central symmetry of Proposition 4 is a straight line with constant direction $(-1, +1, -2)$ in $(\log M_0, \log B_1^+, \log T_1)$; in linear coordinates its orbits are curves with a direction that varies from voxel to voxel. With $D_\theta = \mathrm{diag}(M_0, B_1^+, T_1, T_2, T_2^*, 1, 1)$, $\mathcal{I}_{\mathrm{lin}} = D_\theta^{-1}\mathcal{I}D_\theta^{-1}$, so $\sqrt{(\mathcal{I}^{-1})_{ll}}$ is the local coefficient of variation; for large spreads it is not (CV $= \sqrt{\mathrm{e}^{s^2} - 1}$ for a Gaussian log-error of SD s). For the relaxation times the likelihood is closer to quadratic in log coordinates (cost asymmetry for $\pm 20\%$ steps: T_1 0.58 linear vs 0.87 log; T_2 0.78 vs 1.16), whereas the signal is exactly linear in M_0, so for M_0 the argument for logs is the symmetry and relative errors, not curvature. Correlations, normalised eigenvalues and variance inflation factors are identical in both coordinates (the change of variables has a diagonal Jacobian), so no conclusion below depends on this choice. Positivity is automatic in log coordinates, but the ordering $T_2^* \le T_2$ ($R_2' \ge 0$) is not and must be enforced by an estimator. $\Delta\omega$ and ϕ_0 are not scale parameters and stay linear; $\log T_2^*$ is used rather than $\log R_2'$ because R_2' may be close to zero.

4.3 One scan, and small flip angles

Corollary 1 already shows that one scan cannot separate M_0, B_1^+ and T_1: the scan-1-only information is exactly singular, so the bound on $\log T_1$ with B_1^+ unknown is infinite (a numerical inverse returns values above 10^8, set by rounding). Two scans tied by a common B_1^+ can, unless both flip angles are small.

Proposition 4 (Small-angle scaling invariance)**.** *Fix T_R, T_2 ($E_2 < 1$), a pathway k and $\kappa > 0$. Let $\alpha \to 0$ and $\rho = T_R/T_1 \to 0$ with $c = \alpha^2/(2\rho)$ fixed. Then*

$$a_k(\alpha, T_1) = \alpha\, f_k(c, E_2) + O(\alpha^3 + \alpha\rho), \qquad M_0 a_k(\alpha, T_1) - \tfrac{M_0}{\kappa}\, a_k(\kappa\alpha, T_1/\kappa^2) = O(\alpha^3 + \alpha\rho), \tag{26}$$

with $f_k(c, E_2) = G_k\big(\frac{c-1}{c+1}, E_2, 0\big)/(1+c)$*. Hence, through* (2)*, the signal of every scan with a small flip angle is invariant to leading order under*

$$(M_0,\ B_1^+,\ T_1) \longmapsto (M_0/\kappa,\ \kappa B_1^+,\ T_1/\kappa^2). \tag{27}$$

The remainder is additive and not uniform in κ*;* κ *is fixed as the limit is taken.*

Proof. With $E_1 = 1 - \rho + O(\rho^2)$ and $\cos\alpha = 1 - \alpha^2/2 + O(\alpha^4)$,

$$u = \frac{\alpha^2/2 - \rho + O(\alpha^4 + \rho^2)}{\alpha^2/2 + \rho + O(\alpha^4 + \rho^2 + \alpha^2\rho)} = \frac{c-1}{c+1} + O(\alpha^2 + \rho),$$
$$v = \frac{\alpha\rho + O(\alpha^3\rho + \alpha\rho^2)}{\rho + \alpha^2/2 + O(\alpha^4 + \rho^2 + \alpha^2\rho)} = \frac{\alpha}{1+c}\big(1 + O(\alpha^2 + \rho)\big).$$

$G_k(u, E_2, 0)$ is a Fourier coefficient of the solution of (9), whose matrix $I - E_2 D_u R(\beta)$ is invertible with inverse bounded by $1/(1 - E_2)$ uniformly in $|u| \le 1$ and β, and depends smoothly on u; hence G_k is Lipschitz in u and $G_k(u) = G_k(\frac{c-1}{c+1}) + O(\alpha^2 + \rho)$. By Theorem 1, $a_k = v\, G_k(u) = \alpha f_k(c, E_2) + O(\alpha^3 + \alpha\rho)$. Under $\alpha \to \kappa\alpha$, $\rho \to \kappa^2\rho$, c is unchanged and the leading term scales by κ, which $M_0 \to M_0/\kappa$ compensates; the remainders are of the stated order for fixed κ. Since B_1^+ multiplies every flip angle, $\alpha \to \kappa\alpha$ is $B_1^+ \to \kappa B_1^+$, and the echo factors of (21) do not involve α or T_1. □

A direct expansion of the isochromat solution gives the same leading term in explicit form, $m_\perp(\psi) \simeq -i\alpha/\{(1-E_2 e^{i\psi})[1+cH(\psi)]\}$ with $H = (1-E_2^2)/|1-E_2 e^{i\psi}|^2$ (Appendix A.1). Numerically, with $c = 0.75$, the leading-order formula matches the exact solver for pathways $-3 \dots 2$ to 2.5×10^{-3}, 6.2×10^{-4}, 1.5×10^{-4}, 3.8×10^{-5} (relative to $|a_0|$) at $\alpha = 15°, 7.5°, 3.75°, 1.875°$: the factor four per halving of an $O(\alpha^2)$ relative remainder.

Corollary 2 (Leading-order degeneracy)**.** *If all flip angles are small, the directional derivative of the signal along* $v = (-1, +1, -2, 0, 0, 0, 0)$ *vanishes at leading order (it is* $O(\alpha^3 + \alpha\rho)$ *while the others are* $O(\alpha)$*), so* $\mathcal{I}$ *is ill-conditioned. No number of small-angle acquisitions removes this leading-order degeneracy, because* B_1^+ *scales all of them by the same factor.*

The degeneracy is of leading order only: at finite angles $\mathcal{I}$ is nonsingular and the parameters are identifiable in principle, but separation is very noise-sensitive. For the reference tissue the condition number of $\tilde{\mathcal{I}}$ is 2.6×10^4 for 30°/60°, 5.0×10^5 for 15°/30° and 2.5×10^7 for 3.75°/7.5°. Repeated acquisitions shrink all bounds as $1/\sqrt{N}$ but do not improve conditioning. On the full model with three pathways, holding B_1^+ wrong by ε with 15°/30° shifts the fit by $\delta \log T_1 = -2.06\varepsilon$ and $\delta \log M_0 = -1.02\varepsilon$ (predicted -2, -1).

4.4 How the published protocol escapes

Proposition 5 (Periodicity)**.** $a_k(-\alpha) = -a_k(\alpha)$ *and* $a_k(\alpha + 2\pi) = a_k(\alpha)$ *for all* k*. Hence two protocols whose actual flip angles agree up to sign and multiples of* 2π *in every scan carry identical Fisher information about every parameter when* B_1^+ *is known; in particular* 15°/330° *and* 15°/30° *at* $B_1^+ = 1$*.*

Proof. $R_x(-\alpha) = R_z(\pi) R_x(\alpha) R_z(-\pi)$, an RF phase of π. Precession and relaxation commute with R_z, so the steady state for RF phase π is $R_z(\pi)$ applied to the steady state for phase 0, which multiplies every F_k by -1. Periodicity is that of R_x. A known, scan-wide sign changes neither the likelihood's curvature nor $\mathcal{I}$; with B_1^+ known the actual angles, not the nominal ones, determine the data. □

The equivalence concerns actual angles: $330°B_1^+$ and $-30°B_1^+$ agree modulo 360° only at $B_1^+ = 1$ in the physiological range (at $B_1^+ = 1.1$: 363° vs 33°, and known-B_1^+ T_1 bounds 29.6 vs 29.0). Near $B_1^+ = 1$ the value of the large pulse lies in the B_1^+ derivative of its effective angle $\alpha_2^{\text{eff}} = 330°B_1^+ - 360°$:

$$\frac{\partial \log|\alpha_2^{\text{eff}}|}{\partial \log B_1^+} = \frac{330\,B_1^+}{330\,B_1^+ - 360}, \qquad \frac{\partial \log \alpha_1}{\partial \log B_1^+} = 1, \tag{28}$$

which is -11 at $B_1^+ = 1$, -2.75 at $B_1^+ = 0.8$, and diverges as $\alpha_2^{\text{eff}} \to 0$, i.e. as the actual angle $330°B_1^+$ approaches 360° ($B_1^+ \to 360/330 = 1.09$). Equation (27) needs all angles to scale alike; under a B_1^+ change the second moves much faster and in the opposite direction, and the near-null direction is lifted.

Equation (28) is the small-effective-angle form of an exact statement. By Proposition 2, a scan sees its flip angle through $\ln|\tan(\alpha_i/2)|$, whose log-derivative with respect to B_1^+ is $\alpha_i/\sin\alpha_i$. After T_1 and M_0 are profiled out, only the difference of the two scans remains, the lever $L = \alpha_2/\sin\alpha_2 - \alpha_1/\sin\alpha_1$ of (19): at $B_1^+ = 1$ the scan-2 term is -11.52 (Eq. (28) gives -11), the scan-1 term 1.01, and $L = -12.53$; for 15°/30°, $L = 0.036$. At $B_1^+ = 1$ the two protocols produce the same data up to the sign of scan 2 (Proposition 5), so they differ only in L, and the ratio of their B_1^+ bounds is $|L_{330}/L_{30}| = 351.17$. The computed bounds for WM, 774.40 and 2.2052, have exactly this ratio.

4.5 Bounds and conditioning

Table 6: CRLB per unit σ/M_0 (log-parameter SD; rad/ms for $\Delta\omega$), both scans at full resolution.

Protocol	Tissue ($T_1/T_2/T_2^*$)	M_0	B_1^+	T_1	T_2	T_2^*	$\Delta\omega$
15°/330°	WM (850/66/50)	12.5	2.21	27.3	16.0	27.2	0.30
15°/330°	GM (1350/90/60)	14.7	2.75	33.1	20.7	38.2	0.31
15°/330°	reference (1500/70/60)	19.0	2.91	40.5	23.1	47.3	0.37
15°/330°	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37
15°/30°	reference	1050	1020	2110	23.1	47.3	0.37
15°/30°	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37

Table 7: Conditioning (reference tissue).

	15°/330°	15°/30°
Eigenvalues of $\tilde{\mathcal{I}}$	$0.024, 0.11, 0.22, 0.51,$ $0.52, 1.49, 4.13$	$7.7\times10^{-6}, 0.11, 0.29,$ $0.51, 0.76, 1.49, 3.84$
Condition number	173	5.0×10^5
Variance inflation $M_0/B_1^+/T_1$	16.7/4.6/26.6	51 000/7 700/72 000
Correlations (B_1^+, T_1), (B_1^+, M_0), (T_1, M_0)	$0.72, 0.53, 0.77$	$-0.9999, -0.9999, 0.9999$
$\partial \log T_1/\partial \log B_1^+$, B_1^+ fixed, both scans	$+10.1$	-2.06
$\partial \log T_1/\partial \log B_1^+$, B_1^+ fixed, scan 1 only	-2.02	-2.02

The T_1 bound builds up as groups of parameters are freed: 7.9 (T_1 only), 21.1 ($+T_2, T_2^*, \Delta\omega, \phi_0$), 28.0 ($+M_0$), 40.5 ($+B_1^+$, 15°/330°) and 2110 ($+B_1^+$, 15°/30°); the factors depend on the order, and the variance inflation factors are the order-free summary. With the published protocol B_1^+ is no longer the main source of T_1 uncertainty: at 15° and $T_R = 25$ ms, T_1 is weakly encoded and

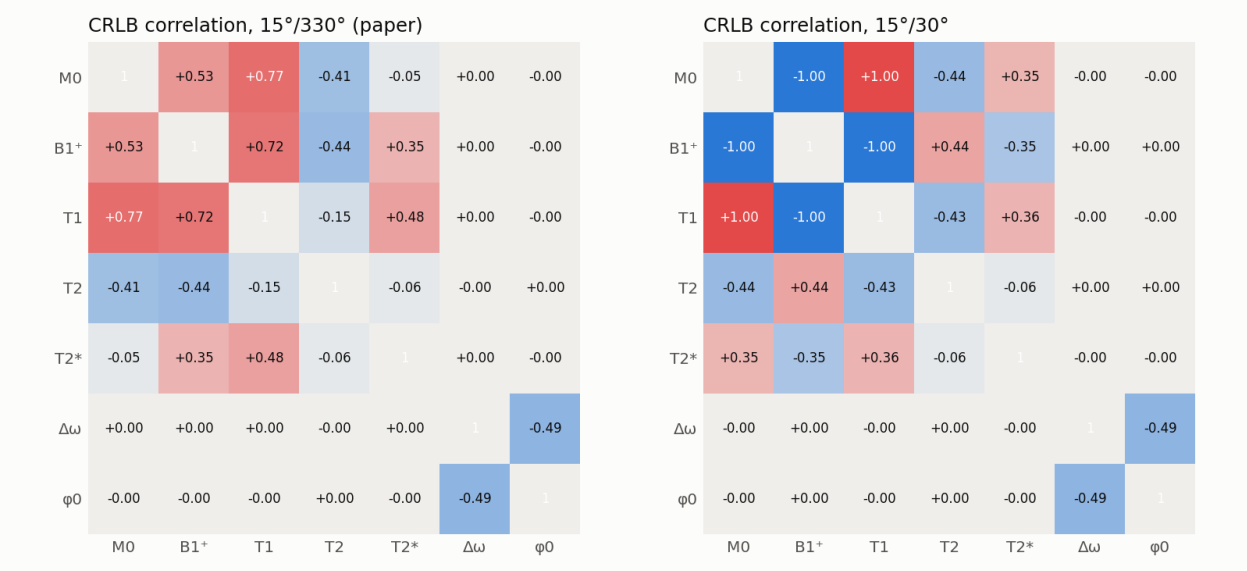


Figure 4: Correlation matrices of $\mathcal{I}^{-1}$. The M_0–B_1^+–T_1 block of the small-angle protocol is perfectly correlated, the signature of (27).

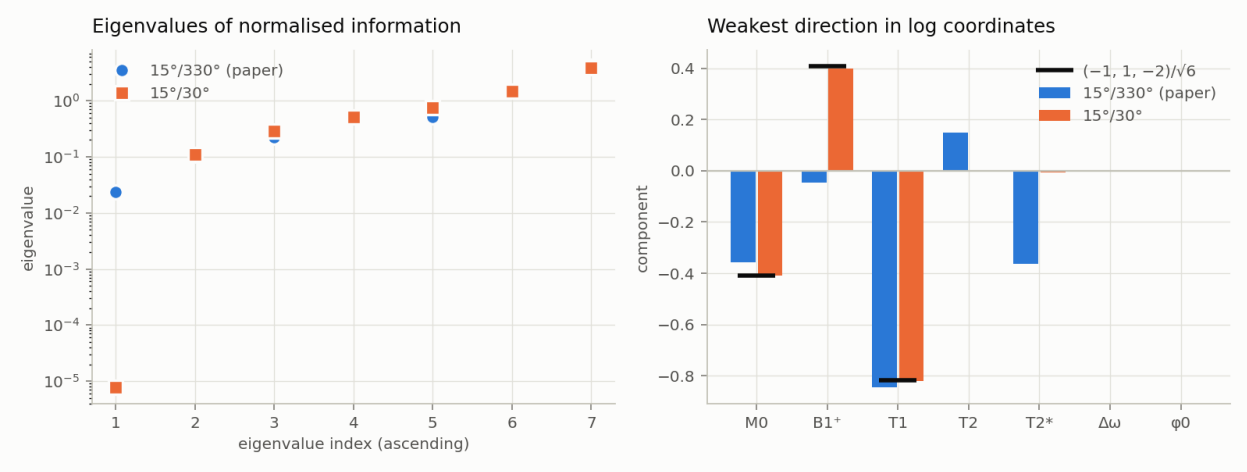


Figure 5: Left: eigenvalues of $\tilde{\mathcal{I}}$. Right: the weakest direction in log coordinates, $w \propto D^{-1/2}u$. For 15°/30° it is the symmetry direction $(-1, 1, -2)/\sqrt{6}$ (black); for 15°/330° it is dominated by T_1 (-0.85), with M_0 and T_2^* (-0.36 each), and its B_1^+ component is only -0.05.

confounded mostly with the transverse relaxation times and M_0. Separating which data are used from whether B_1^+ is known gives T_1 bounds of 40.5 (both scans, B_1^+ unknown), 28.0 (both, known), 57.1 (scan 1, known) and infinity (scan 1, unknown: exactly singular; numerically above 10^8, set by rounding).

The error transfer +10.1 of Table 7 (both scans, B_1^+ held fixed) is a single-point value and depends strongly on the true B_1^+. For the reference tissue it is +2.0, +2.4, +4.1, +10.1, +105, +168, −127, −15 and −10 at $B_1^+ = 0.7$, 0.8, 0.9, 1.0, 1.08, 1.09, 1.10, 1.2 and 1.3: it rises above 100 within about 1% of the crossing $B_1^+ = 360/330$, where the actual scan-2 angle is 360°, and changes sign there. With scan 1 only it is about −2.0 at every B_1^+. Holding B_1^+ fixed while fitting both scans lowers the T_1 bound from 40.5 to 28.0 at $B_1^+ = 1$, but the gain survives only if the error of the supplied B_1^+ is well below $28.0/10.1 \approx 2.8\,\sigma/M_0$, about the voxel's own B_1^+ bound (2.91), and far smaller near the crossing. A supplied B_1^+ map should therefore not be combined with full-resolution scan-2 data in the T_1 step: either fit B_1^+ jointly with both scans, or use the map with scan 1 alone (Algorithm 4), where the transfer is a modest −2.

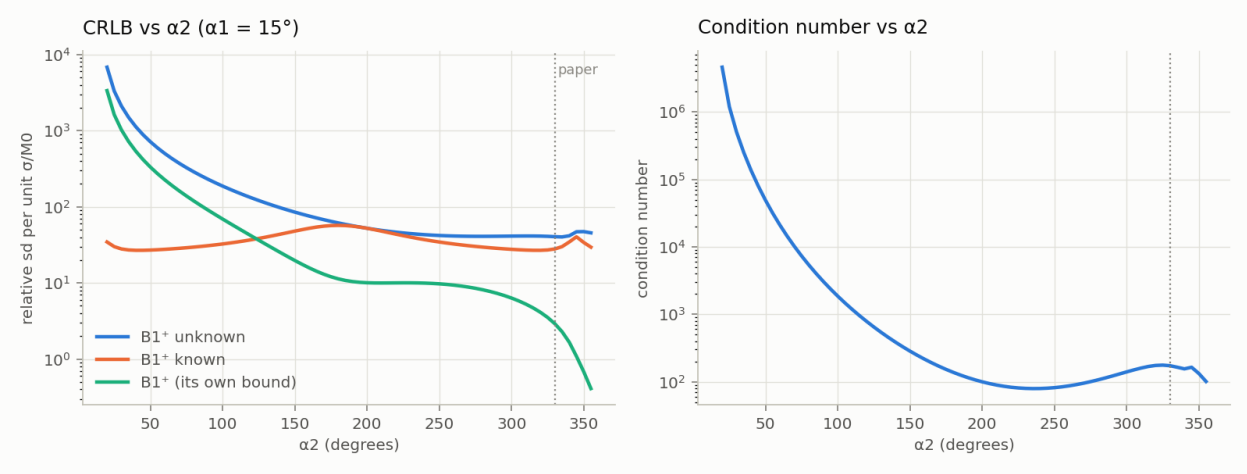


Figure 6: Bounds and condition number versus α_2^{nom} ($\alpha_1^{\text{nom}} = 15°$).

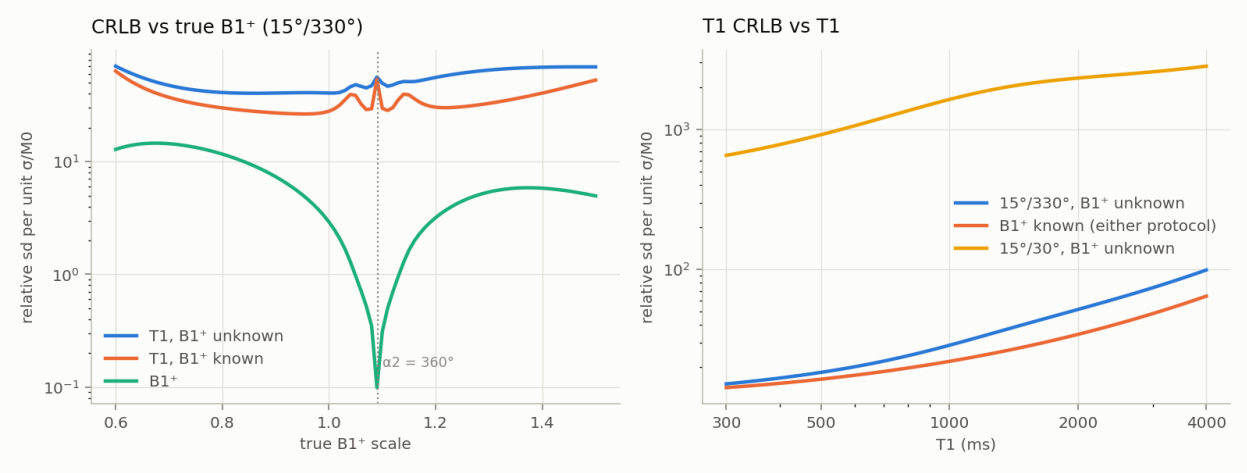


Figure 7: Left: bounds versus the true B_1^+; B_1^+ is locally best determined where the actual scan-2 angle $330° B_1^+$ is 360° ($\alpha_2^{\text{eff}} = 0$). Right: T_1 bound versus T_1.

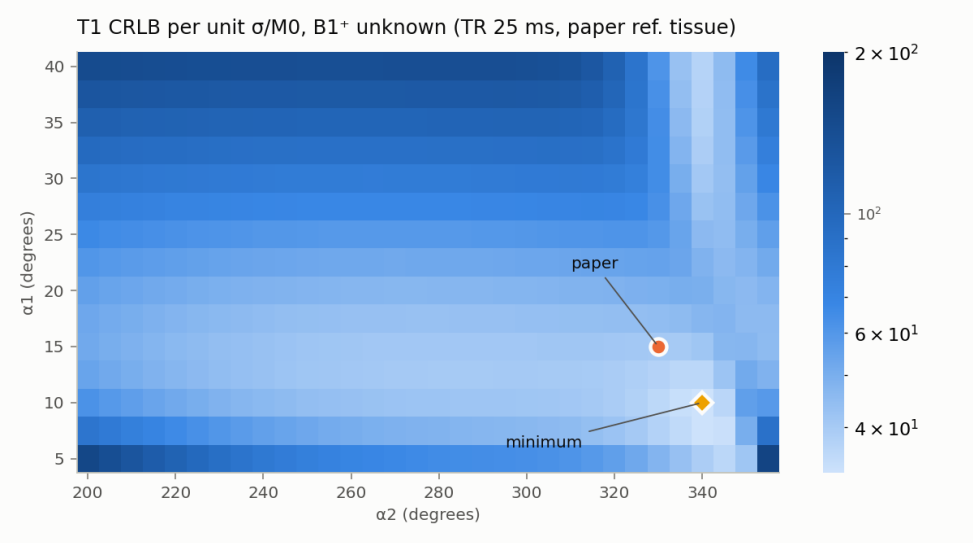


Figure 8: T_1 bound with B_1^+ unknown over nominal (α_1, α_2). The optimum on this grid is 10°/340° (32.9), 19% better than 15°/330°.

Over $\alpha_2^{\text{nom}} \geq 210°$ the B_1^+-unknown T_1 bound stays within 1.5× of the B_1^+-known one. Across $B_1^+ \in [0.7, 1.3]$ the T_1 bound is 40–65, and it grows with T_1 (15 at 300 ms, 100 at 4000 ms). Removing pathway $+1$ raises it from 40.5 to 67; removing the echo pathway -1 raises it to 146 and the T_2 bound from 23 to 208.

4.6 Global aliases

Local bounds say nothing about distant parameter values that produce the same data. Theorem 1 characterises them.

Proposition 6 (Exact data equivalence)**.** *Let both scans have the same T_R, and let T_2, T_2^* and $\Delta\omega$ be fixed. Write u_i, v_i for the invariants of scan i at (B_1^+, T_1) and u_i', v_i' at $(B_1^{+\prime}, T_1')$.*

(a) If $u_i' = u_i$ for $i = 1, 2$ and $M_0' = M_0\sqrt{\tanh(T_R/2T_1)/\tanh(T_R/2T_1')}$, then all magnitudes $|S_{ijk}|$ at $(M_0', B_1^{+\prime}, T_1')$ equal those at (M_0, B_1^+, T_1).

(b) If in addition $\operatorname{sign}(v_1'v_2') = \operatorname{sign}(v_1v_2)$, the complex signals are equal after replacing ϕ_0 by ϕ_0 or $\phi_0 + \pi$.

(c) Given u_1, u_2, the conditions $u_i' = u_i$ are two equations in the two unknowns $(B_1^{+\prime}, E_1')$:

$$\cos(B_1^{+\prime}\alpha_i^{\text{nom}}) = \frac{E_1' - u_i}{1 - u_iE_1'}, \qquad i = 1, 2. \tag{29}$$

Proof. (a) Theorem 1 gives $|a_k| = |v|\,|G_k(u, E_2, \Delta\omega)|$, and Lemma 1 gives

$$|v_i'| = \sqrt{1 - u_i^2}\,\sqrt{\tanh(T_R/2T_1')}.$$

Hence $M_0'|v_i'| = M_0|v_i|$ for both scans with the single stated M_0' (both scans share T_R), and $M_0'|a_k'| = M_0|a_k|$ for every k. The echo factors in (21) depend only on T_2, T_2^*, $\Delta\omega$ and ϕ_0. (b) The complex signal is $M_0v_i\,G_k(u_i, E_2, \Delta\omega)$ times echo factors; v_i is real with $\operatorname{sign} v_i = \operatorname{sign}\sin\alpha_i$ ((6),

$1 - E_1 \cos\alpha > 0$). If $\operatorname{sign} v_i' = \operatorname{sign} v_i$ for both scans the signals are equal; if both signs flip, they differ by a global factor $-1 = \mathrm{e}^{i\pi}$, absorbed by ϕ_0. (c) Inverting the Möbius map (6) in $\cos\alpha$ gives $\cos\alpha_i = (E_1 - u_i)/(1 - u_i E_1)$; with $\alpha_i = B_1^+ \alpha_i^{\mathrm{nom}}$ this is (29). □

Because (29) is a square system, its solutions are generically isolated and each choice of branch of arccos in the two equations ($\pm$, plus multiples of 2π) is a candidate. Magnitude data can then be matched exactly on branches that complex data exclude by the sign condition (b). Solving on each branch from the noise-free data gives Table 8: every alias fits to machine precision.

Table 8: Exact aliases (noise-free; relative residual $< 10^{-14}$). Magnitude alias: α_2 on the other side of 360°. Complex alias: $\alpha_2 > 540°$ with the same sign pattern.

Truth		Magnitude alias		Complex alias	
B_1^+	T_1 (ms)	B_1^+	T_1 (ms)	B_1^+	T_1 (ms)
1.00	1500	1.199	1038	2.008	359
0.90	1200	1.350	527	1.865	271
1.05	850	1.135	726	2.100	204
0.80	2500	1.480	717	1.769	497
1.08	1000	1.102	960	2.160	240

The complex aliases need $B_1^+ \approx 1.8$–2.2 and are excluded by the assumption $\alpha_2^{\mathrm{nom}} B_1^+ \le 540°$ of [1], i.e. $B_1^+ < 1.64$; our fits impose this bound. The magnitude aliases lie at $B_1^+ \approx 1.1$–1.5, inside the physiological range, and require one bit of phase information (Section 7). Each lies on the other side of the crossing $330° B_1^+ = 360°$, and as the actual scan-2 angle approaches 360° ($\alpha_2^{\mathrm{eff}} \to 0$) the truth and its magnitude alias merge (last row); the complex alias does not. None of this is visible in local bounds.

4.7 Information in magnitude data

A magnitude $m = |S + n|$ is Rician with amplitude $A = |S|$. Its Fisher information about A is $g(A/\sigma)/\sigma^2$ with

$$g(a) = \mathbb{E}\Big[x^2 \frac{I_1(xa)^2}{I_0(xa)^2}\Big] - a^2, \qquad x = m/\sigma \tag{30}$$

(Appendix A.2); g increases from 0 to 1 ($g(0.5) = 0.20$, $g(2) = 0.85$, $g(5) = 0.98$). The information about the five magnitude parameters is $\mathcal{I}_{\mathrm{mag}} = \sigma^{-2} \sum_j g(A_j/\sigma) \nabla A_j \nabla A_j^{\mathsf{T}}$ with $\nabla A_j = \mathrm{Re}(\overline{S_j} \nabla S_j)/A_j$. For the published protocol and the reference tissue at $B_1^+ = 1$ the weakest amplitude is $0.0074\, M_0$, so even at $\mathrm{SNR}(M_0) = 300$ every measurement has $A/\sigma \ge 2.2$, and the magnitude bounds equal the complex bounds to within 1.5% for T_1, B_1^+, T_2 and M_0 at SNR 300, 1000 and 3000 (within 0.5% except B_1^+ at SNR 300, 1.4%). Near the 360° crossing the scan-2 amplitudes vanish and the loss grows: at $B_1^+ = 1.09$ the B_1^+ bound is 27% larger at SNR 300 and 1.8% at SNR 1000. Away from the crossing, discarding the phase loses essentially no local information; it loses the global bit of Proposition 6(b).

5 Why two scans are required

A natural idea for a faster protocol is to acquire only the full-resolution scan, fit a smooth (e.g. low-order polynomial) B_1^+ field to the whole image, and estimate T_1, M_0 and the other parameters voxel by voxel. The field has only a handful of coefficients, while the image has 10^5–10^6 voxels, so it

seems that the data should over-determine it. This section shows that they do not: the single-scan degeneracy of Corollary 1 is exact and local, and no spatial model of B_1^+ alone can remove it. A second acquisition in which the flip angle enters differently is required; it can, however, be very short.

A toy model. Suppose each voxel v has an unknown amplitude a_v and the scanner has an unknown gain b that varies smoothly (here, a constant), and one acquisition measures $y_v = a_v\, b$. However many voxels there are, b cannot be found: replacing (a_v, b) with $(a_v/\kappa, \kappa b)$ leaves every y_v unchanged. The smoothness of b does not help, because each a_v is free and absorbs the change locally. A second acquisition that depends on b differently, for example $y'_v = a_v\, b^2$, breaks the degeneracy in every voxel: $b = y'_v/y_v$. The single MPME scan is this situation with $a_v \leftrightarrow (M_0, T_1)$ and $b \leftrightarrow B_1^+$. The data determine only the flip angle relative to the Ernst angle, $\xi = \tan(B_1^+\alpha/2)/\tan(\alpha_E/2)$, and the amplitude $M_0 \tan(\alpha_E/2)$ (Proposition 2 and Table 3). For every trial $B_1^{+\prime}$ the voxel's own T_1' moves its Ernst angle to restore ξ, and its own M_0' restores the amplitude ((16), Table 4). Sharing B_1^+ with other voxels does not change this, because each voxel compensates with its own parameters (Section 5.2).

Counting. With N voxels and a B_1^+ model with p coefficients, one scan supplies $2N$ independent equations for these unknowns (the two invariants per voxel), while the unknowns number $2N + p$ (M_0 and T_1 in each voxel, plus the field). The system is underdetermined by exactly p for every N, so adding voxels never closes the gap. A second flip angle adds N equations (ξ_2 in each voxel, which with ξ_1 gives (18)), giving $3N \geq 2N + p$. With a smooth B_1^+ model only about p of the new equations are needed to fix the field, which is why the second scan can be short (Section 5.6).

5.1 Any B_1^+ field fits a single scan exactly

Proposition 7 (Whole-image degeneracy of a single scan)**.** *Consider one scan with flip angle* α^{nom} *and data generated by* $(M_0(\mathbf{r}), B_1^+(\mathbf{r}), T_1(\mathbf{r}))$ *and arbitrary* T_2, T_2^*, $\Delta\omega$, ϕ_0. *Let* $B_1^{+\prime}(\mathbf{r})$ *be* any *field satisfying*

$$\cos\bigl(B_1^{+\prime}(\mathbf{r})\,\alpha^{\mathrm{nom}}\bigr) > -u(\mathbf{r}) \qquad \textit{for every voxel,} \tag{31}$$

where $u(\mathbf{r})$ *is the invariant* (6) *of the true parameters. Then there are fields* $T_1'(\mathbf{r})$ *and* $M_0'(\mathbf{r}) > 0$ *such that* $(M_0', B_1^{+\prime}, T_1')$*, with the same* T_2, T_2^*, $\Delta\omega$ *and* ϕ_0 *(up to* $\phi_0 \to \phi_0 + \pi$*), produce exactly the same complex signal in every voxel, every pathway and every echo.*

Proof. Fix a voxel and write $c' = \cos(B_1^{+\prime}\alpha^{\mathrm{nom}})$. Inverting the Möbius map (6) in E_1, the value

$$E_1' = \frac{u + c'}{1 + uc'}$$

gives $u(B_1^{+\prime}\alpha^{\mathrm{nom}}, E_1') = u$. Since $|u| < 1$ and $|c'| < 1$, the denominator is positive; $E_1' < 1$ is equivalent to $(1-u)(1-c') > 0$, which always holds; and $E_1' > 0$ is equivalent to (31). Hence $T_1' = -T_R/\log E_1'$ is a valid relaxation time. By Theorem 1 the signal is $M_0\, v\, G_k(u, E_2, \Delta\omega)$ times echo factors that do not involve M_0, B_1^+ or T_1; with u unchanged, choosing $M_0' = M_0\,|v|/|v'|$ makes $M_0'|v'| = M_0|v|$, and if $\operatorname{sign} v' \neq \operatorname{sign} v$ the sign is absorbed by $\phi_0 \to \phi_0 + \pi$ (a single scan has a single phase). The voxel's signal is therefore reproduced exactly. The construction is independent from voxel to voxel, so it applies to any field $B_1^{+\prime}(\mathbf{r})$. □

Condition (31) is no restriction in practice. For the published scan 1 (15°, $T_R = 25\,\mathrm{ms}$), $u = 0.08$ (WM), 0.30 (GM) and 0.69 (CSF), so any field with $B_1^{+\prime} < 6.3$, 7.2 and 8.9, respectively,

is admissible—every physically plausible field. The statement holds at every flip angle because Theorem 1 is exact; it is not a small-angle approximation. In the small-angle limit the compensation reduces to the scaling of Proposition 4, $T_1' \propto (B_1^{+\prime})^{-2}$ and $M_0' \propto (B_1^{+\prime})^{-1}$.

Corollary 3 (A parametric B_1^+ model has no information from one scan)**.** *Let $B_1^+(\mathbf{r}) = \sum_{j=1}^{p} \beta_j \phi_j(\mathbf{r})$ for any basis $\{\phi_j\}$ (polynomials, splines, ...), with T_1, M_0 (and T_2, T_2^*, $\Delta\omega$, ϕ_0) free in every voxel. From one scan, the Fisher information about β, after the per-voxel parameters are accounted for (profiled), is the zero matrix, and the CRLB on every β_j is infinite.*

Proof. At each voxel v, Corollary 1 shows that the three Jacobian columns of S_v with respect to $\log M_0$, $\log B_1^+$ and $\log T_1$ span at most two dimensions; the columns for $\log M_0$ and $\log T_1$ are generically independent, so

$$\frac{\partial S_v}{\partial \log B_1^+} = a_v \frac{\partial S_v}{\partial \log T_1} + b_v \frac{\partial S_v}{\partial \log M_0}$$

for some scalars a_v, b_v. The Jacobian column of a coefficient β_j is the stack over voxels of $\phi_j(\mathbf{r}_v)\, B_1^+(\mathbf{r}_v)^{-1}\, \partial S_v / \partial \log B_1^+$, which therefore lies in the span of the per-voxel $\log T_1$ and $\log M_0$ columns. The profiled information of β is $\mathcal{I}_{\beta\beta} - \mathcal{I}_{\beta\eta}\mathcal{I}_{\eta\eta}^{-1}\mathcal{I}_{\eta\beta}$ with η the per-voxel parameters; it is the Gram matrix of the components of the β-columns orthogonal to the η-columns, which are zero. □

The number of voxels does not matter: each voxel contributes exactly zero information about B_1^+, and a sum of zeros is zero. Smoothness reduces the number of B_1^+ unknowns, but not the per-voxel freedom of T_1 and M_0 to compensate.

Numerical confirmation. Table 9 refits noise-free scan-1 data of the synthetic phantom (4735 voxels across the slice) with three B_1^+ fields held fixed: the true field, the true field scaled by 1.10, and the true field times a smooth tilt and curvature. All three fit to machine precision; the T_1 maps differ by $\mathrm{d}\log T_1 / \mathrm{d}\log B_1^+ = -2.03$ and -2.02, i.e. a 10% B_1^+ error becomes a uniform -19% T_1 error, and the smooth tilt becomes a spurious T_1 gradient from -22% to $+27\%$ across the slice. The profiled information of the six coefficients of a degree-2 polynomial over 60 voxels with varied tissue is zero to rounding (largest eigenvalue 4×10^{-16}) for one scan, and positive definite (smallest eigenvalue 0.89, largest 857) for the two scans of the published protocol.

Table 9: Scan-1 data refitted with fixed B_1^+ fields (noise-free, 4735 voxels).

B_1^+ field used	max relative residual	$\log(\hat{T}_1/T_1)$: median [range]	$\mathrm{d}\log T_1 / \mathrm{d}\log B_1^+$
true	4×10^{-16}	0.000	–
true $\times 1.10$	6×10^{-15}	-0.193 [-0.194, -0.192]	-2.03
true $\times(1 + 0.15x - 0.10y^2)$	8×10^{-15}	$+0.055$ [-0.225, $+0.274$]	-2.02

5.2 Voxels that share B_1^+

The extreme case of a smooth field is a constant one: a group of voxels with the same B_1^+ and otherwise different tissue. Joint estimation with one shared B_1^+ still leaves an exact one-parameter family, but within the group it preserves the ratios of M_0 exactly and the ratios of T_1 almost exactly.

Proposition 8 (Voxels sharing B_1^+)**.** *Let voxels $j = 1, \dots, N$ share B_1^+, with arbitrary $M_{0,j}$, $T_{1,j}$, $T_{2,j}$, $T_{2,j}^*$, $\Delta\omega_j$, $\phi_{0,j}$, and let ξ_j, $\zeta_j = \tan(\alpha_{E,j}/2)$ be their scan-1 quantities (Proposition 2). For every $B_1^{+\prime}$ with $|\tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)| < \min_j |\xi_j|$, the parameters*

$$\zeta_j' = \frac{\tan(B_1^{+\prime}\alpha^{\mathrm{nom}}/2)}{\xi_j}, \qquad T_{1,j}' = \frac{T_R}{2\,\operatorname{artanh}\zeta_j'^2}, \qquad M_{0,j}' = M_{0,j}\frac{\zeta_j}{\zeta_j'}, \tag{32}$$

with all other parameters unchanged, reproduce the scan-1 signal of every voxel exactly. Along this family, for all j, l,

$$\frac{M_{0,j}'}{M_{0,l}'} = \frac{M_{0,j}}{M_{0,l}}, \qquad \frac{\tanh(T_R/2T_{1,j}')}{\tanh(T_R/2T_{1,l}')} = \frac{\tanh(T_R/2T_{1,j})}{\tanh(T_R/2T_{1,l})} \tag{33}$$

exactly, and $T_{1,l}'/T_{1,j}' = (T_{1,l}/T_{1,j})\big(1 + O\big((T_R/2T_{1,\min}')^2\big)\big)$.

Proof. For each voxel, (32) is (16) with the common $B_1^{+\prime}$; it keeps ξ_j and $M_{0,j}\zeta_j$, hence the voxel's signal (Proposition 7), and the condition on $B_1^{+\prime}$ makes every $\zeta_j' < 1$. The voxels' likelihood terms share no other parameter, so the joint data are reproduced. Dividing, $\zeta_j'/\zeta_l' = \xi_l/\xi_j$ does not depend on $B_1^{+\prime}$, and it equals ζ_j/ζ_l at $B_1^{+\prime} = B_1^+$; since $\zeta^2 = \tanh(T_R/2T_1)$ this is the second identity. Then $M_{0,j}'/M_{0,l}' = (M_{0,j}\zeta_j/M_{0,l}\zeta_l)(\zeta_l'/\zeta_j') = M_{0,j}/M_{0,l}$. Finally $\operatorname{artanh} y = y\,(1 + y^2/3 + O(y^4))$ with $y = \zeta^2 \approx T_R/2T_1$ gives the T_1 ratio. □

In words: changing the shared B_1^+ multiplies $\tan(B_1^+\alpha/2)$ by a factor κ, and every voxel keeps its own ξ_j by multiplying its own $\tan(\alpha_{E,j}/2)$ by the same κ. All Ernst angles therefore slide together, all T_1 values change by about κ^{-2} and all M_0 values by exactly κ^{-1}. The shared parameter is never tested, because no voxel's data involve another voxel's parameters. This is Corollary 3 with a single constant basis function: $2N$ equations for $2N + 1$ unknowns.

Table 10 shows the family for a WM voxel ($M_0 = 0.69$, $T_1 = 850\,\mathrm{ms}$) and a GM voxel ($M_0 = 0.82$, $T_1 = 1350\,\mathrm{ms}$) that share $B_1^+ = 1$ and differ in every other parameter as well. Both voxels reproduce their scan-1 signals to machine precision for every $B_1^{+\prime}$ from 0.7 to 1.5, while the absolute T_1 values change by a factor of 4.7. The ratio $M_0^{\mathrm{GM}}/M_0^{\mathrm{WM}}$ stays at $0.82/0.69 = 1.188406$ to all printed digits, and $T_1^{\mathrm{GM}}/T_1^{\mathrm{WM}}$ stays between 1.5882 and 1.5885 (true value 1.5882). Scan 2 again rejects every member but the truth.

Table 10: Two voxels with the same B_1^+ (truth 1) fitted jointly from scan 1: the shared-B_1^+ family (32). Misfit columns as in Table 4, maximum over the two voxels. Generated by `scripts/single_scan_family.py`.

	T_1' (ms)			M_0'			misfit	
$B_1^{+\prime}$	WM	GM	GM/WM	WM	GM	GM/WM	scan 1	scan 2
0.7	1745	2771	1.5882	0.989	1.175	1.188406	1×10^{-15}	0.86
0.8	1334	2118	1.5882	0.864	1.027	1.188406	2×10^{-15}	0.70
0.9	1052	1670	1.5882	0.768	0.912	1.188406	3×10^{-15}	0.43
1.0	850	1350	1.5882	0.690	0.820	1.188406	2×10^{-16}	0.00
1.1	701	1113	1.5883	0.627	0.745	1.188406	5×10^{-16}	1.42
1.2	587	933	1.5883	0.574	0.682	1.188406	4×10^{-16}	2.00
1.5	372	591	1.5885	0.457	0.543	1.188406	1×10^{-15}	1.27

A joint fit can look as if it worked. Table 11 fits the WM and GM voxels of Table 10 jointly, with one shared B_1^+, to noisy data ($\sigma = 10^{-3}$, one realisation). The method is a block Levenberg–Marquardt with exact Jacobians, started from a generic point ($M_0 = 0.8$, $T_1 = 1000\,\mathrm{ms}$, $T_2 = 80\,\mathrm{ms}$, $T_2^* = 56\,\mathrm{ms}$) at five values of B_1^+. From scan 1 alone, every fit reaches the same cost, $22.9687\,\sigma^2$, but the final B_1^+ depends on the start (from 0.90 to 1.00): the optimiser stops at the first point of the solution curve it reaches. A shared-B_1^+ fit that starts near the truth and ends near it is therefore no evidence that the data determine B_1^+. The other diagnostics agree:

- *Profile.* Over $B_1^{+\prime} \in [0.6, 1.6]$ the profile cost is constant to $5 \times 10^{-14}\sigma^2$, and to $5 \times 10^{-13}\sigma^2$ for a set of 20 voxels with random tissue parameters.
- *Fisher information.* The joint Fisher matrix has one zero eigenvalue (below 4×10^{-16} after normalisation), whose eigenvector is the family tangent (17) to 10^{-13}. The profiled information on $\log B_1^+$ is zero to rounding.

With both scans, every start up to 1.1 reaches the minimum at $B_1^+ = 1.005$ (20 voxels: 1.0004). The curvature of the profile there gives $\mathrm{SD}(\log B_1^+) = 2.14 \times 10^{-3}$, equal to the information at the estimate (20 voxels: 5.52×10^{-4} against the bound 5.54×10^{-4}). Starts at 1.2 and above end on the bound 540/330 at a much higher cost, which a multi-start fit discards. Without the bound they reach the complex alias 2.016 at exactly the minimal cost. Because f in (18) is the same in every voxel, sharing B_1^+ does not remove the alias.

Table 11: Joint fit of the WM and GM voxels with a shared B_1^+ (truth 1), $\sigma = 10^{-3}$, from a generic start at five values of B_1^+; fitting box $B_1^+ \in [\mathrm{e}^{-1}, 540/330]$. Generated by `scripts/shared_b1_check.py`.

	scan 1 only		both scans	
start B_1^+	final B_1^+	cost/σ^2	final B_1^+	cost/σ^2
0.6	0.8966	22.9687	1.0050	76.0635
0.8	0.9538	22.9687	1.0050	76.0635
1.0	0.9821	22.9687	1.0050	76.0635
1.3	0.9981	22.9687	1.6364	3.9×10^4
1.6	1.0032	22.9687	1.6364	3.9×10^4

So one scan with a shared (or smooth) B_1^+ determines *relative* T_1 and M_0 maps within a region of constant B_1^+, to within $O((T_R/2T_1)^2)$, but not their common scale: an error $\delta \log B_1^+$ multiplies every T_1 by about $\mathrm{e}^{-2\delta \log B_1^+}$ and every M_0 by about $\mathrm{e}^{-\delta \log B_1^+}$. The scale can be fixed only by information from outside the scan, for example a reference voxel whose T_1 is assumed known: then $B_1^+ = (2/\alpha^{\mathrm{nom}}) \arctan(\xi_{\mathrm{ref}}\zeta_{\mathrm{ref}})$. A relative error δ in the assumed reference T_1 then becomes a B_1^+ error of about $-\delta/2$ and an error of about δ in every T_1 of the region. This is a prior of the kind discussed next, and the accuracy of the maps is that of the assumption.

5.3 What a prior would do instead

Identifiability can be restored by constraining T_1 or M_0 spatially, for example T_1 constant within each tissue class, or $M_0 = C \cdot \mathrm{PD}$ with a smooth C and class-constant PD. The field is then determined by the prior, not by the data: any departure of the true T_1 from the assumed uniformity is absorbed into B_1^+. By the compensation law $\delta \log T_1 = -2\,\delta \log B_1^+$, a real regional T_1 difference δ within a tissue is converted into a B_1^+ change of about $-\delta/2$ and removed from the T_1 map. For

quantitative mapping, whose purpose is to detect such differences, this is circular. Priors are useful on top of a second acquisition (e.g. the hierarchical model of Section 11.3), not in place of one.

5.4 Physics outside the ideal model

The degeneracy belongs to the ideal model: instantaneous pulses, one flip angle per voxel and no diffusion. Real data contain effects that make the signal depend on the flip angle in other ways, and each could in principle give a single scan some information about B_1^+. Table 12 adds each effect to the model and gives the scan-1 bound on $\log B_1^+$, with every other parameter free, against 2.2 (σ/M_0 units, WM) for the ideal two-scan protocol.

Table 12: Information about B_1^+ in scan 1 (15°) when the ideal model is extended: CRLB on $\log B_1^+$ in units of σ/M_0, all other parameters free ($\log M_0$, $\log T_1$, $\log T_2$, $\log T_2^*$, $\Delta\omega$, ϕ_0, plus the extra parameter where one is unknown). "Exactly degenerate": zero information to rounding. Generated by `scripts/beyond_ideal_model.py`.

Effect	Assumption	WM	GM
Ideal model	reference: both scans	2.21	2.75
	scan 1	exactly degenerate	exactly degenerate
Gaussian slice profile, $\lvert z\rvert \le 3\sigma$	shape known	8.4×10^5	5.9×10^5
	width unknown	exactly degenerate	exactly degenerate
Hann-sinc slice profile, TBW 4	shape known	1.2×10^6	8.3×10^5
	width unknown	exactly degenerate	exactly degenerate
Slab-edge ramp, $p \in [0.9, 1]$	shape known	1.2×10^7	1.2×10^7
	width unknown	exactly degenerate	exactly degenerate
Relaxation during the pulse, on resonance	$\tau = 1\,\mathrm{ms}$	exactly degenerate	exactly degenerate
	$\tau = 2\,\mathrm{ms}$	exactly degenerate	exactly degenerate
Off-resonance during the pulse, $\tau = 1\,\mathrm{ms}$	$\Delta\omega/2\pi = 8\,\mathrm{Hz}$	5.0×10^4	5.3×10^4
	$\Delta\omega/2\pi = 48\,\mathrm{Hz}$	8.4×10^3	8.9×10^3
	$\Delta\omega/2\pi = 160\,\mathrm{Hz}$	2.7×10^3	2.8×10^3
Diffusion, $D = 0.7 \times 10^{-3}\,\mathrm{mm^2/s}$	$q = 2\pi/1\,\mathrm{mm}$, D known	4.8×10^3	3.7×10^3
	$q = 2\pi/1\,\mathrm{mm}$, D free	exactly degenerate	exactly degenerate
	$q = 2\pi/0.5\,\mathrm{mm}$, D known	1.5×10^3	1.2×10^3
	$q = 2\pi/0.5\,\mathrm{mm}$, D free	exactly degenerate	exactly degenerate

- *Flip-angle distribution within the voxel* (slice or slab profile), each position in its own steady state. With the profile shape known exactly, the bound is finite but 5.9×10^5–1.2×10^7, five to seven orders of magnitude above the two-scan value. With the profile width unknown, the degeneracy is exact again.
- *Relaxation during an on-resonance pulse* ($\tau = 1$–$2\,\mathrm{ms}$). The signal changes by 0.15–0.8%, but the degeneracy remains exact. As for an ideal pulse, the pathway amplitudes keep a one-ratio structure: $a_k/a_0 = r^k$ for $k \ge 0$ and $a_{-1}/a_0 = (r - E_2)/(1 - E_2 r)$ (verified to 10^{-16}).
- *Off-resonance during a 1 ms pulse* (8–160 Hz). Bound 2.7×10^3–5.3×10^4, carried by small pathway-dependent changes that unmodelled gradient delays or eddy currents would easily mask.

- *Diffusion in the unbalanced gradient* ($D = 0.7 \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$, one cycle of dephasing per 1 or 0.5 mm). With D known the bound is 1.2–4.8×10^3; with D free the degeneracy is exact.

A simple count explains the pattern. The echo decay and phases of a scan determine T_2, T_2^*, $\Delta\omega$ and ϕ_0, which leaves three real pathway amplitudes for (M_0, B_1^+, T_1). In the ideal model they depend on only two combinations. An effect that adds structure supplies a weak third, unless it brings an unknown of its own (a profile width, D), which absorbs it exactly.

Even in the best cases this information is of no practical use. Pooled through a smooth field (34) with $p = 35$ and $N = 10^5$ at SNR 300, the bounds of Table 12 give $\mathrm{SD}(\log B_1^+) \approx 0.09$ (diffusion, 0.5 mm, D known) to 0.17 (160 Hz), against 1.4×10^{-4} for the two-scan protocol. Even these values require the extra physics to be modelled exactly, and the profile shows how strongly a model error biases the answer. If the true Gaussian profile is 2% narrower than assumed, the scan-1 fit converges exactly (residual 10^{-16}) to $B_1^+ = 1.47$, $T_1 = 386$ ms for WM (truth 1 and 850 ms); 5% narrower gives $B_1^+ = 1.89$; for the ramp, 2% gives $B_1^+ = 1.73$; for the Hann-sinc profile the bias has the opposite sign ($B_1^+ = 0.93$). If the true Gaussian or ramp profile is wider than assumed, no exact solution exists. The fit then runs to $B_1^+ \to 0$, $T_1 \to \infty$ with a residual of 10^{-7}–10^{-4} of the signal, far below the noise. A single scan cannot be rescued by modelling more physics, any more than by a smoother B_1^+.

5.5 What a second acquisition does

With a second scan of the same T_R and a different nominal flip angle, $\alpha_2 = r\,\alpha_1$ with a common B_1^+, a voxel provides u_1, u_2 and, by Lemma 1, $M_0\sqrt{\tanh(T_R/2T_1)}$: three numbers for the three unknowns (M_0, B_1^+, T_1). The construction of Proposition 7 now fails: E_1' must satisfy $u(B_1^{+\prime}\alpha_1, E_1') = u_1$ and $u(B_1^{+\prime}\alpha_2, E_1') = u_2$ simultaneously, which are the two equations (29) in two unknowns. In Ernst-angle form, the ratio ξ_2/ξ_1 of the two scans is the function $f(B_1^+)$ of (18), in which T_1 and M_0 cancel. Its solutions are isolated: the per-voxel null curve collapses to discrete points, the true parameters and the aliases of Proposition 6, which the FID phase and the range bound remove. Locally the information is nonsingular (Table 6), and the profiled information of a parametric field is positive definite.

The single-scan family is the familiar B_1^+ sensitivity of variable-flip-angle (VFA, DESPOT1) T_1 mapping. With B_1^+ known, one MPME scan determines T_1 and M_0, as a complete VFA experiment does; with B_1^+ unknown, it inherits the VFA error law $\delta \ln T_1 \approx -2\,\delta \ln B_1^+$, $\delta \ln M_0 \approx -\delta \ln B_1^+$ (exactly (17)). Its pathways and echoes do not change this. Spoiled VFA data give one amplitude per angle: two angles cannot determine B_1^+ (two numbers for three unknowns), and more angles determine it only weakly, because on the axis of Fig. 2 their samples move with B_1^+ at nearly equal rates $\alpha_i/\sin\alpha_i$ and translate almost together with the bump. Each MPME scan also reports its offset from the bump, so two scans suffice in principle; but two small angles are again nearly degenerate (Fig. 2b), and the published design relies on a second angle whose sample moves at a very different rate.

Two variants achieve the same in principle: a second scan with the same flip angle and a different T_R (then E_1 changes while α does not, the principle of actual-flip-angle imaging), or transient (non-steady-state) data, to which Theorem 1 does not apply. The published design uses a second flip angle near 360°, whose B_1^+ lever arm (28) is large.

5.6 How much second scan is needed

Once each voxel carries some information about B_1^+, pooling it through a smooth field is effective, and the second scan can be short and coarse. If scan 2 covers a fraction r of k-space and both scans

are reduced to that resolution, each low-resolution voxel has about $1/\sqrt{r}$ times the per-voxel SNR of a full-resolution voxel, and there are rN of them for N full-resolution voxels. A parametric field with p coefficients fitted to them has

$$\mathrm{SD}(\log \hat{B}_1^+) \approx \mathrm{CRLB}_{B_1^+} \cdot \frac{\sigma}{M_0}\sqrt{r} \cdot \sqrt{\frac{p}{rN}} = \mathrm{CRLB}_{B_1^+} \cdot \frac{\sigma}{M_0}\sqrt{\frac{p}{N}}, \tag{34}$$

independent of r to this order. With $\mathrm{CRLB}_{B_1^+} \approx 2.5$, $p = 35$ and $N = 10^5$–10^6 this is of order 10^{-4} at SNR 300: noise in the smoothed field is negligible, and its accuracy is limited by bias (truncation ringing, partial volume, approach to steady state, model mismatch), which grows as the coverage shrinks. Table 13 summarises the design implications.

Table 13: Design of the second acquisition (bounds per unit σ/M_0, white-matter-like tissue, B_1^+ over $[0.8, 1.2]$; median and worst case over B_1^+).

Choice	Recommendation	Evidence
α_2^{nom}	$\approx 330°$	B_1^+ bound median 2.8 at 330° vs 3.5 (320°), 3.1 (340°), 7.0 (290°) with $\alpha_1 =$ 15°; the worst case (11–13) hardly depends on α_2, so a 360° crossing inside the head is unavoidable
α_1^{nom}	$\approx 20°$	improves both stages: B_1^+ bound median 2.44, worst 8.5 (vs 2.77, 11.7 at 15°); scan-1 T_1 bound with B_1^+ known WM 46, GM 52 (vs 53, 54)
Scan-2 readout	same three windows, same T_R	a one-window scan 2 with $T_R = 12$ ms is less efficient at equal duration (median 3.8 vs 2.8): fewer echoes constrain T_2, T_2^* less
Scan-2 coverage	small, e.g. 24×24 central k_y–k_z lines (≈ 8 mm)	(34): coverage barely affects the smoothed noise; bias limits it (apodise both scans identically; robust, uncertainty-weighted fit)
Preparation	$\geq 5T_1/T_R$ dummy repetitions	a short scan 2 must start in steady state
Phase	no re-shim or receiver change between scans	complex ML and the FID-sign test use the relative phase

With 24×24 lines the second scan takes about 14 s plus 7 s of preparation, under 3% of a whole-brain exam, instead of about 48 s in the published design. The claim that coverage can be reduced this far without bias is argued from (34) and the phantom results of Section 9, not yet demonstrated by a coverage sweep. The two-stage reconstruction that uses this scan is given in Algorithm 4 and evaluated in Section 9.

6 Maximum-likelihood estimation

6.1 Estimator

Stack the real and imaginary parts of all 18 complex measurements of a voxel into $y \in \mathbb{R}^{36}$ and the model (21) into $s(\theta)$. Under the Gaussian noise model the negative log-likelihood is $\|r(\theta)\|^2/(2\sigma^2)$ with $r = s(\theta) - y$, so the ML estimate is the nonlinear least-squares solution $\hat{\theta} = \arg\min_\theta \|r(\theta)\|^2$, independent of σ. The model is the exact steady state (5), so the estimator and the CRLB use the same forward model.

Constraints. The fit is a projected Levenberg–Marquardt (LM) iteration. After every step the parameters are projected onto the physical domain of the Lorentzian model, $\log T_2^* \le \log T_2$ ($R_2' \ge 0$), and onto a box $B_1^+ \in [\mathrm{e}^{-1}, 540/330]$, $T_1 \in [10, 20\,000]$ ms, $T_2 \in [1, 5000]$ ms. The upper B_1^+ bound is the assumption $\alpha_2 \le 540°$ of [1]; it excludes the exact complex alias of Proposition 6 by construction. Active constraints are recorded per voxel; such estimates are not locally unbiased and fall outside the interior CRLB.

Initialisation. Magnitude parameters and $\Delta\omega$ come from a sequential estimator; the global phase has a closed-form optimum for fixed remaining parameters, $\phi_0^{(0)} = \arg\bigl(\sum S_{ijk}\overline{S_{ijk}^{(0)}}\bigr)$, where $S^{(0)}$ is the model at the initial values with $\phi_0 = 0$.

Iteration. With $J = \partial r/\partial\theta$ (exact, Section 8),

$$\bigl(J^{\mathsf{T}}J + \lambda\,\mathrm{diag}(J^{\mathsf{T}}J)\bigr)\Delta = -J^{\mathsf{T}}r, \qquad \theta \leftarrow \Pi(\theta + \Delta), \tag{35}$$

where Π is the projection. A step is accepted if it lowers $\|r\|^2$ ($\lambda \leftarrow 0.3\lambda$), otherwise rejected ($\lambda \leftarrow 10\lambda$); $\lambda_0 = 10^{-2}$, $\lambda \in [10^{-9}, 10^{9}]$, 40 iterations. Marquardt's diagonal scaling makes the step invariant to rescaling individual parameters. A parameter held fixed (e.g. a given B_1^+ map) has its Jacobian column removed and a unit diagonal, so that $\Delta = 0$ for it. For zero-residual (noise-free) problems with a nonsingular Jacobian, Gauss–Newton converges locally quadratically, and noise-free fits recover the parameters to 10^{-15}; with noise the residual at the solution is nonzero and the local convergence is in general linear, with a rate set by the residual and the model curvature.

Multiple starts. Nonlinear least squares can stop at a local minimum: across the 360° branch, at the branch edge where scan 2 vanishes, or along the weak direction of Fig. 5. We run four starts—the model-fit estimate, the analytic inverse where valid, and the model-fit estimate with T_1 replaced by 700 and by 2000 ms—and keep the lowest final cost. In simulation the global optimum can be checked: a final cost above the cost at the true parameters is certainly not the ML estimate.

Is one start enough? On 2000 voxels with random tissue (T_1 600–3000 ms, T_2 40–140 ms, T_2^*/T_2 0.6–0.95, B_1^+ 0.85–1.15), a single start ends at a higher cost than the best of four in 15.8% of voxels at $\mathrm{SNR}(M_0) = 300$ (median $|\Delta \log \hat{T}_1| = 0.025$, maximum 4.1) and 2.3% at SNR 1000. A χ^2 test on the final cost (99.9 % quantile, 29 degrees of freedom) flags only 26% and 22% of these, respectively. One start with a cost check is therefore not sufficient; we use four starts throughout, which the exact Jacobian makes affordable.

6.2 Efficiency

Tables 14 to 16 compare complex ML with the analytic inverse, a sequential model-fit baseline (the same steps with B_1^+, T_1, M_0 fitted to the FID and echo amplitudes of both scans) and the CRLB, on 1000 noise realisations of the reference tissue. All ratios use ordinary standard deviations; the Monte Carlo standard error of an SD from 1000 samples is about 2%. (These runs used the box $B_1^+ \le \mathrm{e}$; for the published protocol no estimate reached it, so the results are the same with the 540/330 cap.)

- **Where its assumptions hold, ML attains the bound.** At SNR 1000 and 3000 the SD/CRLB ratio is 0.99–1.03 for T_1, B_1^+, T_2 and M_0, the bias is below 10^{-3} in log units, at most 0.3% of estimates touch a constraint, and no voxel fails the cost check. The published protocol can separate B_1^+ and T_1.

Table 14: Monte Carlo errors of $\log \hat{T}_1$ (1000 noise realisations, reference tissue, $B_1^+ = 1$). SD, mean (bias), RMSE and robust spread (IQR/1.349) are over voxels with a defined estimate. "Undef.": fraction with no valid T_1; "bound": fraction with a constraint or search limit active (for ML: $R_2' = 0$ or the safeguard box). CRLB: bound on the SD for unbiased estimators using both scans with B_1^+ unknown.

Protocol	SNR(M_0)	Estimator	SD	mean	RMSE	robust	undef. %	bound %
15°/330°	300	CRLB	0.135					
		Joint ML	0.128	-0.007	0.128	0.134	0.0	19.5
		Model fit	0.939	+0.335	0.996	0.907	0.0	17.6
		Analytic inverse	0.959	+0.445	1.056	1.026	1.9	0.2
		Analytic, B_1^+ given	0.526	+0.119	0.539	0.359	1.9	–
15°/330°	1000	CRLB	0.041					
		Joint ML	0.041	+0.001	0.041	0.039	0.0	0.3
		Model fit	0.157	+0.016	0.158	0.114	0.0	0.0
		Analytic inverse	0.207	+0.036	0.210	0.127	0.0	0.0
		Analytic, B_1^+ given	0.108	+0.009	0.108	0.107	0.0	–
15°/330°	3000	CRLB	0.014					
		Joint ML	0.014	+0.000	0.014	0.013	0.0	0.0
		Model fit	0.038	+0.001	0.038	0.040	0.0	0.0
		Analytic inverse	0.043	+0.001	0.043	0.044	0.0	0.0
		Analytic, B_1^+ given	0.035	+0.000	0.035	0.039	0.0	–
15°/30°	1000	CRLB	2.106					
		Joint ML	1.364	+0.409	1.423	2.117	0.0	34.8
		Model fit	1.794	+0.094	1.796	2.876	0.0	66.2
15°/30°	3000	CRLB	0.702					
		Joint ML	0.861	+0.244	0.894	0.771	0.0	10.2
		Model fit	1.492	+0.267	1.515	2.256	0.0	34.7

- **At SNR 300 the comparison is approximate.** The ratio is 1.00 for B_1^+ and 0.95 for T_1, but $R_2' \geq 0$ is active in 19.5% of voxels (true R_2' is small); those estimates are biased, which is how the T_1 SD can fall slightly below the bound.

- **The analytic inverse is 3.2–7.1 times above the bound** for T_1, with bias +0.44 in $\log T_1$ and 1.9% undefined estimates at SNR 300; B_1^+ is 4.8–28 times above. Given the true B_1^+, its T_1 step is 1.9–2.8 times above its matched bound (scan 1, B_1^+ known: 57.1). The model-fit baseline, which shares the sequential structure but not the closed forms, behaves similarly, so the loss comes from the sequential structure.

- **The small-angle control supports no efficiency claim.** For 15°/30°, 10–35% of ML estimates are on a box bound and biased; the bound itself is of order one in log units (2.1 at SNR 1000, a one-sigma factor of about 8), while T_2 is unaffected.

Table 15: Monte Carlo SD (robust spread in parentheses) of $\log \hat{B_1^+}$, $\log \hat{T}_2$ and $\log \hat{M}_0$, and the mean of $\log \hat{B_1^+}$. Same runs as Table 14.

Protocol	SNR	Estimator	B_1^+ SD (rob.)	B_1^+ mean	T_2 SD (rob.)	M_0 SD (rob.)
15°/330°	300	CRLB	0.010		0.077	0.063
		Joint ML	0.010 (0.009)	-0.0017	0.075 (0.074)	0.063 (0.060)
		Model fit	0.060 (0.037)	+0.0011	0.476 (0.368)	0.649 (0.569)
		Analytic inverse	0.270 (0.461)	-0.1699	0.506 (0.370)	0.604 (0.571)
15°/330°	1000	CRLB	0.003		0.023	0.019
		Joint ML	0.003 (0.003)	-0.0001	0.023 (0.025)	0.019 (0.019)
		Model fit	0.017 (0.013)	-0.0025	0.115 (0.110)	0.108 (0.085)
		Analytic inverse	0.066 (0.015)	-0.0135	0.118 (0.112)	0.118 (0.076)
15°/330°	3000	CRLB	0.001		0.008	0.006
		Joint ML	0.001 (0.001)	+0.0000	0.008 (0.008)	0.006 (0.006)
		Model fit	0.004 (0.004)	-0.0003	0.036 (0.038)	0.028 (0.029)
		Analytic inverse	0.005 (0.004)	-0.0005	0.037 (0.037)	0.025 (0.026)
15°/30°	1000	CRLB	1.023		0.023	1.048
		Joint ML	0.663 (1.037)	-0.2099	0.022 (0.022)	0.679 (1.053)
		Model fit	0.862 (1.380)	-0.0678	0.115 (0.110)	0.937 (1.490)
15°/30°	3000	CRLB	0.341		0.008	0.349
		Joint ML	0.422 (0.373)	-0.1228	0.007 (0.007)	0.429 (0.381)
		Model fit	0.723 (1.097)	-0.1443	0.036 (0.038)	0.757 (1.139)

Table 16: Joint ML diagnostics, and the analytic inverse with B_1^+ given compared with matched bounds: both scans with B_1^+ known (28.0 per unit σ/M_0) and scan 1 only with B_1^+ known (57.1), the data its T_1 step actually uses.

		Joint ML			Analytic, B_1^+ given: SD / bound	
Protocol	SNR	$R_2' = 0\,\%$	box %	cost > cost(truth)	both scans	scan 1 only
15°/330°	300	19.5	0.0	0	5.65	2.77
15°/330°	1000	0.3	0.0	0	3.86	1.89
15°/330°	3000	0.0	0.0	0	3.80	1.86
15°/30°	1000	0.1	34.7	0	–	–
15°/30°	3000	0.0	10.2	0	–	–

7 Least squares on magnitude images

The published inverse works with magnitudes, made real-valued by the sign convention of its Eq. 18. A direct alternative is to fit the model magnitudes,

$$\hat{\theta} = \arg\min_{\theta} \sum_{j} \big(|y_j| - |s_j(\theta)|\big)^2, \tag{36}$$

over $\theta = (\log M_0, \log B_1^+, \log T_1, \log T_2, \log T_2^*)$ with the same projected LM iteration, constraints and four starts as complex ML. This is not maximum likelihood for magnitude data, which are Rician: it ignores the noise floor $\mathbb{E}|y| - A \approx \sigma^2/(2A)$, which for the weakest amplitude of the reference tissue is 10% at SNR 300, 1% at SNR 1000 and 0.1% at SNR 3000. Section 4.7 showed that magnitudes carry essentially all the local information; this section concerns the global bit they lose.

7.1 Why a magnitude fit needs the branch test and complex ML does not

1. Magnitudes lose the sign of $\sin\alpha$. By Theorem 1 and Lemma 1 the magnitudes of scan i determine u_i and $M_0|v_i| = M_0\sqrt{1-u_i^2}\sqrt{\tanh(T_R/2T_1)}$, i.e. $\cos\alpha_i$ for any trial E_1, but not the sign of $\sin\alpha_i$. Equivalently, $|a_k(\alpha)|$ is even and 2π-periodic in α, hence symmetric about every multiple of 180°.

2. This produces an exact alias in the physiological range. By Proposition 6(a,c) the magnitude data at (M_0, B_1^+, T_1) are reproduced exactly at every solution $(B_1^{+\prime}, E_1')$ of (29), with M_0' set by part (a). The branch $\alpha_2 = 360° + x'$ admits such a solution: for the reference tissue $(1, 1500\,\text{ms}, 1)$ and $(1.199, 1038\,\text{ms}, 0.832)$ give the same 18 magnitudes to machine precision (Table 8). The alias is close to the small-angle scaling ($1500/1.199^2 = 1043$) but exact.

3. It is a symmetry of the magnitude likelihood, not a noise effect. The alias exists for every parameter value, so the map $\theta \mapsto |s(\theta)|$ takes the same values on two separate sheets of parameter space. Any likelihood that depends on the data only through magnitudes, Gaussian or Rician, has two equal global maxima for every noise realisation. We verified this directly: fitting noisy magnitudes on each side of 360° gives final costs that agree to a median relative difference of 10^{-12} at SNR 300, 1000 and 3000. No SNR resolves it, and the local CRLB, identical on both sheets, cannot detect it. An unconstrained magnitude fit returns the maximum nearest its starting point: in Table 17 it chose the wrong branch in 47, 3 and 0 of 1000 voxels at SNR 300, 1000 and 3000, and in 13, 3 and 0 when the same fits were run with the exact Jacobian, which takes slightly different paths. The counts describe the optimiser, not the data.

4. Complex data carry the missing bit. By Proposition 5, $a_k(-\alpha) = -a_k(\alpha)$, so moving α_2 from $360° - x$ to $360° + x'$ flips the sign of every scan-2 signal relative to scan 1 (Proposition 6(b)). A single ϕ_0 and a single $\Delta\omega$ are shared by both scans, so the relative sign cannot be absorbed. The best complex fit restricted to $360° < \alpha_2 < 540°$ moves to the edge $\alpha_2 = 360°$, where scan 2 vanishes, and leaves the entire scan-2 signal ($0.144\,M_0$) as residual: its cost exceeds the true-branch cost by $1.9 \times 10^3\sigma^2$ at SNR 300 ($2.1 \times 10^4\sigma^2$ at SNR 1000), a likelihood ratio of about e^{-935} or smaller. Complex ML selects the correct branch automatically.

5. The FID-sign test restores the bit. The test of [1],

$$\operatorname{sign} \operatorname{Re} \sum_j S_{1j0}\overline{S_{2j0}} = \operatorname{sign}(\sin\alpha_1 \sin\alpha_2), \tag{37}$$

uses only the sign of the relative phase of the two FID images (it follows from $a_0 \propto v \propto \sin\alpha$ and the shared echo phases). In the magnitude fit it becomes a constraint: B_1^+ is confined to the interval between consecutive zeros of $\sin(B_1^+\alpha_i^{\mathrm{nom}})$ on which $\sin\alpha_1 \sin\alpha_2$ has the measured sign; a start on the wrong side is reflected across the nearest zero, where $|a_k|$ is symmetric, and every step is projected back into the interval (Algorithm 2).

6. What the test cannot do. Branches with the same sign pattern (the complex aliases, $\alpha_2 > 540°$) are invisible to the test and to complex ML; the range bound excludes them. Near the crossing $\alpha_2 = 330° B_1^+ = 360°$ the scan-2 FID vanishes and the sign is unreliable. A wrong decision is cheap only very close to it: the magnitude alias approaches the truth only within about ±2% of the crossing $B_1^+ = 360/330$ (there its B_1^+ is within 4% and its T_1 within 8% of the truth for the reference tissue, against 20% and 31% at $B_1^+ = 1$). Moreover, with scan 2 almost empty the problem approaches the one-scan degeneracy, and the likelihood, complex or magnitude, has distant near-equal maxima at the other scan-2 nulls, $\alpha_2 \approx 180°$ and 540°, where scan 1 is matched by the family of Proposition 7. In the phantom at SNR 1000, all but three of the large T_1 errors left by either fit lie within 3° of 360°, nearly all at such a maximum; for complex ML each has a cost no higher than at the truth (Section 9, Fig. 11). The region needs care: multiple starts and the spatial restart of Section 9 remove optimiser failures there but not these genuine maxima, so the voxels concerned should be flagged, or fitted from scan 1 alone with B_1^+ from a smooth map (Scenario B; not with both scans, Table 7). Both the test and complex ML need the relative phase of the two scans to be preserved: the same receiver phase and coil combination and no motion or B_0 drift between the sequential acquisitions. With an unknown phase per scan, complex ML would lose the relative sign and the alias would reappear for complex data too.

7.2 Comparison on identical data

Table 17 compares the analytic inverse, magnitude least squares without and with the FID-sign constraint, and complex ML on the noise realisations of Table 14.

With the branch test, magnitude least squares has SD/CRLB within the Monte Carlo error of complex ML at every SNR and no wrong-branch voxels; the Rician floor adds a small bias at SNR 300 (-0.005 in $\log T_1$). Without it, wrong-branch voxels dominate the B_1^+ error (SD up to seven times the robust spread) and bias T_1 by about $(B_1^+)^{-2}$. The analytic inverse, which uses the same magnitudes and the same FID sign, is 3–7 times worse for T_1 and up to 28 times for B_1^+: since magnitudes carry essentially all the local information and a joint magnitude fit extracts it, its loss is algorithmic.

Table 17: Magnitude versus complex estimation on identical data (paper protocol, reference tissue, 1000 realisations). SD and mean of the log-estimates; bounds for complex data and for magnitude (Rician) data. Last column: voxels on the wrong side of $\alpha_2 = 360^\circ$ ("wrong").

SNR		T_1 SD	T_1 mean	B_1^+ SD	T_2 SD	wrong
300	CRLB complex/Rician	0.1350/0.1356		0.0097/0.0098	0.0769/0.0771	
	Analytic inverse [1]	0.9587	+0.445	0.2695	0.5060	–
	Magnitude LS	0.1909	-0.030	0.0692	0.0743	47
	Magnitude LS + FID sign	0.1251	-0.005	0.0094	0.0742	0
	Complex ML	0.1281	-0.007	0.0097	0.0749	–
1000	CRLB complex/Rician	0.0405/0.0405		0.0029/0.0029	0.0231/0.0231	
	Analytic inverse [1]	0.2072	+0.036	0.0657	0.1180	–
	Magnitude LS	0.0457	+0.000	0.0104	0.0231	3
	Magnitude LS + FID sign	0.0409	+0.001	0.0029	0.0231	0
	Complex ML	0.0411	+0.001	0.0029	0.0231	–
3000	CRLB complex/Rician	0.0135/0.0135		0.0010/0.0010	0.0077/0.0077	
	Analytic inverse [1]	0.0425	+0.001	0.0046	0.0373	–
	Magnitude LS	0.0136	+0.000	0.0010	0.0076	0
	Magnitude LS + FID sign	0.0136	+0.000	0.0010	0.0076	0
	Complex ML	0.0136	+0.000	0.0010	0.0076	–

8 Exact Jacobians by implicit differentiation

Both least-squares estimators spend nearly all their time forming Jacobians: 36×7 per voxel for complex data and 18×5 for magnitudes. Reverse-mode differentiation (back-propagation) produces one *row* per backward pass and would need 36 or 18 passes; forward mode produces one *column* per pass, 7 or 5 passes, and forward differences are an inexact version of it. The structure of the steady state allows something cheaper.

8.1 Differentiating the steady state

With RF phase 0 the post-pulse steady state of one isochromat solves (5),

$$A\mathbf{M} = \mathbf{b}, \qquad A = I - RQ, \qquad Q = R_z(\psi)\,\mathrm{diag}(E_2, E_2, E_1), \qquad \mathbf{b} = (1 - E_1)R\hat{\mathbf{z}}, \tag{38}$$

with $R = R_x(\alpha)$ and $\Delta\omega = 0$ (Proposition 1). For $x \in \{\alpha, E_1, E_2\}$,

$$\frac{\partial \mathbf{M}}{\partial x} = A^{-1}\Big(\frac{\partial \mathbf{b}}{\partial x} - \frac{\partial A}{\partial x}\mathbf{M}\Big), \tag{39}$$

so the inverse used for the solution serves all three derivatives. With $R' = \partial R_x/\partial\alpha$, $\Pi_z = \mathrm{diag}(0,0,1)$, $\Pi_{xy} = \mathrm{diag}(1,1,0)$ and $R_z\Pi_z = \Pi_z$,

$$\begin{aligned}
\frac{\partial A}{\partial \alpha} &= -R'Q, & \frac{\partial \mathbf{b}}{\partial \alpha} &= (1 - E_1)R'\hat{\mathbf{z}},\\
\frac{\partial A}{\partial E_1} &= -R\Pi_z, & \frac{\partial \mathbf{b}}{\partial E_1} &= -R\hat{\mathbf{z}},\\
\frac{\partial A}{\partial E_2} &= -RR_z(\psi)\Pi_{xy}, & \frac{\partial \mathbf{b}}{\partial E_2} &= \mathbf{0},
\end{aligned} \tag{40}$$

and the three right-hand sides of (39) are

$$(1 - E_1)R'\hat{\mathbf{z}} + R'Q\mathbf{M}, \qquad -R\hat{\mathbf{z}} + R\hat{\mathbf{z}}\,M_z, \qquad RR_z(\psi)\Pi_{xy}\mathbf{M}. \tag{41}$$

There are $N \times n_{\mathrm{iso}}$ systems per scan; batched general solvers carry a large per-matrix overhead at size 3×3, so $A^{-1} = \mathrm{adj}(A)/\det A$ is formed explicitly (nine cofactors and one determinant, elementwise over the batch). A is invertible whenever $E_1, E_2 < 1$. Pathway amplitudes are discrete Fourier coefficients over the isochromats, and the transform is linear, so $\partial a_k/\partial x$ is the same FFT of $\partial m_\perp/\partial x$; for magnitudes $\partial|a_k|/\partial x = \mathrm{Re}(\overline{a_k}\,\partial a_k/\partial x)/|a_k|$.

8.2 Chain rule

With $\tau = t + kT_R$, $\Phi = \phi_0 + \Delta\omega\tau$, $D = \exp(-t/T_2 - R_2'|\tau|)$, $R_2' = 1/T_2^* - 1/T_2$, $S = M_0 a_k D\mathrm{e}^{i\Phi}$ and $\partial\alpha/\partial \log B_1^+ = \alpha$, $\partial E_1/\partial \log T_1 = E_1 T_R/T_1$, $\partial E_2/\partial \log T_2 = E_2 T_R/T_2$:

$$
\begin{aligned}
\frac{\partial S}{\partial \log M_0} &= S, & \frac{\partial S}{\partial \log B_1^+} &= M_0 D\mathrm{e}^{i\Phi}\alpha\frac{\partial a_k}{\partial \alpha}, \\
\frac{\partial S}{\partial \log T_1} &= M_0 D\mathrm{e}^{i\Phi}\frac{E_1 T_R}{T_1}\frac{\partial a_k}{\partial E_1}, & \frac{\partial S}{\partial \log T_2} &= M_0 D\mathrm{e}^{i\Phi}\frac{E_2 T_R}{T_2}\frac{\partial a_k}{\partial E_2} + S\frac{t-|\tau|}{T_2}, \\
\frac{\partial S}{\partial \log T_2^*} &= S\frac{|\tau|}{T_2^*}, & \frac{\partial S}{\partial \Delta\omega} &= i\tau S, \qquad \frac{\partial S}{\partial \phi_0} = iS.
\end{aligned}
\tag{42}
$$

The second term of $\partial S/\partial \log T_2$ combines e^{-t/T_2} and the coupling through R_2' ($\partial R_2'/\partial \log T_2 = 1/T_2$). For magnitudes, $|a_k|$ replaces a_k and the phase terms are dropped.

8.3 Levenberg–Marquardt with exact Jacobians

A forward-difference iteration costs one trial evaluation plus one per parameter (8 for complex data, 6 for magnitudes). With exact Jacobians each trial evaluation returns the residual and its Jacobian; an accepted step thus already carries the Jacobian of the next iteration, and a rejected one keeps the current Jacobian. Each iteration costs one explicit inverse and four right-hand sides per isochromat. Exact derivatives also make single precision usable: forward differences in float32 would need steps of order 10^{-3}, whereas the implicit Jacobian is accurate to about 10^{-6}. A small Jacobian error changes the convergence path but not the solution, which is set by the residual.

8.4 Verification and benchmark

The implicit Jacobians agree with automatic differentiation of the reference model to 6×10^{-15} (magnitude) and 3×10^{-16} (complex) in double precision and to 10^{-6} in single precision. Table 18 lists wall times on 4096 voxels with random tissue (T_1 600–3000 ms, T_2 40–120 ms, B_1^+ 0.8–1.1), $\mathrm{SNR}(M_0) = 1000$, one start and 15 iterations (15 and 40 iterations differ by less than 10^{-9} in $\log T_1$), on a laptop CPU with 8 threads.

The exact Jacobian alone gives a factor of 10–12, more than the reduction from 6–8 evaluations to one would suggest, because the explicit inverse also avoids per-matrix solver overhead. Halving the isochromats and moving to float32 each roughly halve the time again, changing estimates by at most 10^{-3} in $\log \hat{T}_1$ ($< 1\%$ of the noise spread). With the full Monte Carlo scripts, the implicit Jacobian reproduces the reference results to every displayed digit for the published protocol and cuts a 25-minute run to 88 s (22 s in float32 with 128 isochromats and 15 iterations).

8.5 Implications

- **Whole-brain feasibility.** With four starts a 256×256 slice takes about 2–3 minutes in float64 and about 1 minute in float32 on this CPU; a whole brain (about 160 slices, masked)

Table 18: Fitting time and agreement with the forward-difference float64 reference (largest $|\Delta \log \hat{T}_1|$ over 4096 voxels; the noise-induced spread is 0.164 for magnitudes and 0.163 for complex data).

Fit	Jacobian, precision, isochromats	ms/voxel	256^2 slice	speed-up	max $\lvert\Delta \log \hat{T}_1\rvert$
Magnitude	finite differences, f64, 256	15.0	16.4 min	1.0	reference
	implicit, f64, 256	1.41	1.5 min	10.7	1.6×10^{-5}
	implicit, f64, 128	0.73	0.8 min	20.6	1.6×10^{-5}
	implicit, f32, 256	0.74	0.8 min	20.2	1.2×10^{-3}
	implicit, f32, 128	0.44	0.5 min	33.7	1.0×10^{-3}
Complex ML	finite differences, f64, 256	19.2	21.0 min	1.0	reference
	implicit, f64, 256	1.55	1.7 min	12.4	1.4×10^{-5}
	implicit, f64, 128	0.83	0.9 min	23.2	1.4×10^{-5}
	implicit, f32, 256	0.90	1.0 min	21.3	1.2×10^{-3}
	implicit, f32, 128	0.56	0.6 min	34.4	1.2×10^{-3}

a few hours, and minutes on a GPU, since every operation is a batched elementwise or FFT operation.

- **Uncertainty maps at no extra cost.** The Jacobian at the solution gives the local CRLB, $\sqrt{\mathrm{diag}(J^\top J)^{-1}}\,\sigma$, per voxel. Section 9 shows these maps are calibrated where the estimator is efficient.
- **Protocol optimisation.** The Fisher information of a protocol is one Jacobian evaluation; gradients with respect to protocol parameters (TR, flip angles, echo times) can be obtained the same way, enabling direct CRLB-based design rather than grid searches.
- **Training data and learned estimators.** Simulated signals with exact Jacobians and bounds are cheap, for neural networks (Section 11.3) and for evaluating them against the CRLB.
- **Hierarchical estimation.** An empirical-Bayes EM algorithm (Section 11.3) needs a MAP fit and a Laplace covariance per voxel and class at every iteration; both come directly from this machinery.
- **Robust starts.** Because one start is not enough (Section 6), multiple starts are the dominant cost; making each fit 10–34 times cheaper is what makes four starts routine.

9 Digital phantom

9.1 Phantom and simulated acquisition

The phantom is a 256×256 axial-slice-like arrangement of tissue classes (Fig. 9): an outer CSF rim, cortical grey matter with a gyrated boundary, white matter, two lateral ventricles, two deep grey-matter nuclei and a white-matter lesion, in total 37876 voxels (Table 19). Each class carries 3 T-typical relaxation values, modulated by a smooth random texture of 3% (log scale) so that tissues are not uniform. Smooth fields model the transmit field (centre brightening, $B_1^+ = 0.87$–1.15, so that about a quarter of the voxels have $\alpha_2 > 360°$ and a ring of voxels has $\alpha_2 \approx 360°$), the off-resonance (a quadratic background of ± 20 Hz and a frontal hot spot of about $+50$ Hz), the receive sensitivity ($\pm 15\%$) and the receive phase.

Table 19: Phantom tissue classes (before the 3% texture).

Tissue	T_1 (ms)	T_2 (ms)	T_2^* (ms)	PD	voxels
CSF	4000	1500	900	1.00	5614
GM	1350	90	60	0.82	14 782
WM	850	66	50	0.69	16 196
deep GM	1130	58	40	0.75	1104
lesion	1500	120	80	0.85	180

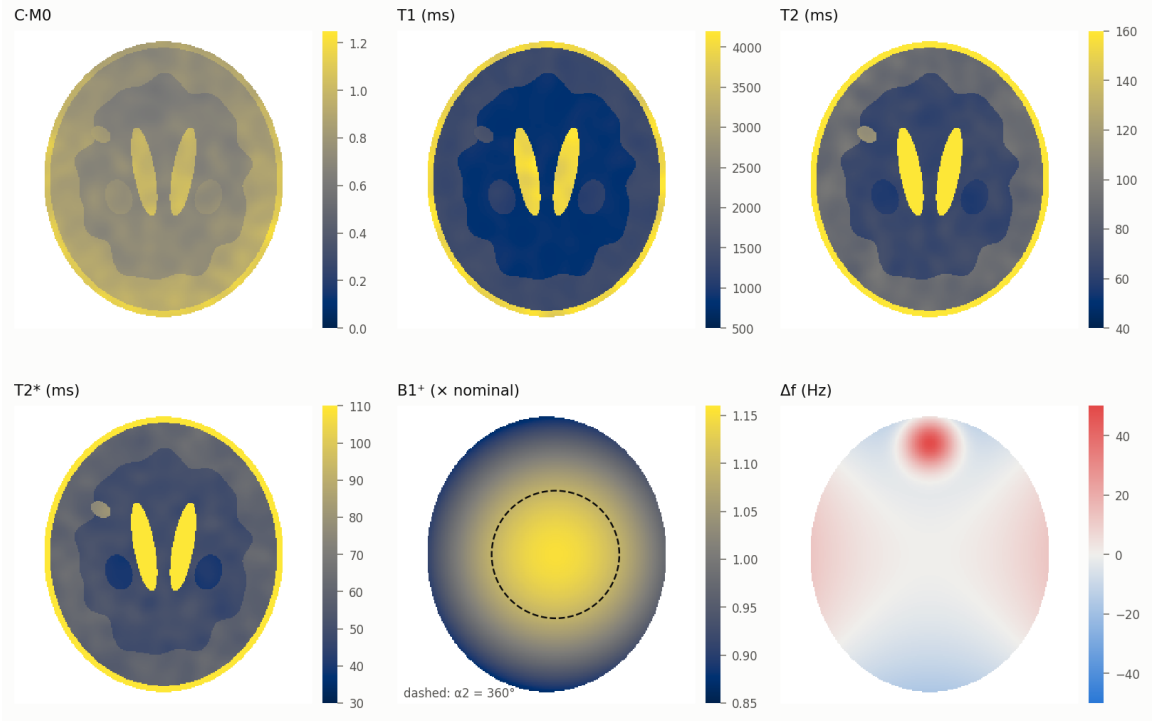


Figure 9: True parameter maps. The dashed contour marks $\alpha_2 = 330°$ $B_1^+ = 360°$.

Images are generated voxel by voxel with the forward model (21) (512 isochromats; no partial volume within a voxel) and complex Gaussian noise with $\sigma = 1/\mathrm{SNR}(M_0)$, for $\mathrm{SNR}(M_0) = 1000$ and 300 (M_0 ranges from 0.6 to 1.2). Two acquisitions are simulated (Fig. 10):

- **Scenario A**: both scans at full resolution, the per-voxel setting of Sections 4, 6 and 7;
- **Scenario B**: the published design, in which scan 2 covers only the central 25% of the phase-encoding lines (k_y), with zero filling. (In 3D the published design keeps $25\% \times 25\%$ of k_y–k_z, a factor 16; in 2D only one phase-encoding axis can be truncated, a factor 4.) Noise is added before truncation, i.e. the same per-sample noise with fewer samples.

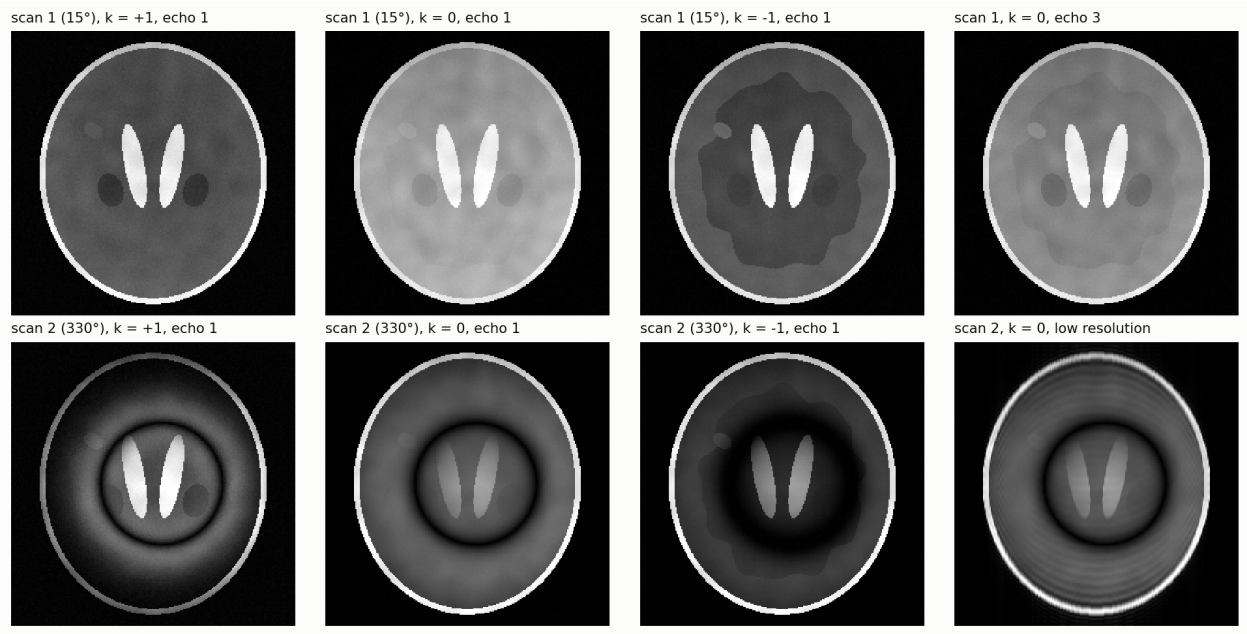


Figure 10: Simulated MPME images (magnitude, $\mathrm{SNR}(M_0) = 1000$). The scan-2 images show the dark ring where $\alpha_2 = 330° B_1^+ = 360°$ and the signal vanishes; bottom right: the low-resolution scan 2 of Scenario B, with truncation ringing.

9.2 Reconstructions

All fits use the exact Jacobian in single precision with 128 isochromats, 25 iterations, the $B_1^+ \leq 540/330$ bound and four starts (Algorithm 3), processed in chunks of 4096 voxels.

- **Scenario A.** The analytic inverse (voxel-wise flip angle, no smoothing); complex ML; magnitude least squares with the FID-sign constraint. For the last two, a fifth *spatial-restart* start is added after the first pass: the first-pass B_1^+ map is smoothed by a robust polynomial fit, and each voxel is restarted from $(M_0\kappa,\ B_{1,\mathrm{smooth}}^+,\ T_1\kappa^2)$ with $\kappa = B_1^+/B_{1,\mathrm{smooth}}^+$, i.e. moved along the near-degenerate direction of Proposition 4; the lowest cost is kept. The neighbourhood only proposes a start, so every estimate remains a per-voxel likelihood maximum.
- **Scenario B.** The paper pipeline (Algorithm 4): the flip angle from the two low-resolution scans by Eq. 15, a degree-4 robust polynomial fit, then T_1, T_2 and M_0 from full-resolution scan 1 by Eqs. 1, 16 and 17. The ML pipeline: B_1^+ from the two low-resolution scans by complex ML, the same polynomial fit, then complex ML on full-resolution scan 1 with B_1^+ fixed.

Why the spatial restart. Without it, 324 voxels (SNR 1000) had $|\log(\hat{T}_1/T_1)| > 0.3$; 321 of them lie within 3° of $\alpha_2 = 360°$, where scan 2 carries almost no signal and the problem approaches the one-scan degeneracy of Corollary 1. In 222 of the 324 the final cost exceeded the cost at the true parameters (optimiser failures: all four starts derive from estimators that are unreliable at 360°). The smoothed B_1^+ map is accurate to 0.03% there, and the restart removes all optimiser failures (no voxel ends above the true-parameter cost). The remaining large errors (53 at SNR 1000, 50 of them within 3° of 360°) all have a cost at or below the true one: they are genuine maxima of the likelihood, a property of the protocol at $\alpha_2 \approx 360°$ rather than of the estimator.

9.3 Results

Table 20: Phantom, SNR(M_0) = 1000: median and SD of $\log(\hat{\theta}/\theta)$ over each tissue (bias and spread), and the fraction of voxels with an undefined estimate. Truth includes a 3% smooth texture.

Tissue	Method	T_1 median	T_1 SD	T_2 median	T_2 SD	B_1^+ median	undef. %
WM	A, analytic inverse	+0.006	0.298	-0.000	0.145	-0.000	0.0
	A, complex ML	+0.001	0.094	+0.001	0.041	+0.000	0.0
	A, magnitude LS + FID	-0.000	0.124	+0.003	0.041	+0.000	0.0
	B, paper pipeline	+0.001	0.101	-0.000	0.145	+0.001	0.0
	B, ML pipeline	+0.001	0.073	-0.000	0.055	-0.000	0.0
GM	A, analytic inverse	+0.003	0.403	+0.001	0.160	+0.000	0.0
	A, complex ML	+0.001	0.042	+0.001	0.030	+0.000	0.0
	A, magnitude LS + FID	+0.002	0.042	+0.000	0.030	+0.000	0.0
	B, paper pipeline	+0.043	0.117	+0.001	0.160	-0.015	0.0
	B, ML pipeline	+0.001	0.065	+0.001	0.051	+0.000	0.0
deep GM	A, analytic inverse	+0.011	0.321	-0.009	0.141	-0.000	0.0
	A, complex ML	-0.000	0.115	+0.000	0.041	+0.000	0.0
	A, magnitude LS + FID	-0.002	0.131	+0.003	0.041	+0.000	0.0
	B, paper pipeline	+0.006	0.154	-0.009	0.141	-0.001	0.0
	B, ML pipeline	+0.001	0.084	-0.000	0.052	-0.000	0.0
lesion	A, analytic inverse	-0.004	0.119	-0.009	0.167	-0.000	0.0
	A, complex ML	-0.007	0.050	-0.005	0.031	-0.000	0.0
	A, magnitude LS + FID	-0.008	0.050	-0.004	0.031	-0.000	0.0
	B, paper pipeline	-0.008	0.118	-0.009	0.167	+0.001	0.0
	B, ML pipeline	+0.005	0.059	-0.004	0.050	+0.000	0.0
CSF	A, analytic inverse	+0.015	0.650	-0.019	0.802	-0.000	0.1
	A, complex ML	-0.001	0.079	-0.000	0.080	+0.000	0.0
	A, magnitude LS + FID	-0.001	0.079	-0.000	0.080	+0.000	0.0
	B, paper pipeline	+0.085	0.356	-0.019	0.802	-0.015	0.1
	B, ML pipeline	-0.004	0.099	+0.003	0.114	-0.000	0.0

Scenario A. The maps reproduce the voxel-wise analysis (Figs. 11 to 13, Table 20). The analytic inverse is unbiased in the median but noisy: at SNR 1000 its T_1 spread is 0.30 in white matter and 0.40 in grey matter, against 0.042 for complex ML in grey matter, and its errors grow towards the periphery (lower B_1^+) and form a sharp ring at $\alpha_2 = 360°$. Complex ML and magnitude least squares with the FID sign are nearly indistinguishable, with negligible bias in every tissue; their white-matter spread (0.094 and 0.124) is inflated by the genuine outliers on the 360° ring, which lies in white matter, while the grey-matter spread (0.042) is close to the voxel-wise bound. No voxel farther than 3° from 360° ends on the wrong branch for any method.

Table 21: Phantom, SNR(M_0) = 300: median and SD of $\log(\hat{\theta}/\theta)$ over each tissue (bias and spread), and the fraction of voxels with an undefined estimate. Truth includes a 3% smooth texture.

Tissue	Method	T_1 median	T_1 SD	T_2 median	T_2 SD	B_1^+ median	undef. %
WM	A, analytic inverse	+0.015	0.652	+0.003	0.771	-0.002	1.3
	A, complex ML	+0.000	0.230	+0.005	0.138	+0.000	0.0
	A, magnitude LS + FID	-0.009	0.250	+0.026	0.139	+0.000	0.0
	B, paper pipeline	-0.017	0.452	+0.003	0.771	+0.001	1.3
	B, ML pipeline	-0.006	0.225	+0.008	0.176	-0.000	0.0
GM	A, analytic inverse	+0.015	0.913	+0.004	0.879	+0.000	1.6
	A, complex ML	+0.002	0.139	+0.004	0.102	+0.000	0.0
	A, magnitude LS + FID	+0.013	0.137	+0.001	0.100	+0.002	0.0
	B, paper pipeline	+0.027	0.508	+0.004	0.879	-0.016	1.6
	B, ML pipeline	-0.000	0.208	+0.009	0.168	+0.000	0.0
deep GM	A, analytic inverse	-0.018	0.728	-0.023	0.762	-0.002	5.4
	A, complex ML	+0.002	0.287	+0.003	0.170	+0.000	0.0
	A, magnitude LS + FID	-0.026	0.301	+0.033	0.143	+0.000	0.0
	B, paper pipeline	-0.048	0.656	-0.023	0.762	-0.000	5.4
	B, ML pipeline	-0.003	0.276	+0.004	0.173	+0.000	0.0
lesion	A, analytic inverse	-0.032	0.747	-0.028	0.933	-0.001	1.7
	A, complex ML	-0.026	0.161	-0.006	0.100	-0.000	0.0
	A, magnitude LS + FID	-0.030	0.160	+0.007	0.100	-0.000	0.0
	B, paper pipeline	-0.037	0.593	-0.028	0.933	+0.001	1.7
	B, ML pipeline	-0.009	0.179	-0.014	0.160	-0.000	0.0
CSF	A, analytic inverse	+0.047	1.011	-0.054	1.228	-0.010	17.8
	A, complex ML	-0.015	0.275	+0.003	0.280	+0.000	0.0
	A, magnitude LS + FID	-0.015	0.275	+0.002	0.280	+0.000	0.0
	B, paper pipeline	+0.050	0.757	-0.054	1.228	-0.020	17.8
	B, ML pipeline	-0.035	0.343	+0.031	0.393	+0.000	0.0

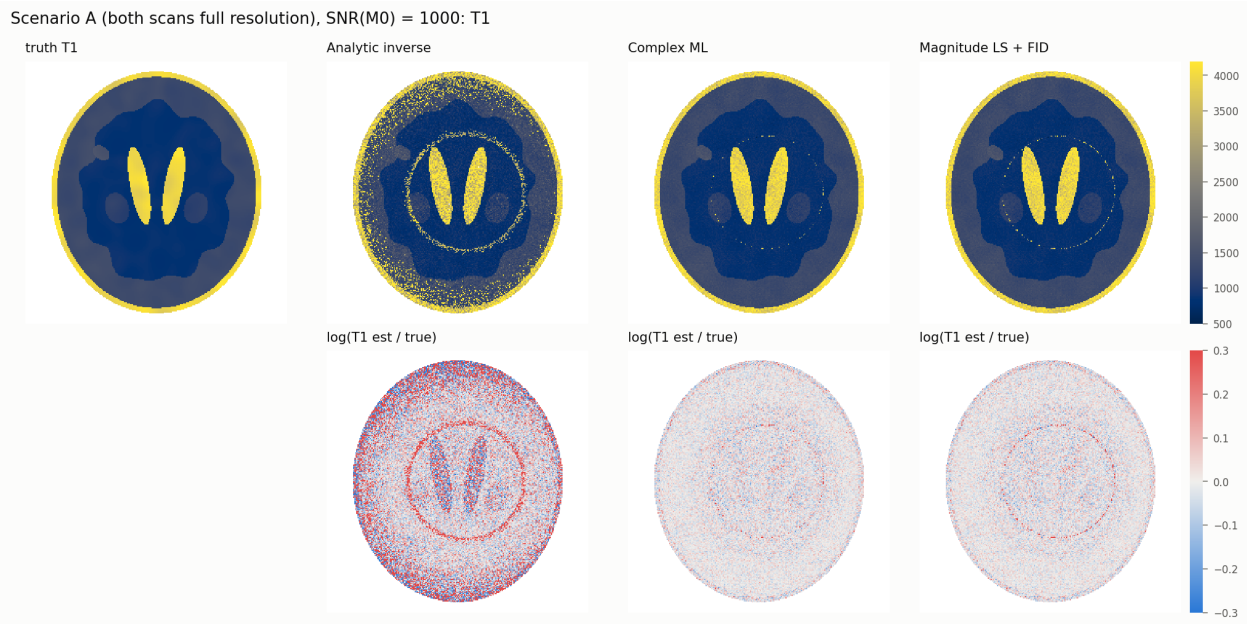


Figure 11: Scenario A, SNR(M_0) = 1000: T_1 maps and log-ratio error maps.

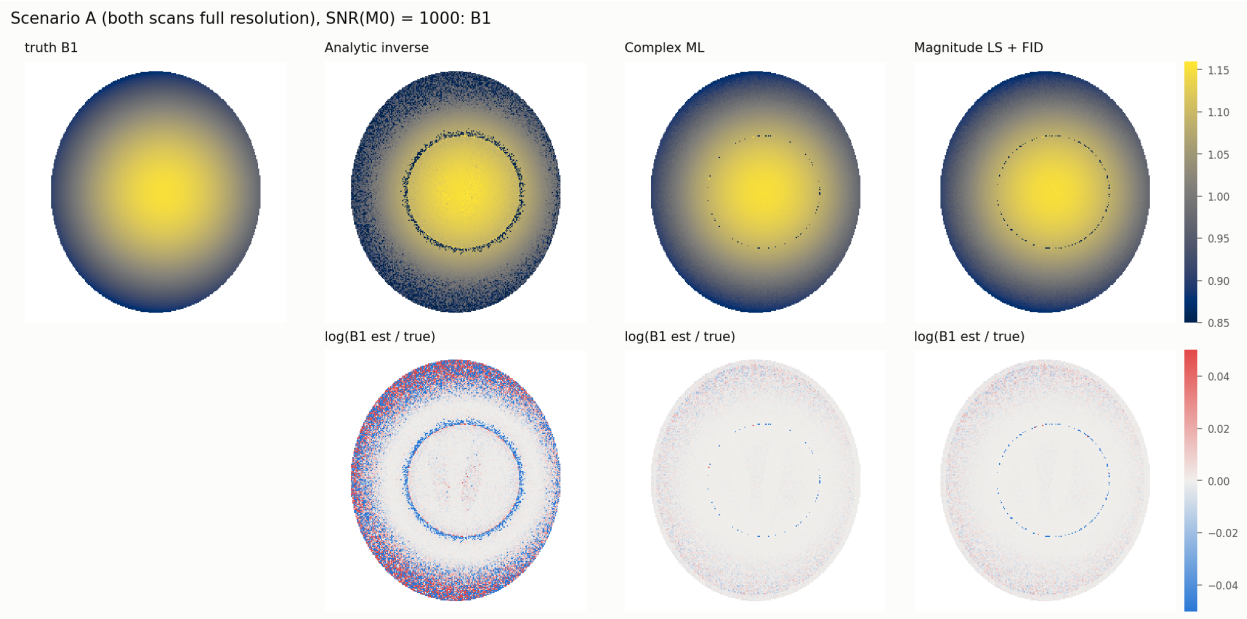


Figure 12: Scenario A, SNR(M_0) = 1000: B_1^+ maps and log-ratio error maps.

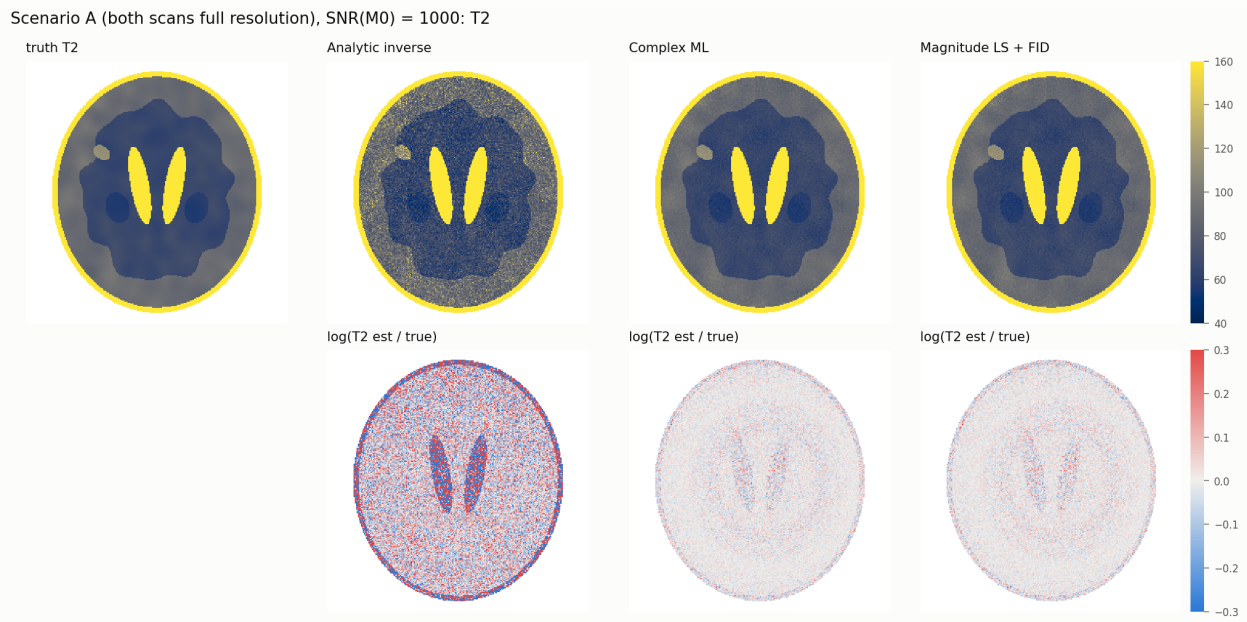


Figure 13: Scenario A, SNR(M_0) = 1000: T_2 maps and log-ratio error maps.

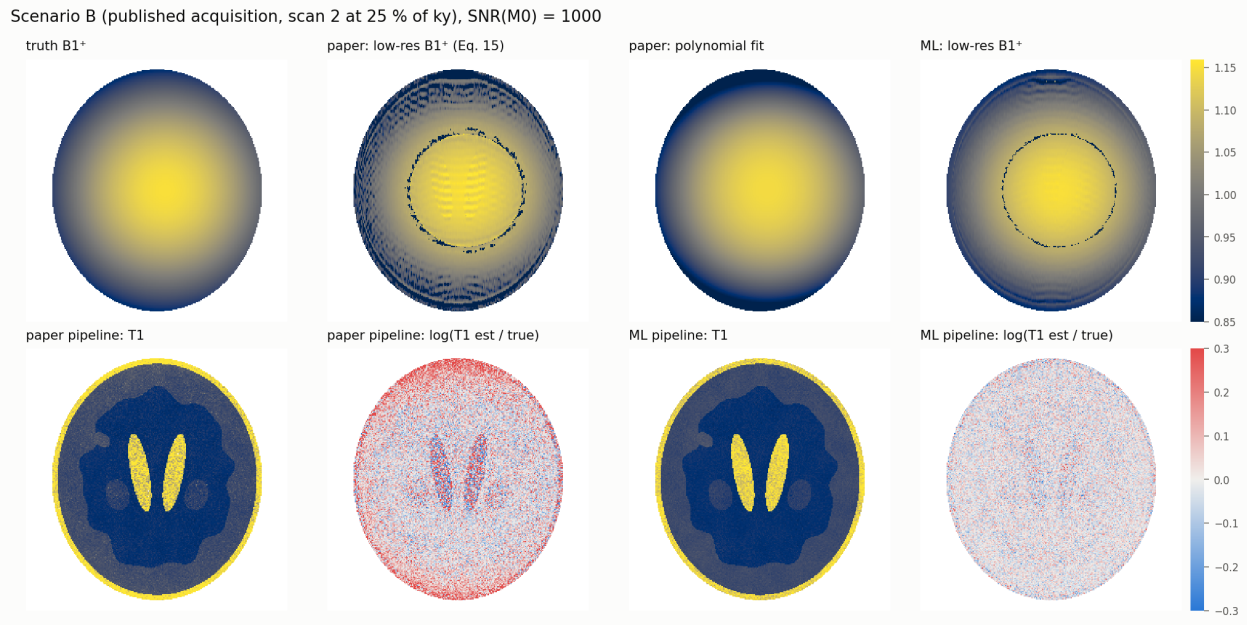


Figure 14: Scenario B (scan 2 at 25% of k_y), SNR(M_0) = 1000. Top: true B_1^+; low-resolution B_1^+ from Eq. 15; its polynomial fit; low-resolution B_1^+ from complex ML. Bottom: T_1 and its error for the two pipelines.

Scenario B. The low-resolution flip-angle map of the paper pipeline shows truncation ringing, errors at the 360° ring and outliers at the periphery (Fig. 14); the polynomial fit absorbs much of this but remains biased by −1.5% in grey matter (up to 15% at the edge). Through the error transfer $\partial \log T_1 / \partial \log B_1^+ \approx -2$ of the scan-1 T_1 step (Table 7) this produces a T_1 bias of +4.3% in grey matter and +8.5% in CSF; with the true B_1^+, the same T_1 step is unbiased (−0.06% in grey matter). The ML pipeline's smoothed map is accurate to 0.2% everywhere and its T_1 is unbiased in every tissue. Its T_1 spread is larger than in Scenario A in grey matter (0.065 vs 0.042), as expected from the scan-1-only bound (57.1 vs 40.5), but smaller in white matter (0.073 vs 0.094), because a smooth, fixed B_1^+ removes the 360° ambiguity. Its T_2 spread is larger than in Scenario A because stage 2 uses only scan 1.

SNR 300. Table 21 gives the same statistics at SNR(M_0) = 300 (image SNR about 26). The analytic inverse breaks down: its T_1 spread is 0.65 in white matter and 0.91 in grey matter, and T_1 is undefined in 1.3–1.7% of tissue voxels and 17.8% of CSF voxels. Complex ML and magnitude least squares with the FID sign remain nearly unbiased, with spreads of 0.14 (grey matter) to 0.28 (CSF), no undefined estimates and no wrong-branch voxels away from the 360° ring (Fig. 15). In Scenario B the paper pipeline has spreads of 0.45–0.51 in tissue and the same undefined fractions, whereas the ML pipeline is nearly unbiased with spreads of 0.21–0.23; its grey-matter spread is 1.5 times that of Scenario A, close to the bound ratio 57.1/40.5 = 1.4. At this SNR, errors above 0.3 in log units are expected from noise alone in long-T_1 tissue (the local bound for CSF is of that order), and only 4 of 37876 complex-ML voxels end above the true-parameter cost.

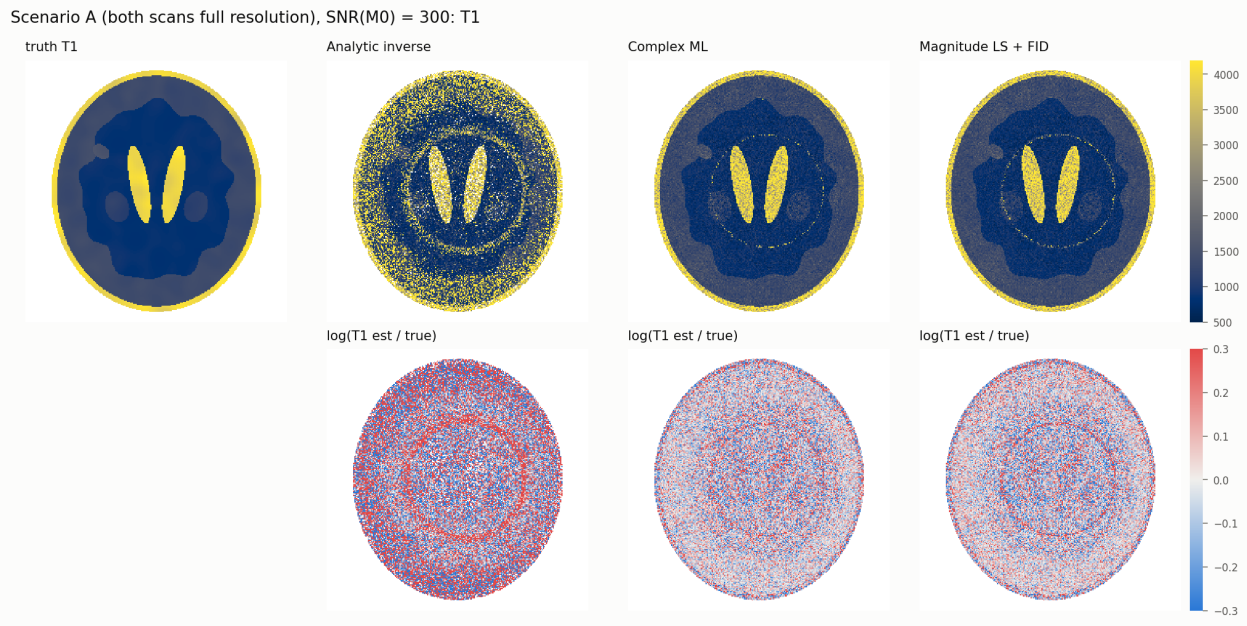


Figure 15: Scenario A, SNR(M_0) = 300: T_1 maps and log-ratio error maps.

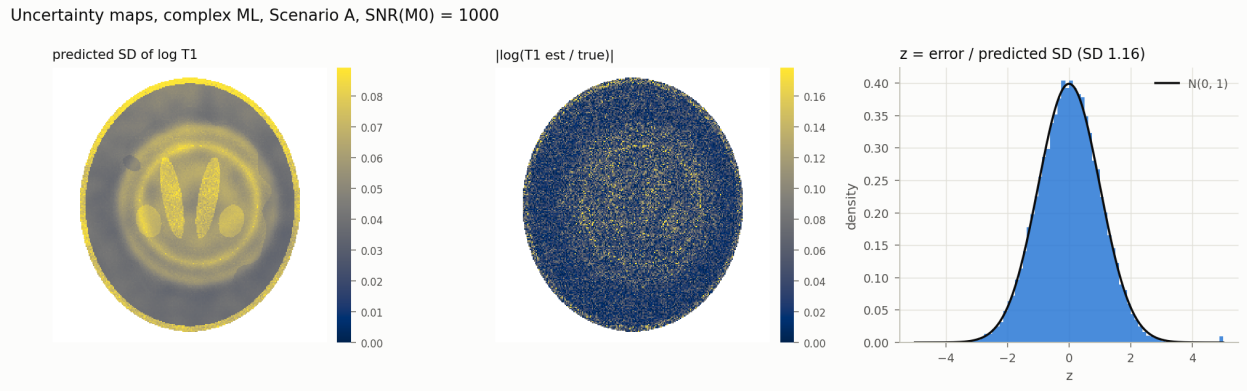


Figure 16: Complex ML, Scenario A, SNR(M_0) = 1000: predicted SD of $\log T_1$ (local CRLB at the estimate), absolute error, and the distribution of the standardised error z against $N(0,1)$ (z clipped to ± 5 for display).

9.4 Uncertainty maps

The local CRLB evaluated at the estimate gives a predicted standard deviation for every voxel at the cost of one Jacobian (Fig. 16). The map shows the structure expected from Section 4: higher uncertainty in CSF and long-T_1 tissue, in deep grey matter, at low B_1^+, and on the 360° ring. Standardised errors are close to $N(0,1)$. For complex ML in Scenario A, 95% intervals cover the truth in 95.2% of voxels at SNR 1000 (95.4% at SNR 300); the SD of z is 1.16 (1.01), above 1 only because of the genuine outliers at 360°, which a local bound cannot describe. For the ML pipeline of Scenario B (with B_1^+ treated as known), coverage is 95.5% and 97.7% and the SD of z is 0.98 and 0.87: slightly conservative at SNR 300, consistent with the constraint $R_2' \geq 0$ being active in many voxels at that SNR (a constrained estimator can have a variance below the interior bound, Section 6).

9.5 Run times

Table 22: Reconstruction times for the phantom (voxels in the mask; laptop CPU, 8 threads, implicit Jacobian, float32, 128 isochromats, 25 iterations).

Step	SNR 1000	SNR 300
A: analytic inverse	3.1 s	3.0 s
init (model fit)	8.7 s	8.8 s
A: complex ML (4 starts)	130.3 s	143.4 s
A: complex ML spatial restart	32.1 s	34.2 s
A: magnitude LS + FID sign (4 starts)	116.5 s	126.0 s
A: magnitude LS spatial restart	29.8 s	29.3 s
A: uncertainty maps	3.3 s	2.8 s
B: paper pipeline	3.8 s	3.9 s
B: ML pipeline	207.3 s	204.5 s

A full set of reconstructions for one SNR level takes about 9 minutes on the laptop CPU (Table 22); each full-slice fit with four starts takes about 2 minutes, and the uncertainty maps a few seconds.

10 Partial volume: the BrainWeb phantom

The forward model assumes one tissue per voxel. Real voxels at 1–1.2 mm often contain several, and the cortex, two to three voxels thick, borders CSF almost everywhere. This section measures the consequences with the BrainWeb anatomical model and the unchanged estimators of Sections 3, 6 and 7, and explains them from the structure of the model.

10.1 The mixed-voxel signal

If a voxel contains compartments c with volume fractions f_c, proton densities PD_c and relaxation parameters θ_c, and shares the location-dependent quantities B_1^+, $\Delta\omega$, ϕ_0 and the receive sensitivity C, its signal is

$$S_{ijk}^{\mathrm{mix}} = C \sum_c f_c \,\mathrm{PD}_c \; s_{ijk}(\theta_c), \qquad s_{ijk}(\theta_c) = v_{i,c}\, G_k(u_{i,c}, E_{2,c}, \Delta\omega)\, \mathrm{e}^{-t/T_{2,c}} \mathrm{e}^{-R'_{2,c}|t+kT_R|} \mathrm{e}^{i\Phi}, \quad (43)$$

by Theorem 1 applied to each compartment. A single-compartment estimator returns the $\hat{\theta}$ whose model $S(\hat{\theta})$ is closest to S^{mix} (in the sense of its own cost function). Two questions follow: how far S^{mix} lies from the set of single-compartment signals (the model manifold), which determines bias and detectability, and where on the manifold the closest point lies, which determines the reported values.

Nearby compartments: a second-order effect. If the compartments are close in parameter space, write $\theta_c = \bar{\theta} + \delta_c$ with $\bar{\theta}$ the weighted mean (weights $w_c = f_c\text{PD}_c / \sum f\text{PD}$, in log coordinates). Then

$$\frac{S^{\text{mix}}}{C \sum f\text{PD}} = \sum_c w_c S(\theta_c) = S(\bar{\theta}) + \tfrac{1}{2} \sum_c w_c \, \delta_c^{\mathsf{T}} \nabla^2 S(\bar{\theta}) \, \delta_c + O(|\delta|^3), \tag{44}$$

because the first-order terms cancel ($\sum w_c \delta_c = 0$). The mixture is a single-compartment signal at the weighted mean, up to a second-order term whose size is set by the curvature of the model and the spread of the compartments. White and grey matter are close (T_1 850 vs 1350 ms, T_2 66 vs 90 ms): a WM/GM mixture is fitted to within 0.01σ per measurement at SNR 1000 and reported at the weighted mean (median deviation -0.013 in $\log T_1$, Table 23). Such voxels are harmless: they look like an intermediate tissue, which is what they are.

CSF: a first-order mismatch. CSF is far from tissue ($T_1 = 4000$ ms, $T_2 = 1500$ ms), so (44) does not apply, and its signal contribution is strongly pathway- and scan-dependent. Per unit volume, CSF produces 1.4–1.7 times the grey-matter signal in the FID of scan 1, but 2.6–3.4 times in the echo pathways of scan 1 and 6.5–7.8 times in pathway +1 of scan 2 (proton-density ratio 1.22): long T_2 preserves the higher-order configuration states that the echo pathways sample, and the large second pulse feeds them. A voxel with 10% CSF by volume therefore takes about 13% of its FID but up to about 46% of its scan-2, $k = +1$ signal from CSF. No single (T_1, T_2, T_2^*) reproduces this pattern: the mixture lies off the manifold, the residual is first order in the CSF fraction, and the fitted values depend on which measurements an estimator weights most.

Consequences. (i) Apparent parameters are not any simple average: relative to the PD-weighted mean they are biased in a direction and by an amount that depend on the parameter and the estimator (Fig. 18 and Table 23). (ii) Different estimators no longer agree, because they project the same off-manifold point differently: complex ML weights all 36 measurements, the analytic inverse estimates T_2 from scan 1 alone and B_1^+ from closed-form ratios. (iii) The shared B_1^+ absorbs part of the mismatch: the two scans see the CSF admixture differently, and an estimator that ties them by a common B_1^+ may shift B_1^+ to reconcile them. (iv) The residual grows only linearly with the CSF fraction while the noise is fixed, so detectability depends on SNR: the noise-free cost $\|r\|^2$ makes the residual χ^2 non-central with parameter $\lambda = \|r\|^2/\sigma^2$, which grows as SNR^2. (v) Uncertainty maps describe noise, not model error, so their coverage must drop wherever the bias is comparable to the noise.

10.2 Phantom and protocol

We use the BrainWeb normal-brain anatomical model [6, 7, 8]: fuzzy tissue-membership volumes at 1 mm isotropic resolution ($181 \times 217 \times 181$). An axial slice through the lateral ventricles (index 95, Talairach $z = 23$ mm) is zero-padded to 256×256. Voxels whose discrete label is CSF, grey or white matter are kept (18 668 voxels); fat, skin and skull are excluded, since fat would require a chemical-shift model. Each compartment takes the tissue values of Table 19 with a 3% smooth

texture, and the fields B_1^+, $\Delta\omega$, C, ϕ_0 are those of Section 9, rescaled to the head. Signals follow (43). Of the brain voxels, 13 258 are pure (one tissue $\geq 95\%$), 2818 are WM/GM mixtures and 2592 contain 5–95% CSF (Fig. 17).

Three cases are reconstructed with the estimators unchanged (four starts plus the spatial restart for the two least-squares fits): *noise-free with partial volume* (model error alone), *SNR 1000 with partial volume*, and a *control* at SNR 1000 in which each voxel is assigned entirely to its dominant tissue (same anatomy, no partial volume). For a mixed voxel we report errors against the PD-weighted geometric mean $\theta_{\text{eff}} = \exp(\sum_c w_c \log \theta_c)$, and show the dominant-tissue value alongside, since a mixed voxel has no single true value.

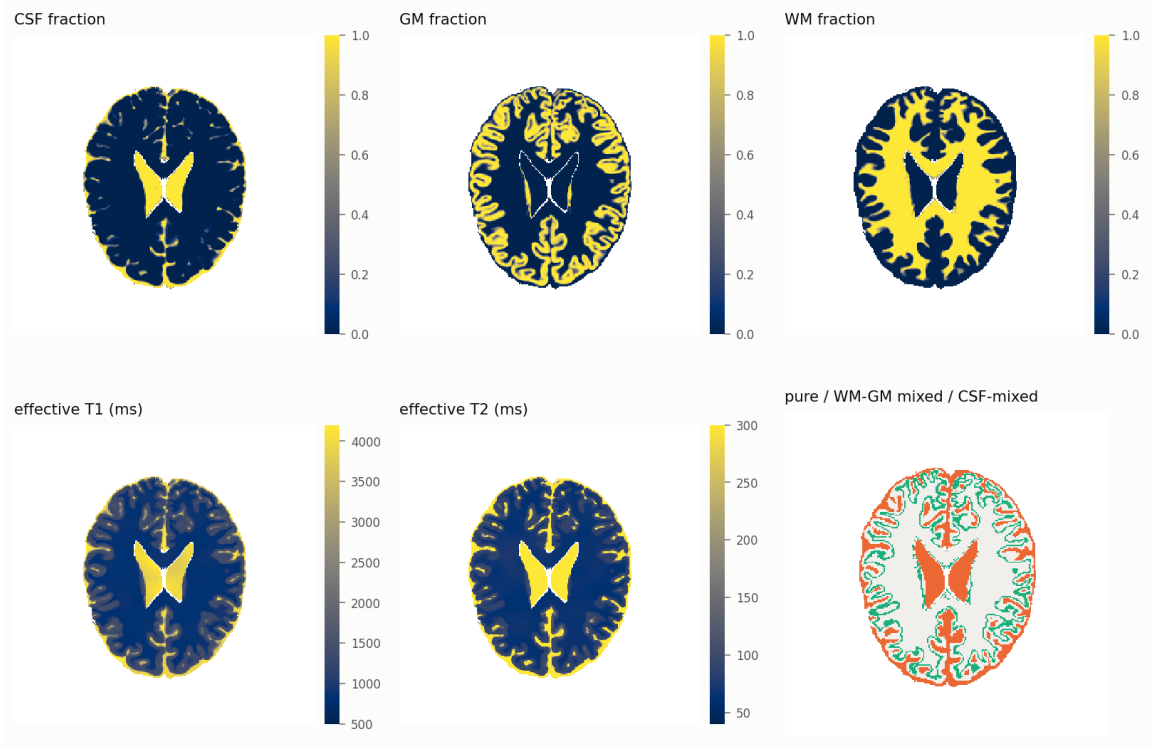


Figure 17: BrainWeb slice: tissue fractions, the PD-weighted effective T_1 and T_2, and voxel categories (pure; WM/GM mixed; containing $\geq 5\%$ CSF).

10.3 Results

Pure tissue and WM/GM mixtures. Pure voxels are reconstructed without bias by all estimators, and complex ML has the same spread and 94.6–94.9% coverage as in the control (Table 24). WM/GM mixtures are fitted to 0.01σ per measurement, reported at the PD-weighted mean to within -0.013 in $\log T_1$ and -0.009 in $\log T_2$ for complex ML, and their coverage (93.8%) is close to nominal: the second-order picture of (44) holds.

CSF-mixed voxels: biased, confidently. In the 2592 voxels with 5–95% CSF all estimators are biased by far more than the noise (Table 23 and Figs. 18 and 19). For complex ML the deviation of $\log T_1$ from the PD-weighted mean is -0.13, -0.32 and -0.24 in the three CSF bins; T_2 deviates by -0.04 to -0.11 from the mean but lies far above the grey-matter value (Fig. 18: at 10% CSF the apparent T_2 is about 40% above that of grey matter). Against the dominant tissue, the cortex

Table 23: BrainWeb, noise-free: median of $\log(\hat{\theta}/\theta_{\text{eff}})$ per voxel category, where θ_{eff} is the PD-weighted geometric mean of the compartment values (B_1^+: relative to the true field). Last columns: RMS mismatch residual per measurement in units of σ at SNR 1000, and the probability that a χ^2 test (99.9 %) flags the voxel at SNR 1000 and 3000.

		Complex ML			Analytic inverse				
Category	voxels	T_1	T_2	B_1^+	T_1	T_2	B_1^+	RMS/σ	flagged 1000 / 3000
pure CSF	1823	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	0.00	0.1 % / 0.1 %
pure GM	3931	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	0.00	0.1 % / 0.1 %
pure WM	7504	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	0.00	0.1 % / 0.1 %
WM/GM mixed	2818	-0.013	-0.009	+0.000	-0.017	-0.006	+0.000	0.01	0.1 % / 0.1 %
CSF 5-25%	1069	-0.133	-0.038	+0.005	-0.042	-0.142	-0.007	0.35	3.4 % / 59.2 %
CSF 25-50%	580	-0.324	-0.103	+0.009	-0.113	-0.428	-0.032	0.61	12.4 % / 95.7 %
CSF 50-95%	943	-0.251	-0.113	+0.002	-0.029	-0.503	-0.012	0.29	2.2 % / 48.1 %

Table 24: BrainWeb at SNR(M_0) = 1000, complex ML: SD of $\log(\hat{T}_1/T_{1,\text{eff}})$ with partial volume and in the control without it, coverage of 95% uncertainty intervals, and the fraction of voxels flagged by the χ^2 test.

Category	T_1 SD (PV)	T_1 SD (control)	median (PV)	coverage (PV)	flagged (PV)
pure CSF	0.077	0.077	-0.003	94.7 %	0.0 %
pure GM	0.060	0.060	+0.000	94.6 %	0.2 %
pure WM	0.084	0.084	+0.000	94.9 %	0.1 %
WM/GM mixed	0.057	0.049	-0.012	93.8 %	0.1 %
CSF 5-25%	0.085	0.063	-0.138	21.9 %	4.0 %
CSF 25-50%	0.058	0.052	-0.318	0.5 %	15.2 %
CSF 50-95%	0.139	0.079	-0.243	26.8 %	3.1 %

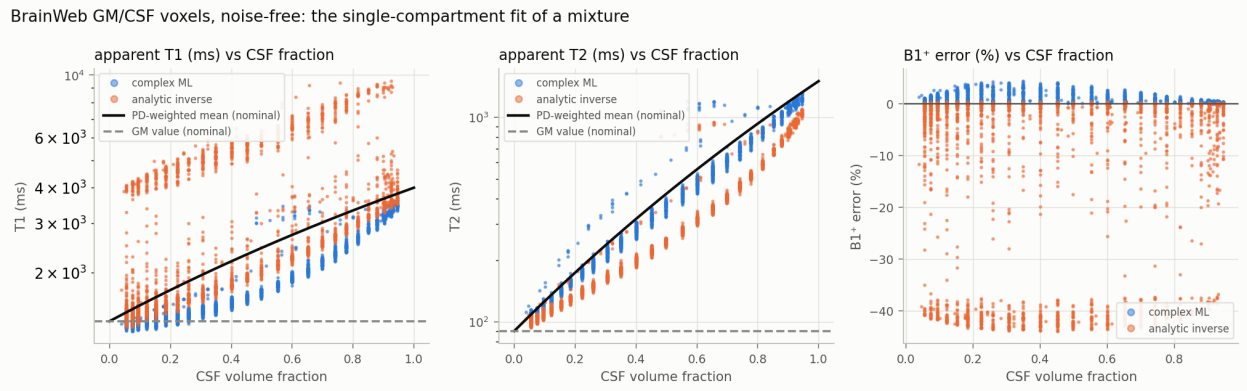


Figure 18: Noise-free GM/CSF voxels (< 5% WM): values returned by the single-compartment estimators versus the CSF volume fraction, with the PD-weighted mean and the grey-matter value (nominal, without texture). Right: B_1^+ error.

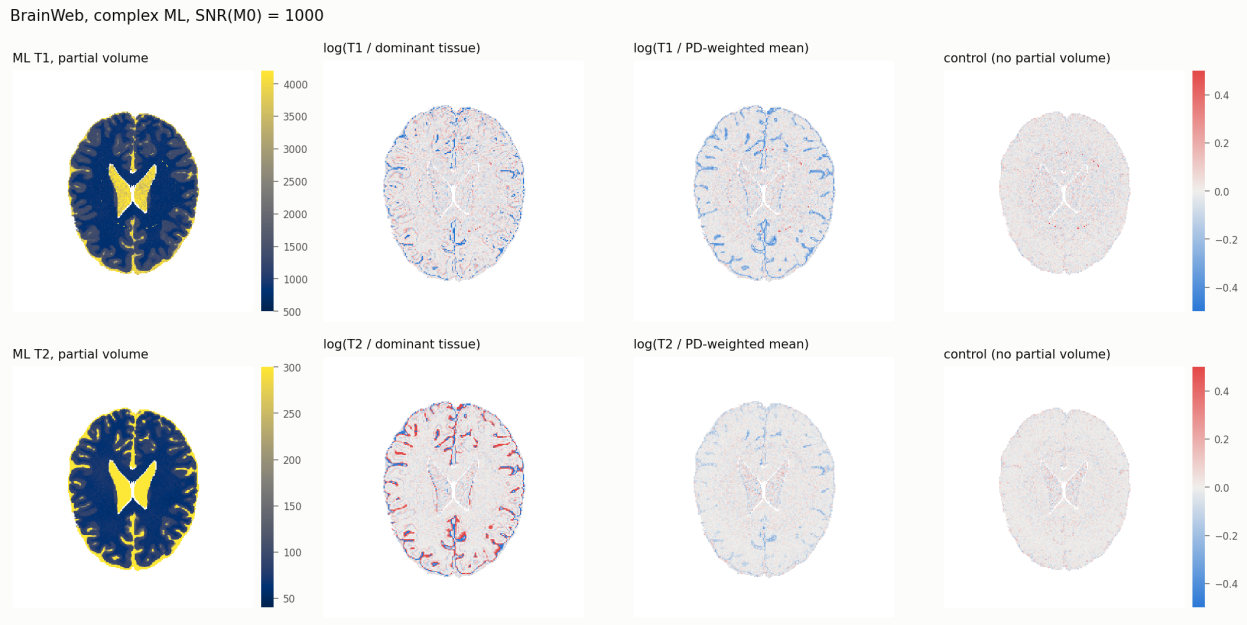


Figure 19: Complex ML at SNR(M_0) = 1000. From left: estimate with partial volume; log-ratio to the dominant tissue's value; log-ratio to the PD-weighted mean; the same anatomy without partial volume (control). Rows: T_1 and T_2.

therefore appears with elevated T_2 and, where CSF dominates, depressed T_1 (Fig. 19). The noise spread in these voxels is similar to the control (Table 24), so the uncertainty maps report small intervals around a biased value: coverage of 95% intervals drops to 21.9%, 0.5% and 26.8%, against 94–95.5% in the control. The residual does not reveal the problem at this SNR: the mismatch is 0.29–0.61σ per measurement, and a χ^2 test at 99.9% flags 3–15% of these voxels at SNR 1000 (predicted 2.2–12.4% from the noise-free residual), rising to 48–96% only at SNR 3000.

The estimators disagree. Complex ML and magnitude least squares return the same values (both fit all measurements), but the analytic inverse does not: its T_2 deviates by -0.14 to -0.50 from the mean (it uses scan 1 only), and in 19.5–38% of the CSF-mixed voxels the mismatch moves its flip-angle equation onto a different root, with B_1^+ about 40% too low and T_1 of 4–9 s even without noise (Fig. 18, right). Complex ML's B_1^+ stays within +4% in every mixed voxel; its error is confined to the relaxation times.

10.4 What partial volume does to the model and the estimators

1. **Partial volume is a model error, not noise.** It biases every single-compartment estimator, does not average out with more data, and is invisible to the CRLB, which assumes the model is correct. Efficiency comparisons (Section 6) remain valid in pure voxels, where they were made.

2. **Its size is governed by distance on the model manifold.** Mixtures of nearby tissues are, to second order, single-compartment signals at the weighted mean (44); they are harmless and should be read as intermediate tissue. Mixtures with CSF are far from the manifold because CSF's long T_2 inflates its contribution to the echo pathways and to the second scan,

up to eight times its volume share: the reported values then depend on the measurements an estimator weights most, and no estimator returns a well-defined average.

3. **It is hard to detect per voxel.** The mismatch residual grows linearly with the CSF fraction while the noise is fixed, so a χ^2 test detects partial volume only at high SNR or after averaging. Per-voxel uncertainty maps become falsely confident where it acts; they should be reported together with a partial-volume indicator, e.g. the distance to a tissue boundary from a segmentation, or the fit of a mixture model.

4. **The shared B_1^+ matters.** Because B_1^+ couples the two scans, an estimator that uses the scans in separate closed-form steps (the analytic inverse) can resolve the mismatch by a large B_1^+ error, whereas the joint fit keeps B_1^+ nearly unbiased and pushes the error into the relaxation times. For the two-stage design this favours estimating B_1^+ from robustly smoothed maps, which the polynomial fit already provides: partial-volume errors are local and are rejected as outliers.

5. **Remedy.** A two-compartment model, $S = C\,[f\,\mathrm{PD}_a s(\theta_a) + (1-f)\,\mathrm{PD}_b s(\theta_b)]$ with shared B_1^+, $\Delta\omega$, ϕ_0, is the natural correction. Fitting both compartments freely is ill-conditioned (13 parameters, near-collinear signals), but with class parameters fixed from pure voxels or a population estimate (the hierarchical model of Section 11.3) the signal is linear in the fractions given B_1^+, $\Delta\omega$, ϕ_0, and variable projection reduces the fit to a small nonlinear problem. CSF fractions should then be well determined, because CSF's signature is distinct, whereas WM/GM fractions will be poorly determined—and, by item 2, need not be. The implicit Jacobian, the CRLB machinery and the FID-sign test carry over unchanged (every compartment's FID has the sign of $\sin\alpha$).

11 Discussion

11.1 Forward model or inverse?

The published protocol is sound from an information standpoint: the near-360° pulse lifts the leading-order invariance that makes B_1^+/T_1 separation hopeless with small angles, and the data contain the information that the bounds promise. The published inverse does not extract it efficiently; joint estimation does, for complex data and, with one bit of phase, for magnitudes. The phantom (Section 9) confirms these conclusions spatially and quantifies the cost of the low-resolution second scan.

No unbiased voxel-wise estimator can beat the bounds of Section 4. Further gains require prior information. Spatial smoothness of B_1^+ pools the B_1^+ information that a second scan has created; from a single scan it supplies none (Section 5, Proposition 7 and Corollary 3). Treating the pooled B_1^+ as known lowers the T_1 bound by a factor of about 1.45 (40.5 → 28.0, reference tissue, $B_1^+ = 1$) only under two conditions. First, the T_1 step must use the full-resolution scan-2 data: with scan 1 only, as in the published low-resolution design, the B_1^+-known bound is 57.1, worse than the 40.5 of joint estimation from both full-resolution scans ($\alpha_1 = 17$–$19°$ lowers it by 5–7% in the worst case and mean over $B_1^+ \in [0.7, 1.3]$ and five tissues, according to the companion flip-angle design note, `docs/flipnote`). Second, the supplied B_1^+ must be accurate to well below $28.0/10.1 \approx 2.8\,\sigma/M_0$ (about 0.9% at SNR 300, 0.3% at SNR 1000), because with both scans and B_1^+ fixed a B_1^+ error enters $\log T_1$ with a factor of +10.1 at $B_1^+ = 1$ that exceeds 100 near the 360° crossing (Table 7 and the text after it). The smoothed maps of Section 9 are accurate to 0.2% (ML pipeline) and biased by 1.5% in grey matter (paper pipeline), so the factor 1.45 is an upper limit, not an expected gain.

Table 25: Origin of each difficulty.

Difficulty	Origin	Remedy
One scan cannot separate M_0, B_1^+, T_1 (Corollary 1)	forward model (exact)	two scans with a common B_1^+
Small flip angles: leading-order invariance (27), severe ill-conditioning	forward model + protocol	a pulse whose effective angle does not scale with B_1^+ (near 360°, as published)
Residual T_1 bound ($40.5\,\sigma/M_0$): confounding with T_2, T_2^*, M_0, B_1^+	forward model + protocol	SNR, averaging, spatial priors; protocol tuning ($\leq 20\%$ for flip angles)
Exact magnitude alias $\alpha_2 = 360° \mp x$ at physiological B_1^+	forward model: magnitudes lose sign $\sin\alpha$	FID-sign test, or complex data with a shared phase
Exact complex alias $\alpha_2 > 540°$	forward model: 2π periodicity	range bound $B_1^+ < 1.64$
Error 3–7× the bound; bias and undefined T_1 at moderate SNR	analytic inverse (sequential, closed form)	joint ML or magnitude least squares with the FID sign
T_1 bound 57.1 instead of 28.0 with B_1^+ known; T_1 bias -2ε per B_1^+ error ε	design: scan-1-only T_1 step, low-resolution scan 2	joint estimation, from both scans at full resolution if affordable (40.5; not with B_1^+ fixed, Table 7)
Estimates on $R_2' = 0$ at low SNR	model domain + small true R_2'	report; Bayesian or constrained bounds
One start insufficient	nonconvex likelihood	multiple starts (affordable with exact Jacobians)
Partial volume with CSF: biased, confidently wrong estimates; estimators disagree (Section 10)	forward model (single compartment)	two-compartment model with class parameters; PV indicator
In-vivo calibration $\beta = 1.24$	model mismatch (not variance)	richer forward model (Section 11.3)

The confounding with M_0 and the transverse relaxation times can only be reduced by priors on the parameter maps themselves. Such estimators are biased by design, and their bias must be measured rather than ignored.

11.2 Limitations

The forward model assumes instantaneous RF pulses, a Lorentzian intravoxel frequency distribution, no diffusion, no magnetisation transfer, no flow or motion, a single compartment per voxel (whose consequences Section 10 quantifies), and perfect steady state; the time from the pulse to the first echo is assumed (4.5 ms). The CRLB is local and assumes an exact model, Gaussian noise, an interior true parameter and locally unbiased estimators. The digital phantom is noise-only and model-consistent: it validates estimators and uncertainty maps but not the model itself.

11.3 Future work

Model mismatch. The in-vivo calibration $\beta = 1.24$ of [1], unexplained by the authors and absent in phantoms, indicates that the forward model misses something in vivo. Candidates, each of which can be added to the simulator and assessed by fitting mismatched data with the current model: (i) finite hard pulses with off-resonance during the pulse (their Eq. 8 nutation function) and an inaccurate ratio of the actual flip angles; (ii) non-Lorentzian intravoxel frequency distributions (e.g. Gaussian or susceptibility-driven), which change $\exp(-R_2'|\tau|)$; (iii) diffusion attenuation of the higher-order pathways under the unbalanced gradient (already available in the EPG simulator); (iv) magnetisation transfer, which lowers the effective longitudinal magnetisation under the large pulse; (v) partial volume, quantified in Section 10, where a two-compartment remedy is outlined; (vi) B_0 drift or motion between the sequential scans, which corrupts the relative phase used by the branch test and by complex ML. The measure of interest is the bias each effect induces in T_1 and B_1^+ relative to the noise spread, and whether a single global factor like β can absorb it. None of (i)–(iii) gives a single scan useful information about B_1^+ (Section 5.4); their importance lies in the bias they cause.

Rician maximum likelihood. Magnitude least squares ignores the Rician noise floor. At the published protocol the induced bias is small (Section 7), but it grows with lower SNR, smaller flip angles or protocols with weaker pathways. Two routes are natural. (i) Direct ML with the Rician likelihood $\log p(m \mid A) = \log m - \log \sigma^2 - (m^2 + A^2)/(2\sigma^2) + \log I_0(mA/\sigma^2)$, whose gradient and Gauss–Newton approximation follow from the exact Jacobian of $A = |s(\theta)|$. (ii) An EM algorithm that treats the lost phase as missing data: with $\kappa = m|s_t|/\sigma^2$, the E-step gives $\mathbb{E}[|z| \mid m] = m\, I_1(\kappa)/I_0(\kappa)$ and the M-step is magnitude least squares on these shrunk pseudo-data, which is the existing fit. Either should be combined with the FID-sign constraint, which the alias of Proposition 6 requires for any magnitude likelihood. The Rician Fisher information of Section 4.7 provides the reference bound.

Hierarchical (empirical-Bayes) estimation. A hierarchical model with tissue classes (a Gaussian mixture in $(\log T_1, \log T_2, \log T_2^*)$) and a smooth parametric B_1^+ field, whose hyperparameters are estimated by EM with a Laplace E-step (MAP fit plus $(J^{\mathsf{T}}J + \Sigma_k^{-1})^{-1}$ per voxel and class), would use the subject's own data as the prior. It targets exactly the weak direction of Fig. 5 and makes the B_1^+ field effectively known; its bias would be evaluated on the phantom, including the lesion.

Learned estimators. Networks trained on simulated signals [3] are biased estimators whose prior is the training distribution. The tools here—exact bounds, efficient ML as a reference, calibrated uncertainty, and the phantom—are what is needed to measure their bias and variance against the CRLB rather than only their average error.

Acquisition. Direct CRLB-based optimisation of T_R, flip angles, echo placement and scan-2 coverage, now cheap (Section 8); and the joint use of a low-resolution scan 2 inside a single likelihood (modelling its k-space truncation) instead of the two-stage pipeline.

A Proofs and derivations

The proofs of Proposition 1, Theorem 1, Lemma 1, Corollary 1, Proposition 3, Proposition 4, Proposition 5 and Proposition 6 are given in the main text. This appendix contains the remaining derivations.

A.1 Direct small-angle expansion of the isochromat solution

This gives the leading term of Proposition 4 in explicit form, independently of Theorem 1. Use the notation of the proof of Theorem 1 with $\varphi = \Delta\omega = 0$, write $m_\perp$ and m_z for the post-pulse state, $m_\perp^- = E_2 \mathrm{e}^{i\psi} m_\perp$ and $Z = m_z^- = E_1 m_z + 1 - E_1$ for the pre-pulse state. The pulse gives

$$m_\perp = m_\perp^- - i \sin\alpha \, Z + i(\cos\alpha - 1) \operatorname{Im} m_\perp^-, \qquad m_z = \sin\alpha \, \operatorname{Im} m_\perp^- + \cos\alpha \, Z.$$

In the steady state $Z = O(1)$ and $m_\perp = O(\alpha)$, so the last term in $m_\perp$ is $O(\alpha^3)$ and

$$m_\perp = \frac{-i\alpha Z}{1 - E_2 \mathrm{e}^{i\psi}} + O(\alpha^3), \qquad |1 - E_2 \mathrm{e}^{i\psi}| \geq 1 - E_2 > 0. \tag{45}$$

With $E_1 = 1 - \rho + O(\rho^2)$, $m_z = Z - \rho(1 - Z) + O(\rho^2)$, and with $\cos\alpha = 1 - \alpha^2/2 + O(\alpha^4)$ the longitudinal equation becomes $\rho(1 - Z) = -\alpha \operatorname{Im} m_\perp^- + \frac{\alpha^2}{2} Z + O(\alpha^4 + \rho^2 + \alpha^2\rho)$. From the transverse result, $\operatorname{Im} m_\perp^- = -\alpha Z (E_2 \cos\psi - E_2^2)/|1 - E_2 \mathrm{e}^{i\psi}|^2$, and

$$\frac{E_2 \cos\psi - E_2^2}{|1 - E_2 \mathrm{e}^{i\psi}|^2} + \frac{1}{2} = \frac{1}{2} H(\psi), \qquad H(\psi) = \frac{1 - E_2^2}{1 - 2E_2 \cos\psi + E_2^2}.$$

Hence $Z = [1 + cH(\psi)]^{-1}$ with $c = \alpha^2/(2\rho)$ to leading order, and

$$m_\perp(\psi) = \frac{-i\alpha}{(1 - E_2 \mathrm{e}^{i\psi})\,[1 + cH(\psi)]} + O(\alpha^3 + \alpha\rho), \tag{46}$$

uniformly in ψ because $1 + cH \geq 1$. Its Fourier coefficients are $\alpha f_k(c, E_2)$ plus a remainder bounded by the uniform remainder, which proves (26) for every k. (Consistency with Proposition 4: $u \to (c-1)/(c+1)$ and $v \to \alpha/(1+c)$.)

A.2 Fisher information of a Rician amplitude

Let $m = |A + n|$ with n circular complex Gaussian of variance σ^2 per component. Then

$$p(m \mid A) = \frac{m}{\sigma^2} \exp\Big(-\frac{m^2 + A^2}{2\sigma^2}\Big) I_0\Big(\frac{mA}{\sigma^2}\Big), \qquad \frac{\partial \log p}{\partial A} = -\frac{A}{\sigma^2} + \frac{m}{\sigma^2} R\Big(\frac{mA}{\sigma^2}\Big), \quad R = \frac{I_1}{I_0},$$

using $I_0' = I_1$. The score has zero mean, so $\mathbb{E}[mR(mA/\sigma^2)] = A$, and

$$\mathcal{I}_A = \mathbb{E}\Big[\Big(\frac{\partial \log p}{\partial A}\Big)^2\Big] = \frac{\mathbb{E}[m^2R^2]}{\sigma^4} - \frac{2A\,\mathbb{E}[mR]}{\sigma^4} + \frac{A^2}{\sigma^4} = \frac{\mathbb{E}[m^2R^2] - A^2}{\sigma^4}.$$

With $x = m/\sigma$ and $a = A/\sigma$, $\mathcal{I}_A = g(a)/\sigma^2$ with g as in (30). As $a \to \infty$, $R \to 1$ and m is approximately Gaussian with mean A and variance σ^2, so $g \to 1$; as $a \to 0$, $R(xa) \approx xa/2$ and $g = O(a^2)$. For several independent magnitudes depending on θ through $A_j(\theta)$, the chain rule gives $\mathcal{I}_{\text{mag}} = \sum_j \mathcal{I}_{A_j} \nabla A_j \nabla A_j^{\mathsf{T}}$, and $\nabla|S| = \mathrm{Re}(\overline{S}\nabla S)/|S|$. We evaluate g by trapezoidal quadrature over $x \in [\max(0, a-14), a+14]$ with exponentially scaled Bessel functions.

A.3 Implicit differentiation

If $A(x)\mathbf{M}(x) = \mathbf{b}(x)$ with A invertible and differentiable, differentiating gives $A'\mathbf{M} + A\mathbf{M}' = \mathbf{b}'$, hence (39). The derivatives of $A = I - R_x(\alpha)R_z(\psi)\mathrm{diag}(E_2, E_2, E_1)$ follow from the product rule; for E_1, $R_z(\psi)\,\mathrm{diag}(0,0,1) = \mathrm{diag}(0,0,1)$ because R_z fixes $\hat{\mathbf{z}}$. The explicit inverse of a 3×3 matrix is $\mathrm{adj}(A)/\det A$ with $\det A$ expanded along the first row.

A.4 The pathway amplitudes are imaginary; the FID-sign identity

Lemma 2. *For $\varphi = 0$ and $\Delta\omega = 0$, every a_k (equivalently every $G_k(u, E_2, 0)$) is purely imaginary.*

Proof. Let $m(\beta)$ be the unique solution of (9) and set $\tilde{m}(\beta) = -\overline{m(-\beta)}$. With $w = E_2 \mathrm{e}^{-i\beta} m(-\beta)$ we have $E_2 \mathrm{e}^{i\beta}\tilde{m}(\beta) = -\overline{w}$, so the right-hand side of (9) evaluated at $\tilde{m}$ is $-\mathrm{Re}\, w - iu\,\mathrm{Im}\, w - iv$. The left-hand side is $-\overline{m(-\beta)} = -\overline{\mathrm{Re}\, w - iu\,\mathrm{Im}\, w - iv} = -\mathrm{Re}\, w - iu\,\mathrm{Im}\, w - iv$. Hence $\tilde{m}$ solves the same equation and, by uniqueness, $m(\beta) = -\overline{m(-\beta)}$. With $\Delta\omega = 0$, $\beta = \psi$ and

$$\overline{a_k} = \frac{1}{2\pi}\int_0^{2\pi} \overline{m(\psi)}\,\mathrm{e}^{ik\psi} d\psi = -\frac{1}{2\pi}\int_0^{2\pi} m(-\psi)\,\mathrm{e}^{ik\psi} d\psi = -a_k.$$

□

By Theorem 1, $a_0^{(i)} = v_i G_0(u_i)$ with $G_0 = -i\,g_0(u)$, g_0 real. The echo factors of the $k = 0$ pathway at echo j are the same for both scans (same echo times, T_2, T_2^*, $\Delta\omega$, ϕ_0; with $\Delta\omega \neq 0$ both scans acquire the same phase $\mathrm{e}^{ik\Delta\omega T_R}$, which is 1 for $k = 0$). Hence

$$S_{1j0}\overline{S_{2j0}} = M_0^2\, v_1 v_2\, g_0(u_1) g_0(u_2)\, |D_j|^2,$$

which is real with sign $\mathrm{sign}(v_1 v_2)\ \mathrm{sign}(g_0(u_1)g_0(u_2))$. Numerically $g_0 > 0$ (the phase of a_0/v is exactly $-\pi/2$) for all 126 combinations of $\alpha \in [2^\circ, 359^\circ]$, $T_1 \in \{300, 1500, 4000\}$ ms and $T_2 \in \{30, 70, 1500\}$ ms that we tested, so the sign is that of $v_1 v_2$, i.e. of $\sin\alpha_1 \sin\alpha_2$: this is the test (37), and its agreement with $\mathrm{sign}(\sin\alpha_1 \sin\alpha_2)$ is a unit test over $B_1^+ \in [0.9, 1.3]$. Lemma 2 also explains why the published method can work with real-valued states (their Eq. 18): all configuration amplitudes share the phase $\pm\pi/2$.

B Algorithms

Algorithm 1 Projected Levenberg–Marquardt with exact Jacobian (batched over voxels)

Require: data y, start θ_0, model-and-Jacobian $\mathcal{S}(\theta) \mapsto (s, J)$, projection Π, fixed mask $\mathcal{F}$

$\theta \leftarrow \Pi(\theta_0)$; $(s, J) \leftarrow \mathcal{S}(\theta)$; $J_{:,\mathcal{F}} \leftarrow 0$; $r \leftarrow s - y$; $c \leftarrow \|r\|^2$; $\lambda \leftarrow 10^{-2}$
for $n = 1, \ldots, n_{\text{iter}}$ **do**
  $H \leftarrow J^\mathsf{T} J + \lambda \operatorname{diag}(J^\mathsf{T} J) + \operatorname{diag}(\mathbb{1}_{\mathcal{F}})$; $g \leftarrow J^\mathsf{T} r$
  $\theta' \leftarrow \Pi(\theta - H^{-1} g)$; $(s', J') \leftarrow \mathcal{S}(\theta')$; $J'_{:,\mathcal{F}} \leftarrow 0$; $r' \leftarrow s' - y$
  if $\|r'\|^2 < c$ **then**
    $(\theta, r, J, c) \leftarrow (\theta', r', J', \|r'\|^2)$; $\lambda \leftarrow 0.3\lambda$
  else
    $\lambda \leftarrow 10\lambda$
  end if
  $\lambda \leftarrow \min(\max(\lambda, 10^{-9}), 10^{9})$
end for
return θ, c, constraint flags

Algorithm 2 FID-sign branch constraint for the magnitude fit

Require: complex images (for the sign only), start $B_1^{+(0)}$

$\sigma_{\text{FID}} \leftarrow \operatorname{sign} \operatorname{Re} \sum_j S_{1j0} \overline{S_{2j0}}$ ▷ $= \operatorname{sign}(\sin \alpha_1 \sin \alpha_2)$
zeros $\mathcal{Z} \leftarrow \{n\pi / \alpha_i^{\text{nom}}\}$, sorted
if $\operatorname{sign}(\sin B_1^{+(0)} \alpha_1^{\text{nom}} \sin B_1^{+(0)} \alpha_2^{\text{nom}}) \neq \sigma_{\text{FID}}$ **then**
  reflect $B_1^{+(0)}$ across the nearest $z \in \mathcal{Z}$: $B_1^{+(0)} \leftarrow 2z - B_1^{+(0)}$
end if
$[\ell, h] \leftarrow$ interval between consecutive zeros containing $B_1^{+(0)}$
run Algorithm 1 with Π augmented by $B_1^{+} \leftarrow \min(\max(B_1^{+}, \ell), h)$

Algorithm 3 ML estimation with multiple starts

starts: model-fit estimate; analytic inverse (where defined); model fit with $T_1 = 700$ and 2000 ms
initialise ϕ_0 per start in closed form; set Π: $R_2' \geq 0$, box, $B_1^{+} \leq 540/330$
run Algorithm 1 from each start; keep, per voxel, the lowest final cost
optionally: uncertainty $\sqrt{\operatorname{diag}(J^\mathsf{T} J)^{-1}}\, \sigma$ at the solution

Algorithm 4 Two-stage reconstruction for a low-resolution scan 2

low-pass scan 1 to the k-space support of scan 2
stage 1: estimate B_1^{+} voxel-wise from the two low-resolution scans (analytic inverse, or Algorithm 3)
smooth: fit a 2D (3D) polynomial to B_1^{+} inside the mask, rejecting outliers (3×MAD)
stage 2: estimate $T_1, T_2, T_2^{*}, M_0, \Delta\omega, \phi_0$ from full-resolution scan 1 with B_1^{+} fixed (Eqs. 1, 16, 17; or Algorithm 1 with $\mathcal{F} = \{B_1^{+}\}$)

C Reproducibility

All results are produced by the code in the accompanying repository (`github.com/scjoshi/MPME`). From the repository root,

```
make technote
```

reruns every script in the code map except the reference Monte Carlo and the benchmark, regenerates every table, figure and macro with the fast (implicit-Jacobian) settings, and compiles this note. Numbers quoted in the running text are taken from the outputs of these scripts (`results/*.json` and standard output) and must be updated by hand after a rerun. `make technote-reference` reruns the Monte Carlo experiments with the forward-difference reference settings used for Tables 14 to 17.

Table 26: Code map.

Component	File
Protocol, echo timing	`src/mpme/sequence.py`
Steady state (isochromat, EPG)	`src/mpme/epg.py`
Voxel signal (21)	`src/mpme/signal.py`
Published inverse	`src/mpme/paper.py`
Model-fit baseline	`src/mpme/analytic.py`
Complex ML, magnitude LS, branch test	`src/mpme/mle.py`
Exact Jacobians	`src/mpme/fastjac.py`
Fisher information, CRLB, Rician, uncertainty maps	`src/mpme/crlb.py`
Digital phantom, low-resolution scan 2, polynomial fit	`src/mpme/phantom.py`
BrainWeb partial-volume phantom	`src/mpme/brainweb.py`
CRLB tables and figures (Section 4)	`scripts/crlb_analysis.py`
Estimator efficiency (Tables 14 to 16)	`scripts/estimator_efficiency.py`
Magnitude comparison (Table 17)	`scripts/magnitude_comparison.py`
One start vs four (Section 6)	`scripts/start_count.py`
Single-scan family examples (Tables 4 and 10 and Fig. 2)	`scripts/single_scan_family.py`
Shared-B_1^+ joint fit (Table 11)	`scripts/shared_b1_check.py`
Physics outside the ideal model (Table 12)	`scripts/beyond_ideal_model.py`
Single-scan degeneracy, scan-2 design (Section 5)	`scripts/two_scan_design.py`
Jacobian benchmark (Table 18)	`scripts/benchmark_fit.py`
Phantom experiment (Section 9)	`scripts/phantom_experiment.py`
Partial volume (Section 10)	`scripts/brainweb_experiment.py`
Figures and tables of this note	`scripts/technote_figures.py`
Unit tests	`tests/` (`pytest`)

Settings. Noise seeds are fixed in every script (Monte Carlo: the SNR value; phantom: seed 1 for noise, seed 0 for the phantom). Software: Python 3.12.2, PyTorch 2.6.0, NumPy 2.3.5, SciPy 1.16.3, Matplotlib 3.10.7; laptop CPU with 8 threads, no GPU. Run times quoted in the note refer to this machine.

BrainWeb data. The four BrainWeb volumes (normal brain, 1 mm, fuzzy CSF, grey and white matter, and the discrete model; raw bytes, gzip) are downloaded from the BrainWeb site (`brainweb.bic.mni.mcgill.ca`) into `data/brainweb/`; they are not redistributed with the repository.

Assumptions not stated in [1]. The time from the RF pulse to the first echo (4.5 ms), instantaneous RF pulses, and the polynomial degree used to smooth the flip-angle map (4 here).

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